United Confederation Navy

Academy Training Manual

v. 1.0

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IMPORTANT

Despite what it may say on later pages - **NONE** of this information is vital or critical to be able to play and enjoy Bridge Command! Please ensure **your** experience is what **you** want it to be! The only wrong ways to play Bridge Command are ways that make the game less fun for your fellow officers. Therefore:

Do not pass this information on to people who don't want to know it - not everyone wants to play the game in the same way.

Do not criticise other people for not knowing this information - if they're having fun, they're good players, and they're playing the game well. Don't force your opinion of what is "right" on them... that includes what's in here. This is *unofficial*, remember!

Do not share this manual with first-timers - Bridge Command should be first experienced in person to truly understand its magic. If anyone wants a copy of this manual AFTER, please direct them to the link/QR code so they can agree to the T&Cs.

Who is this for - What is this about?

As soon as I completed my first mission at Bridge Command, I knew I wanted to return. Not merely return, but be good at it [1]. I wanted a crew of professionals… officers who could step aboard, take their stations, and be good at what they do.

There is far more depth to each station than can be covered in the limited time available to Flight Controllers and Trainers. Even Empty Epsilon (which I promptly downloaded upon discovering its existence) reveals little of its inner workings. For example, there are no official advanced guides explaining what those delightful squiggly lines on the Radar actually mean. There are, however, some excellent deep dives scattered across the Bridge Command Discord - and I wanted to bring them together, along with my own investigations into the Empty Epsilon code... so I did.

This guide is intended for those who wish to learn as much as possible about crewing a ship in Bridge Command, without peering too far behind the curtain. It has been deliberately written as a manual issued to cadets upon entry to the Academy. This manual is an interpretation of what new recruits to the United Confederation Navy would receive on Day 1 of their training. None of the data should (in my opinion) be unknown to an Academy graduate.

It is highly recommended that you experience Bridge Command completely fresh, and learn your role from the excellent training officers. Official training videos are also available here: https://bridgecommand.space/training-videos/ These should be your first port of call - they cover the fundamentals superbly.

To reiterate - **if you have never crewed on a Bridge Command ship, stop reading immediately!**

This does not teach you **how** to play Bridge Command, or operate any of the crew stations. However, once you've flown, if you find yourself wanting a little more insight into how the ship truly functions, a deeper understanding of its systems, and a stronger connection to the world around you, then I invite you to enjoy this labour of love. Go Beyond!

SLIGHTLY LESS IMPORTANT

Numbers are drawn from the open-source code of LARP Empty Epsilon. Bridge Command has almost certainly adjusted these at the core - and rightfully reserves the right to meddle further on a whim. This is not gospel, merely an approximation. Treat it accordingly: this is a guide, not a perfectly precise technical manual! Also, be aware, the limits placed on players are not the same as the options available to Flight Controllers.

Icons are likewise taken from the same source, so may not match the ones used by Bridge Command… you have been warned!

ALSO SLIGHTLY LESS IMPORTANT

This is our concept of official nomenclature, slang and standard operating procedures (SOPs). If you use different colour schemes, naming conventions, etc, that is totally fine. This is an *unofficial* Academy Manual. If you want to use what has been amalgamated here in your own crews/games, that would be lovely, and we would love to hear if it's been useful. If you are in a mixed group, and people are not using the same SOPs, then *this is fine*!

[1] "Good" of course is entirely subjective. This is my definition of "good". Yours may be different and equally valid!

Contributors and Thanks

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First and foremost, the wonderful Ymbirtt, whose remarkable deep dives into the technical aspects of EE, BC, and (of course) the superb SA-01 Angel Directive Operating Procedures aka "The Bad Book" truly set this project in motion. Thank you for your patience, technical expertise, and unwavering generosity with your time and knowledge. Sir, I cannot thank you enough for your help.

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Lt. Flamingo

(Apologies if anyone has been promoted since this Manual has been published!)

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And finally, my wife, Emilie - who has indulged my hyperfixation on this project with remarkable grace and affection, and who is the finest gorram engineer and proofreader a Captain could ask for.

- Callum "Face" Fraser

Change Log

Version	Date	Author	Description
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If reading a physical copy, please check the version number.

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Glossary of Terms

The terms below are often used by officers for concise communication, or appear in this manual to explain technical concepts.

° heat	The unit of heat measurement in a system. When it reaches 100°, any further increases will instead cause the system to take damage.
Bearing	The direction of an object that is visible on Long-Range Sensors or Tactical Display (0° to 359.9°).
Boost	A sustained increase of power to a specified system. Maintained until released or reduced due to power or overheating constraints.
Contact / Bogey	An unidentified entity.
Front/Rear Gen	The shield generators located fore and aft. Refers to the *system*, not the shield field.
Friendly / Ally	A confirmed allied contact.
Front/Rear Shields	The active protective field surrounding the ship.
Heading	The bearing a ship is facing.
Hostile / Tango	A confirmed enemy contact (threat).
Impulse	The impulse drive system.
MJ	Megajoules - the standard UCN unit of energy output and consumption. Engineering consoles calculate loss in MJ/min (Megajoules in a minute).
Neutral	A confirmed non-hostile contact.
Reset/Resetting	Returning a system to baseline (100%) power.
Reciprocal Bearing	The opposite bearing from a given direction, calculated by adding or subtracting 180°. If a ship bears 040° from you, your reciprocal bearing from that ship is 220°.
Sector	An area of space represented by the gridline on the Strategic Map. Each Sector is a 20U square (each subdivided into an 2.5U 8x8 grid represented by dots).
Slug	High Velocity Lead Impactor (HVLI).
Spike	A temporary burst of power to a specified system. Duration is limited to a few seconds and must be followed by an immediate reset to prevent overheating. Unless otherwise called, spikes are assumed to be to 300% power.
Thrusters	The manoeuvring system.
Tracker	A Torpedo.
Tractor	The tractor beam system.
U	Unit. The standard distance measurement scale used by all UCN craft. A Sector is 20U. Long-Range Sensors have a maximum view of 30U whilst the Tactical Display (Helm, Weapons and Drones) has a range of 5U.
Warp	The warp drive, or a measurement of warp speed.

Foreword

Cadet,

It is my distinct privilege to welcome you to the United Confederation Naval Academy.

You arrive at a moment previous generations could scarcely have imagined. Humanity stands not on the edge of survival, but on the threshold of expansion - united across worlds, sustained by discovery, and safeguarded by those who choose to serve.

You have chosen to serve. That choice alone sets you apart.

But understand this: the uniform you now wear is not a reward. It is a responsibility. It carries the weight of all those who came before, and of all those who now rely upon this Navy to keep them safe.

Everything you learn here rests on a simple truth: humanity endured not because of technology alone, but because individuals chose discipline over chaos, duty over self-interest, and unity over division. That is what is now expected of you.

During your time at the Academy, you will be tested. Intellectually, physically, and morally. You will be placed under pressure. You will be asked to lead, to follow, and, at times, to stand alone.

You will not be expected to be perfect. But you will be expected to rise.

Beyond what you believe yourself capable of.

Beyond what others demand of you.

Beyond the limits that defined those who came before.

Because that is what service requires.

This manual is your foundation.

It reflects the reality of a Confederation vessel, where no role exists in isolation, and no action occurs without consequence.

You will begin with **Core Principles**: communication, prioritisation, alert levels, and system effectiveness. These apply to every officer.

You will then study how to **read the battlespace** - the Strategic Map, long-range sensors, and the Tactical Display - because no one view can provide all the information you will need.

From there, you will learn the **shared mechanics** of the ship itself: power and heat, shields, scanning, beams, ordnance, tractor operations, hacking, and how to repair the systems that govern how a vessel survives and fights.

You will then study each officer role in depth.

First is **Command**: Captain and First Officer. How decisions are made and carried out.

Next is **Operations**: Helm and Weapons. Position, survival, and force. Here is where battles are won and lost.

Then **Science**: Radar, Navigation, and Comms. Awareness, information, and control of the engagement.

Finally, **Engineering**: Chief/Manual Engineers and Drones. How to control the power, heat, and stability of the systems, and how to project the ship's will further afield.

Reference material and advanced doctrine follow. Not required for competence, but essential for mastery.

If you intend to command, study all of it. If you intend to specialise, master your station - but never mistake specialism for completeness.

A ship does not function in parts. Understand the ship, then understand your role. Because when the moment comes, you will not rise to the occasion… you will fall to the level of your preparation.

Ensure that level is worthy of the uniform you wear.

Welcome to the Academy, Cadet. Go Beyond!

- Admiral Helena Voss

June 2181
Commandant,
United Confederation Naval Academy

1.0 - Core Principles

1.1 - Communication

"Good communication can turn a good crew into a great crew. Bad communication can turn a good crew into their constituent atoms."

Communication is key to any successful UCN crew and mission. This section does not refer to the Comms station, but rather the communication between crew members during operations.

Communication differs significantly between combat and non-combat situations. When beams and missiles are flying, it is vital that the right information reaches the right people at the right time.

When communicating, remember your ABCs:

Accuracy - factually correct
Brevity - as concise as possible
Clarity - easily understood and unambiguous

Whilst most communication should be in plain language, combat introduces stress and unpredictability. A crew must remain composed and coordinated - nothing enables this more than disciplined communication.

> We lost the initiative before we lost the fight. Three contacts came out of the dark, and instead of clean reports, the bridge filled with noise - "shields low," "taking fire," everyone talking, no one saying anything useful. Engineering didn't know where to put power, and Helm moved without warning.
>
> By the time someone finally gave a proper report, the damage was already done. Nothing wrong with the ship or the crew... just… poor communication.
>
> *-Cmdr. Iella Wessiri*
> *UNCA - Command School*

1.1.2 - When to communicate

When you have:

- Information another station needs.
- A request of another station.

Avoid contextless statements such as

"Shields low."

This adds noise and forces others to interpret meaning under pressure. Instead, provide usable information:

"Forward shields at 20%, failing."

1.1.3 - Type of communication

Contact Report

Typically used by the Radar station to report newly detected or observed external entities of tactical relevance. For more details see **6.1 - Radar** on page 84.

Action Request

Used when a crew member requires another station to perform a specific action.

Most commonly, this is a request for a temporary power increase to a system, referred to as a "Spike." For a full breakdown of Spikes, see **Appendix E: Power Spikes** on page 129.

Status Report

Provides the current condition of systems, crew, or tactical situation.

The content of a status report varies depending on whether the crew is in combat or not. Each section of this manual outlines expected reporting standards.

Acknowledgement

Confirms receipt of a request, and whether it will be complied with.

Where possible, include a reason for refusal.

Example:

Helm: "*Spike thrusters.*"
Engineering: "*Deny, thrusters overheating.*"
Helm: "*Spike front shields.*"
Engineering: "*Confirm. Spike front shields.*"

Advisory/Warning

- Used to highlight developing or immediate issues.
- Often delivered as truncated status reports.

1.1.4 - Communication Discipline

During combat, a significant volume of communication will be occurring across the bridge.

Officers should address their intended recipient by station or name before delivering their message, unless using established standard callouts.

Crew members are expected to listen for their own station or name and filter out unrelated traffic. This is particularly important for the Captain, who must prioritise incoming information rapidly.

Some stations must also listen for key terms in addition to direct address. For example, Engineering should remain alert for terms such as "Spike" or "Boost", even if not directly called.

At times, transmissions will overlap. If a message is missed, officers should clearly request repetition, addressing the original speaker by station or name.

When the Captain is speaking, they are not to be interrupted except in cases of immediate danger. Similarly, information being delivered to the Captain generally takes priority over other communications.

All officers are expected to exercise sound judgement when determining whether to speak, interrupt, or defer.

1.1.5 - Hierarchy of Information

Not all information is equal. The following hierarchy should be used to determine priority when speaking:

1. **Information critical to the ship's immediate survival.**
   (e.g. incoming fire, shield failure, imminent collision, reactor instability).
2. **Information required for immediate command decisions.**
   (e.g. new threats, target changes, mission-critical developments).
3. **Time-sensitive tactical information.**
   (e.g. firing solutions, manoeuvre windows, system effectiveness changes).
4. **Strategic or situational updates.**
   (e.g. objective locations, asset status, developing contacts).
5. **Routine status updates.**
   (e.g. system levels, confirmations, non-urgent reports).
6. **Background or informational detail.**
   (e.g. contextual data, analysis, or observations not requiring action).

We started picking up odd readings, nothing dramatic, just… wrong.

Turns out those readings matched the ship we were looking for! There was an update to the briefing mid-mission, but no-one thought to tell the science team! If we'd heard it, we'd have known exactly what we were looking at.

Instead, we spent half the engagement looking for a ghost we could already see!

- Lt. Jesmin Ackbar
UCNA - Radar Training School

1.2 - Alert States

"An alert state is a single order given to the whole ship at once."

Alert States are a form of shorthand communication. Rather than briefing every station individually, an alert state tells the whole bridge what posture the ship is adopting and what level of readiness is expected.

They do not replace the Captain's orders, but provide a shared starting point: if every officer understands what Yellow Alert or Red Alert requires of their station, the crew saves time and avoids repeated instructions.

Both Yellow Alert and Red Alert may be triggered or reset from either the Captain's console or Comms. Normally, the Captain will order the alert state, though Comms may trigger one immediately if urgent external information makes delay unsafe. Whenever an Alert State is changed, the officer triggering it should state the reason clearly.

Examples:

- "Yellow Alert. Unknown contacts ahead."
- "Red Alert. Incoming ordnance."
- "Reset Alert. Return to normal operations."

An alert is only useful if the crew understands what it means. Coloured lights are not a substitute for officers knowing what to do next.

1.2.1 - Normal Operations

Normal Operations indicate no immediate threat to the ship. The vessel may be travelling, patrolling, docked, recovering, investigating, negotiating, or conducting mission tasks without confirmed tactical danger.

Under Normal Operations, officers maintain routine awareness while prioritising efficiency, recovery, and mission progress. Shields are normally down unless conditions require otherwise, weapons remain safe unless directed, and Engineering should use the opportunity to recover energy, reduce heat, and restore systems. It is not inattention; it is simply not heightened readiness.

1.2.2 - Yellow Alert

Yellow Alert indicates possible danger, uncertainty, or increased readiness.

It should be used when the ship is not yet in active combat, but may need to respond quickly.

Common reasons for Yellow Alert include unknown contacts, suspicious sensor readings, approach to hostile or uncertain areas, hazardous space, concerning mission intelligence, dangerous communications traffic, or preparation for boarding, docking, interception, or investigation.

Yellow Alert means: **prepare to act**.

It is not a combat order. It is a readiness order. Officers should reduce uncertainty, prepare their systems, and ensure the ship can respond quickly if the situation deteriorates.

1.2.3 - Red Alert

Red Alert indicates immediate danger, active combat, or emergency action.

It should be used when the ship is under direct threat, about to engage, receiving incoming fire, facing imminent collision or ordnance impact, or dealing with a critical internal emergency.

Common reasons for Red Alert include confirmed hostile engagement, incoming ordnance, active weapons fire, mine detonation risk, collision danger, critical system failure, warp-core instability, boarding threat, nuclear/EMP/area-effect danger, or the Captain's decision to enter combat posture.

Red Alert means: **act now**.

Under Red Alert, all stations shift to combat discipline. Reports must be concise, orders acknowledged, and non-essential chatter stopped. The bridge should become quieter, not louder.

Panic is not improved by amplification.

1.2.4 - Officer Responsibilities by Alert State

Role	Yellow Alert	Red Alert
Captain	State the reason for the alert, clarify current intent, decide whether shields or weapons should be prepared, and set engagement restrictions.	Set the immediate priority: fight, withdraw, protect, disable, delay, or survive. Authorise escalation and keep the mission objective visible.
First Officer	Confirm key stations have understood the alert state. Follow up on unacknowledged tasks and reduce unnecessary bridge noise before pressure increases.	Enforce communication discipline, chase missed acknowledgements, coordinate secondary problems, and protect the Captain from avoidable clutter.
Helm	Shift from routine movement to cautious positioning. Maintain room to turn, dodge, withdraw, or present shields. Confirm escape routes where appropriate.	Maintain survivable positioning, protect weak shield facings, keep weapons arcs useful, avoid ordnance and hazards, and disengage if ordered.
Beams	Check shield status, prepare to raise shields, confirm frequencies, prepare tractor settings, and identify likely targets if visible.	Manage shields, report shield thresholds, engage designated targets, use tractor defensively or tactically, and respond to heat warnings.
Missiles	Check loaded ordnance and reserves, monitor firing lines, prepare tactical options, and avoid restricted weapons without authorisation.	Deliver authorised ordnance at the correct moment, call firing-line hazards, manage non-lethal beams, and confirm nuclear authorisation before loading or firing.
Radar	Identify unknown contacts, monitor signal traces, scan likely threats, link likely targets, and warn before contacts enter tactical range.	Report of the current target's status, report and scan new threats, call shield weaknesses, warn of reinforcements, hazards, and incoming ordnance.
Navigation	Mark hazards, objectives, and contacts. Plot safe routes, withdrawal routes, and probe routes. Advise if current course increases risk.	Maintain strategic context, update routes as the fight moves, mark threats and safe withdrawal options, and warn if the ship is being drawn away from the mission. Hack targets if part of Tactical Strategy.
Communications	Monitor external traffic, request clarification, warn friendly assets, prepare authorisation requests, and pass new locations or objectives to Navigation.	Relay priority traffic, request or confirm authorisations, warn allied assets and shuttles, pass firing restrictions, and record critical changes. Perform PsyOps on enemy leaders. Hack targets if part of Tactical Strategy.
Engineering	Prepare power and coolant posture. Ensure energy is available for likely spikes and pre-cool key systems where appropriate.	Prioritise combat capability and survival. Support shields, movement, weapons, scanners, or warp as needed. Deny unsafe requests clearly.
Damage Control	Check system conditions, pre-position SPANRs near likely critical systems, and prepare for manual repairs.	Repair systems by priority. Report critical failures, move SPANRs decisively, and handle manual repairs immediately.
Dock and Drone	Check drone readiness, prepare scouting or interception options, coordinate reconnaissance, and warn if drones may affect firing lines.	Deploy, recover, intercept, distract, scout, or support as ordered. Warn Helm and Missiles of drone positions near danger-close effects.

1.2.5 - Resetting Alert States

When the immediate threat has passed, the alert state should be reset deliberately.

Resetting from **Red Alert** to **Yellow Alert** may be appropriate when combat has paused but danger remains possible.

Resetting from **Yellow Alert** to **Normal Operations** may be appropriate when contacts are identified, the area is secure, the ship has disengaged, or the Captain determines that heightened readiness is no longer required.

The officer resetting the alert should state the new posture clearly.

Examples:

- *"Reset to Yellow Alert. Hostiles destroyed, but remain cautious."*
- *"Reset Alert. Return to normal operations."*
- *"Remain at Yellow Alert until Radar clears the nebula."*

Resetting the alert does not mean officers stop paying attention. It means the ship's posture has changed. Recovery, repair, reloading, cooling, scanning, and route correction should begin as soon as the situation allows.

[[Behind the Curtain]]

Bridge Command is a game, and the alert lights, sirens, and sound effects are part of the atmosphere. However, they can also be uncomfortable or distracting for some players, especially those with sensory sensitivities. Even players without specific sensory issues may find repeated alarms irritating or tiring after a while. There is only so much dramatic klaxon a person can take before "immersion" becomes "please make it stop".

If you think this may be an issue, speak to your Flight Controller before the mission starts. If the alerts become too much during play, it is completely reasonable to ask for them to be reduced, reset, or turned off.

The important part is the crew posture, not the flashing lights. You can still call "Yellow Alert" or "Red Alert" and have everyone respond appropriately without needing the room to behave like a nightclub having a crisis!

1.2.6 - Summary

Alert States exist to align the crew.

Normal Operations means the ship is not under immediate threat. Officers maintain routine awareness, efficiency, and mission focus.

Yellow Alert means possible danger. Officers prepare systems, reduce uncertainty, and ensure the ship can respond quickly.

Red Alert means immediate danger. Officers act under combat discipline, prioritising survival, tactical execution, and mission-critical communication.

A well-used alert state gives every officer a shared expectation before orders become urgent. It reduces hesitation, focuses attention, and helps the ship shift from routine operation to combat readiness without requiring the Captain to personally remind each station how not to die.

1.3 - System Effectiveness & Power Allocation

"Capability comes from systems that are powered, intact, and ready. Anything less is optimism with a control panel."

Every system aboard a UCN vessel operates according to a single governing principle: System Effectiveness.

System Effectiveness is the true measure of a system's operational capability, derived from power allocation, damage, and interference.

While officers may assign power freely, it is Effectiveness, not raw power, that determines performance.

A talented engineering crew will be able to get the most effective systems for the least available power.

1.3.1 - Determining System Effectiveness

System Effectiveness is influenced by three primary inputs:

Input	Effect on System Effectiveness
Power Allocation	Increases effectiveness
Damage	Reduces effectiveness
Hacking	Temporarily suppresses effectiveness

Table 1-2: System Effectiveness Influences

All systems default to 100% power, representing normal operation.

Effectiveness can be calculated as:

Power Allocation - **Damage** - **Hacking** = Overall effectiveness.

Power allocation comes from Engineering. **Damage** and **hacking** come from enemy action. Both enemy actions can be mitigated using SPANRs.

For Engineering and Power - See page 14
For System Damage - See page 33
For Hacking - see page 40
For System Repair/SPANRs - See page 52

1.2.1 - Effects of Increased System Effectiveness

Below is an overview of each system. For a more in depth table, see **Table 3-13: Main System Reference Table** on page 21.

System	Effectiveness Increase
Reactor	Power generation increases effectiveness with **diminishing returns**.
Beam Weapons	Rate of fire scales **linearly** with effectiveness.
Missiles	Reload speed scales **linearly** with effectiveness.
Manoeuvring	Rotation speed scales **linearly** with effectiveness.
Impulse	Forward/backward speed scales **linearly** with effectiveness.
Warp	**Linear** for power drain, **non-linear** in practical use.
Shield Generators	Recharge speed scales **linearly** with effectiveness. **Non-linear** defensive return.
Scanner	Scanner range and scanner alignment time scales **linearly** with effectiveness. Increase in power increases drone range **dramatically** .
Cargo Docks	Inter dock transfer speed and tractor beam movement speed scales **linearly** with effectiveness.

Table 1-3: System Effectiveness Increase Summary

Linear scaling means 200% power means the system is 200% effective.

1.3.3 - Summary

- Power allocation sets the maximum capability.
- Effectiveness determines the performance.
- Most systems scale linearly - but not all.

2.0 - Reading the Battlespace

"No single screen shows the whole fight. Survive by knowing what yours misses."

A UCN vessel does not see the battlespace through a single window.

Different stations view the same engagement through different displays, each built for a different purpose. A wise officer understands not only what their own screen shows, but what it does not.

As a general rule:

- The Strategic Map provides context.
- Long-Range Sensors provide warning.
- The Tactical Display provides immediacy.

No single display is sufficient on its own.

2.1 - Strategic Map

"A crew that doesn't read the Strategic Map may still win a fight. It may even be the right fight, if fortune is feeling charitable."

Officer Roles

- Navigation
- Communications

The Strategic Map is the ship's broad operational picture, showing the mission area in Sectors.

It is used to understand where the ship sits in relation to mission objectives, friendly assets, hazards, and the wider battlespace. It is the primary tool for planning movement beyond the immediate engagement, and for maintaining awareness of how the current fight relates to the mission as a whole.

A Sector is the standard large-scale unit of reference. Each **Sector** is a **20U** square. Officers using the Strategic Map should therefore think in terms of position, route, and intent, rather than immediate weapons employment.

The Strategic Map is particularly valuable because it displays scan coverage around friendly assets. This includes the ship, probes, drones, shuttles, and other allied units where applicable. Increased scanner power extends that coverage, allowing the ship to see farther through *all* friendly assets with scanning capabilities simultaneously.

This makes the Strategic Map the preferred display for:

- Route planning.
- Objective tracking.
- Probe deployment.
- Monitoring wider movement.
- Understanding where the fight may develop next.

It is, however, not the best display for immediate ship handling. It tells you where you are in the galaxy. It does not tell you anything about what will hit you in the next few seconds.

Neither the Strategic Map nor Long-Range Sensors will show missiles or beam arcs of ships.

White circles on the Strategic map show a 5U radius view around your ship, allied ships (including Drones), and probes that *your* ship has launched (for more, see **3.10 - Probes and Drones** on page 47).

Boosting scanner effectiveness increases this radius. For a visual reference see **Figure 3-5: Scanner Range at different power scales** on page 20.

If something is in range of your current sensor radius, then this will show up in your Strategic Map. Your Strategic Map is not blocked by nebulae.

Note: Boosting scanner effectiveness may allow the Strategic Map to see objects in nebulae that Long-Range sensors will be blocked by.

For examples of the Strategic Map, see **Appendix I - Battlespace Examples** on page 139.

2.2 - Long-Range Sensors

"Long-Range Sensors turn distant uncertainty into useful warning. Ideally before it starts shooting."

Officer Roles

- Radar
- Dock and Drone (view only)

Long-Range Sensors provide the ship's most detailed picture of the external battlespace at range. The display has a maximum range of **30U.**

Where the Strategic Map is concerned with context, Long-Range Sensors are concerned with detection, identification, and warning. It is the display most likely to tell the bridge that a problem exists before anyone else can see it clearly.

Long-Range Sensors can detect, classify, and track distant contacts beyond the view of Tactical stations. On standard displays, shipboard Long-Range Sensors extend vastly farther than Tactical, and can also make use of probes to extend the ship's awareness into remote areas.

Long-Range Sensors are therefore the bridge's principal tool for:

- Early warning.
- Identifying incoming threats.
- Tracking contacts outside local view.
- Linking targets for analysis.
- Monitoring unusual signatures, including warp activity.
- Supporting Helm and Weapons before the enemy is locally visible.

Long-Range Sensors see farther than Tactical. It does not, however, dictate the ship's movements. It provides awareness, not control.

For examples of the Long-Range sensors, see **Appendix I - Battlespace Examples** on page 139.

2.3 - Tactical Display

"The Tactical Display shows what matters now. Regrettably, what matters now is often armed."

Officer Roles

- Helm
- Weapons (Beams and Missiles)
- Dock and Drone (Drone piloting)

The Tactical Display is the immediate battlespace around the ship.

This is the display used most directly by Helm, Weapons, and Drones to manage positioning, arcs, local hazards, weapon employment, and close-range threats. It is the screen concerned most with immediate combat.

The Tactical Display used by Helm, Weapons, and Drones has a range of **5U**. That is enough to fly the ship, dodge immediate threats, place ordnance, manage arcs, and track nearby contacts. It is not enough to provide the whole picture.

If something is within 5U of your ship, it will be visible on the Tactical Display, including if it is hiding in a nebula.

This makes the Tactical Display the principal display for:

- Immediate manoeuvring.
- Shield-facing decisions.
- Local firing lines.
- Close hazard avoidance.
- Tractor use.
- Local drone control.
- Missile timing and evasion.

2.4 - Visual Identification

"Sensors are excellent. Eyes remain stubbornly difficult to jam."

The main viewscreen should not be ignored. Sensors have defined ranges and limitations, but visual observation can reveal ships, hazards, mines, warp jammers, and objects inside nebulae, that may not be clearly represented on a console display.

Lighting, angle, colour, and background all affect visibility, so visual contact should be treated as supporting evidence rather than certainty. In unclear conditions, especially around nebulae or suspected minefields, deliberate visual sweeps can be invaluable. The ship has windows for a reason.

"Had a cadet once who trusted his console more than his own face. Lovely lad. Polished boots, perfect callouts, and absolutely determined to be killed by a glowing blue dot.

"We were crossing a nebula pocket. Sensors were lying, Radar was swearing, and Helm was threading us between old mines and debris.

"So I looked up.

"Just off the bow: a ship sitting cold. No transponder. No clean return. No helpful marker saying 'ambush here.'

"The cadet asked how I knew. I told him the Navy had issued him two advanced optical detection systems at birth, and it would be kind to use them.

"We turned before they fired. That is why I am old, and several pirates are not.

"Scanners are marvellous tools, cadets. They are not permission to stop looking out the bloody window."

- Cmdr. Wedge Antilles
UCNA Starfighter Training

2.5 - How the Displays Fit Together

"The Strategic Map, Sensors, and Tactical Display do not disagree. They are simply answering different questions."

A ship fights best when these three views are understood as complementary.

The Strategic Map answers **where are we, and where do we need to be?**

Long-Range Sensors answer **what is out there, and what is approaching?**

The Tactical Display answers **what is about to matter right now?**

An officer who relies only on the Strategic Map may understand the mission, but miss the immediacy of the fight.

An officer who relies only on Long-Range Sensors may understand the threat, but not the handling required to survive it.

An officer who relies only on Tactical may fight well in the moment, while remaining blind to what is developing beyond local view.

This is why no single console provides a complete picture, and why information flow across the bridge is not an optional courtesy, but a necessity. The ship survives when these views are combined into one shared understanding.

Practical Guidance for Cadets

When learning a new station, ask yourself three questions:

- What does my display show clearly?
- What does it show poorly, or not at all?
- Which station sees the rest?

That habit alone will make you a better officer.

Do not assume that because something is not on your screen, it does not exist. Do not assume that because you can see it, everyone else can.

And do not forget that the most dangerous misunderstanding on a bridge is often not bad information, but incomplete information mistaken for the whole truth.

3.0 - Shared Systems & Combat Mechanics

"Every console is connected to consequences. Understand the systems, and the ship starts making sense."

A UCN vessel is not a collection of isolated stations. It is a single system, governed by shared mechanics no officer can afford to ignore.

Shields, scanners, ordnance, tractor operations, hacking, and system damage affect the whole bridge. A Missiles officer who misunderstands shields may mistime a strike. A Helm officer who misunderstands ordnance may manoeuvre poorly. A Captain who misunderstands hacking, tractors, or scan limits may make decisions based on false assumptions.

This section provides the common foundation: not role duties, but the mechanics that allow the ship to survive, fight, and function as one. Learn them, and every station becomes clearer. Neglect them, and even capable crews will work harder, slower, and to less effect.

The role chapters explain who does what. This section explains what everyone must understand.

Power, Heat, Coolant & Repairs

A ship is controlled by power, power is constrained by heat, heat is managed by coolant, and repairs fix it when it goes wrong. This section explains how all these systems work together to operate the ship to the best of its capabilities or constraints.

Scanning, Detection & Signals

A ship that does not understand what it can see is already at a disadvantage. This section explains how contacts are detected, how scan information is built, and what sensor signals can reveal before and after a target is identified.

Propulsion: Warp, Impulse & Manoeuvring

A ship that doesn't move is called a space station. This section covers all the ways that UCN vessels can move through space, and how to use them to the best of their abilities.

Defensive Capabilities: Shields & Frequencies

Shields are more than a pool of protection. Their strength, calibration, frequency, and timing all shape how well a ship survives under fire. This section covers the defensive mechanics every officer should understand.

Offensive Capabilities: Damage, Beams & Ordnance

A ship does not become ineffective only when it is destroyed. Damage to individual systems can reduce its threat long before the hull is lost. This section explains how system damage works, and why disabling an enemy is a useful tactic.

Additionally, not all weapons behave alike. This section explains the characteristics and practical employment of the ship's beams, launched ordnance and other common weapon effects.

Tractor Beam Operations

The tractor beam is not simply a recovery tool. Used well, it can control the battlespace. This section covers how tractor operation works, what constrains it, and why coordination matters when employing it.

Hacking & Electronic Warfare

Hacking is a means of shaping the fight by reducing enemy capability at the moment it matters most. This section explains its effects, limitations, risks, and practical value, so that officers understand both what it can achieve and what it cannot.

Probes & Drones

A vessel's influence extends beyond its own hull. Probes and Drones expand vision, project force, and create opportunities the main ship alone cannot. This section explains their shared mechanics and why they must be understood as part of the ship's wider capability, not merely as specialist tools.

3.1 - Power, Heat, Coolant, & Repairs

"Get it right and no one notices. Get it wrong, and the darkness and surplus smoke rather give the game away."

Primary Role: Chief Engineer
Secondary Roles: Manual Engineer

Every system aboard a UCN vessel is governed by three linked resources: power, heat, and coolant. Power improves performance, but increases demand on the ship's reserves and may generate heat. Heat is the consequence of pushing systems beyond comfortable limits; if left unchecked, it degrades performance and causes damage. Coolant controls that risk, allowing key systems to run harder or recover after heavy use.

Engineering controls these resources directly, but every officer affects them. Every spike request, sustained burn, shield raise, reload push, scanner boost, or beam volley places strain somewhere in the ship. Good crews do not simply ask for more power; they understand what it costs, when it matters, and when to stop. Power wins moments. Coolant buys time. Heat sends the bill.

3.1.1 - Power

All systems draw power continuously based on their assigned level. Power is not unlimited. It must be allocated, not assumed.

Running all systems at high power simultaneously will result in:

- Energy depletion
- Reduced operational flexibility
- Inability to respond to spikes

Power Redistribution Timing

Power does not change instantly.

- Approximate rate: 30% per second.
- Large reallocations take several seconds to take effect.
- Late power adjustments are often ineffective.

See **Table 3-5** on page 18 for full timings.

Spikes

A Spike is a temporary increase in power to a system. For a full breakdown, see **Appendix E: Power Spikes** on page 129.

Use spikes:

- To counter burst damage (e.g. shields)
- To enable rapid manoeuvres
- To accelerate key system performance

Spikes must be timed early and reset quickly.

Additional Resources

A UCN ship will often carry a shuttle as part of its loadout. This shuttle will have its own reactor, which can be used to charge depleted fuel cells if required. Coordinate with your Manual Engineer.[2]

Drones can also be used to provide additional energy. See **Drones** on page 49.

Reactor Efficiency

Unlike many ship systems, the reactor does not scale linearly with increased power. While raising reactor power will always increase total energy output, gains above normal performance (100%) are subject to diminishing returns.

In practical terms, this means that additional power becomes progressively less efficient the further it is pushed. Moderate increases provide reliable and sustainable gains, but aggressive over-powering yields proportionally less benefit while still increasing strain on coolant and system management.

[2] [[Behind the Curtain]]

Currently, when inserting a powercell into the shuttle's reactor, the powercell is rapidly charged. This is an exploit in the software. Honourable officers don't abuse this exploit!

With 230% reactor power and 100% coolant, a dead fuel cell can be recharged by the shuttle in ~2m 30s (all other systems at 0%) or ~8 mins (all other systems are left at 100%).

If you want to use this resource, be aware of how long it *should* take and use a watch to keep yourself honest!

3.1.2 - Heat and Coolant

All ship systems generate heat passively if their assigned power level is above 100%, even if they are not active. Coolant reduces this heat buildup, allowing systems to operate safely at elevated power.

Some systems also generate additional heat during active use. If heat exceeds safe limits, the system begins to take damage, reducing effectiveness and risking failure.

Coolant therefore serves a single purpose, **to control heat accumulation and extend safe operating limits.**

Heat Generation

Heat is driven primarily by power level, with additional contributions from system use.

- At 100% power, systems do not generate passive heat.
- Above 100%, heat increases.
- At higher power levels, the rate of heat generation increases non-linearly.

Power Level	Behaviour
≤100%	Systems cool naturally
150%	Stable with modest coolant
200%	Requires significant coolant
>230%	Overheating unavoidable
300%	Rapid overheating

Table 3-1: Power level cooling behaviour

This means:

- Small increases in power are manageable.
- Large increases become progressively more difficult to sustain.

Active Systems

Some systems (Warp, Combat Manoeuvring, Hacking and Beams) will generate heat when active, regardless of their power settings.

Coolant Allocation

Coolant is a shared and limited resource.

- Total available: 10 Litres (100%)
- Maximum per system: 10 L (100%)

Coolant reduces heat generation linearly:

- More coolant leads to slower heat buildup.
- Sufficient coolant means system remains thermally stable.
- Excess coolant means system cools over time.

3.1.3 - Thermal Equilibrium

A system is thermally stable when it neither heats nor cools.

This defines a practical limit:

Around 230% power is the highest level that can be sustained indefinitely.

Approximate coolant required to hold a system steady:

Power Level	Coolant Required
100%	0%
150%	~30%
200%	~70%
230%	100%
>230%	Unsustainable

Table 3-2: Power level coolant requirements

For full breakdown, see **Tables 3-4**, **3-6** and **3-8** on page 17 and page 18.

Cooling Behaviour

Cooling is gradual.

- Systems below equilibrium will cool over time.
- Cooling without coolant is slow.
- Coolant significantly accelerates heat reduction.

Repositioning coolant also takes time:

- Full redistribution takes ~8 seconds

Heat from System Use

Some systems generate additional heat when active:

- Warp Drive: generates heat while active.
- Manoeuvring Thrusters/Impulse: generate significant heat during Combat Manoeuvres.
- Hacking: generates heat into the scanner system when completed past a certain threshold.
- Beam Weapons: generate heat when firing.

Other systems (such as shields) generate heat primarily from their assigned power level, not whether they are actively engaged.

For full details see the relevant sections.

Overheating

Systems operate safely up to their thermal limit. If heat *exceeds* this limit:

- The system begins taking damage, therefore the effectiveness degrades.
- Continued overheating risks failure.

Coolant prevents damage. Power discipline prevents overheating.

SPANRs can be deployed to counteract the damage caused by minor overheating.

3.1.4 - Additional Guidance

Depowering Systems

Depowering systems is a valid Engineering tactic when their full performance is not required, either partially depowering, or reducing a system to 0% power.

This can help conserve or recover energy, particularly during post-combat recovery (see **Appendix D: Post Combat Recovery** on page 124), and may also allow overheated systems to cool naturally without committing coolant.

However, this should be done with care. Scanners should normally remain at 100% power, as reducing them can limit the ship's awareness at precisely the wrong moment. Unless the ship is in a secure location, any depowered system should also be capable of being restored without delay if circumstances change suddenly.

If a damaged system is being repaired, reducing power to that system does not slow the repair. Depowering a damaged system can therefore save energy and reduce further heat strain while repairs are underway, provided the system is not immediately required.

3.1.5 - Summary

Passive power draw scales linearly with power allocation: doubling a system's power doubles its passive energy cost. However, the benefit gained from that extra power may not scale linearly.

- Systems at 100% power will cool gradually over time without coolant.
- Use 150% power (with 30% coolant) for sustainable performance gains.
- Use 200% power only on priority systems.
- The highest power you can sustain is 230% with 100% coolant.
- Treat >230% power as temporary.
- Allocate coolant deliberately to key systems.
- Don't leave coolant across multiple systems if it's no longer needed there. If a system doesn't need cooling at this moment in time, see if the coolant could be better used elsewhere, and let the system cool naturally.

[[Behind the Curtain]]

Obviously, the best way to save power to is turn off all "non-essential systems" - the definition of "non-essential" being a whole conversation in itself!

However, if a console is unpowered, it means its operators cannot do anything. And whilst it might be more efficient, it's not fun for anyone to sit there and stare at a screen they can do nothing with.

Don't prevent people from playing the game for the sake of efficiency. If you REALLY need to cut power, talk about it.

3.1.6 - Engineering Tables and Graphs

Table 3-3: Power vs Temp Increase

Power Level	Heat gained per min. (no coolant)	Time to overheat (from 0° to 100°)
100%	Slight cooling	Will not overheat
150%	~88°/min	~ 1 min 08 sec
200%	~207°/min	~29 sec
230%	~295° /min	~20 sec
250%	~362°/min	~17 sec
300%	~564°/min	~11 sec

Figures assume the system is already at the listed power level with no coolant. Power changes ramp over several seconds, so real spike timings may be slightly longer.

Table 3-4: Coolant Required for Thermal Stability

Power Level	Coolant Required to Hold Steady	Practical Meaning
100%	0%	Naturally stable
150%	~30%	Sustainable
200%	~70%	Heavy commitment
~230%	100%	The highest power allocation that can be thermally stable
250%	Not achievable	Temporary only
300%	Not achievable	Spike only

Figure 3-1: System Power vs Passive Heat Change (0% Coolant)

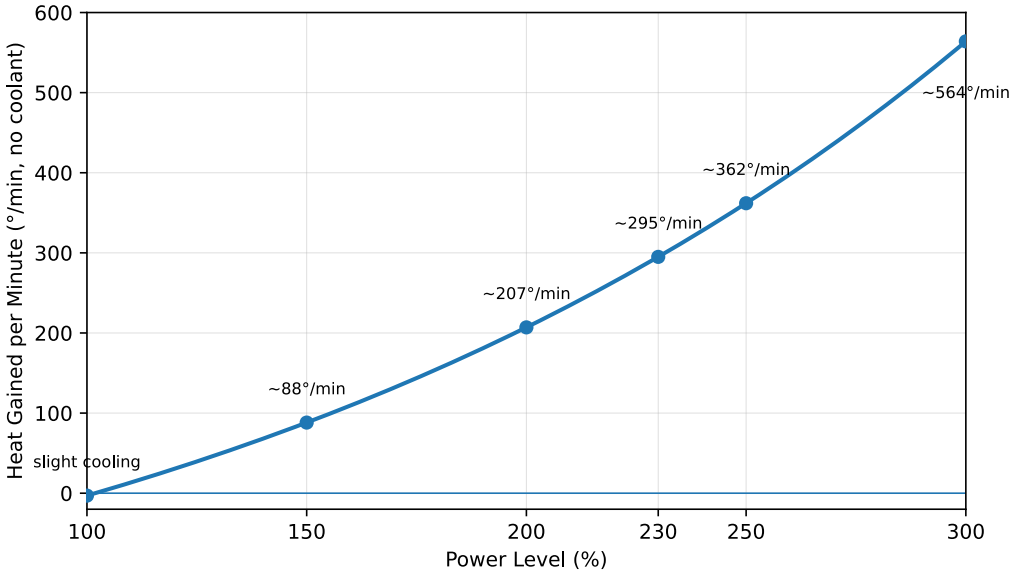


Figure 3-2: Coolant Required to hold system temperature stable

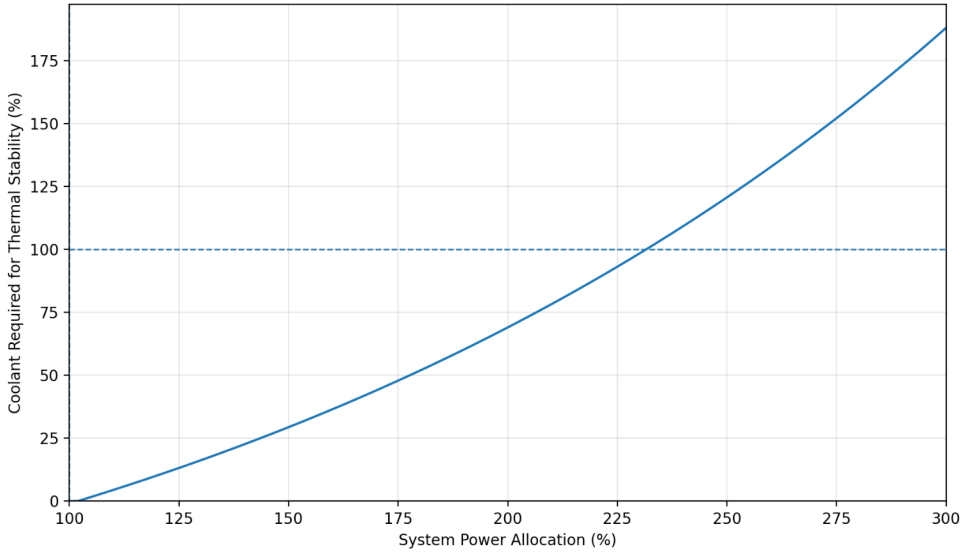


Table 3-5: Power redistribution timing

Seconds it takes to redistribute power.

From/To	0%	50%	100%	150%	200%	250%	300%
0%	-	1.67	3.33	5.00	6.67	8.33	10.00
50%	1.67	-	1.67	3.33	5.00	6.67	8.33
100%	3.33	1.67	-	1.67	3.33	5.00	6.67
150%	5.00	3.33	1.67	-	1.67	3.33	5.00
200%	6.67	5.00	3.33	1.67	-	1.67	3.33
250%	8.33	6.67	5.00	3.33	1.67	-	1.67
300%	10.00	8.33	6.67	5.00	3.33	1.67	-

Table 3-6 - Maximum stable power by coolant assigned

Amount of power that can be assigned based on coolant allocation.

Coolant Assigned	Max Stable Power
0%	100%
10%	~120%
20%	~135%
30%	~150%
40%	~165%
50%	~175%
60%	~190%
70%	~200%
80%	~210%
90%	~220%
100%	~230%

Table 3-7: Time to cool from 100°

Assuming coolant only, with no passive heat gain, active system heat, or other effects.

Coolant Assigned	Cooling Rate	Time to reduce to 0°
0%	None	N/A
10%	0.5°/s	3m 20s
20%	1.0°/s	1m 40s
30%	1.5°/s	1m 07s
40%	2.0°/s	50s
50%	2.5°/s	40s
60%	3.0°/s	33s
70%	3.5°/s	29s
80%	4.0°/s	25s
90%	4.5°/s	22s
100%	5.0°/s	20s

Table 3-8: Coolant shortfall

Time to overheat a system based on lack of coolant.

Coolant Shortfall	Heat Gain	Time to overheat
5%	0.25°/sec	6 mins 40 secs
10%	0.5°/sec	3 min 20 secs
15%	0.75°/sec	2 min 13 sec
20%	1.0°/sec	1 min 40 secs
25%	1.25°/sec	1 min 20 secs
30%	1.5°/sec	1 min 7 secs
40%	2.0°/sec	50 secs
50%	2.5°/sec	40 secs
60%	3.0°/sec	33 secs
70%	3.5°/sec	29 secs
80%	4.0°/sec	25 secs
90%	4.5°/sec	22 secs
100%	5°/sec	20 secs

Figure 3-3: Coolant shortfall vs Time to overheat

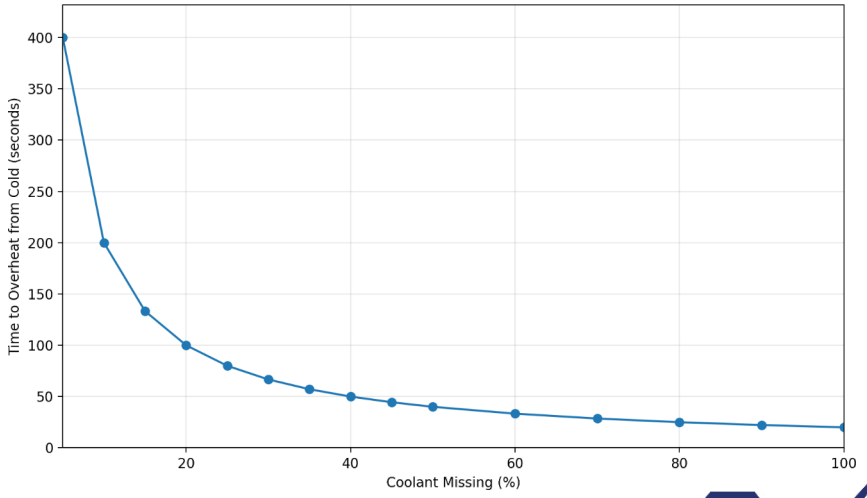


Table 3-9: Beam Weapons - Energy and Heat with continuous fire

The below table calculates the MJ loss and heat gain for a variety of power settings with the standard UCN beam configurations (two lethal, two non-lethal). This includes the base energy.

- **MJ Loss** is how much MJ is lost per minute with **MJ Time** being how long it will take to drain 1500 MJ energy.
- **° heat gain** is how much heat the system will gain in a minute, **° heat Time** is how long it will take to reach 100° heat from 0°.

Beam Type	100%				150%				200%				250%				300%			
	MJ		° heat		MJ		° heat		MJ		° heat		MJ		° heat		MJ		° heat	
	Loss	Time	Gain	Time	Loss	Time	Gain	Time	Loss	Time	Gain	Time	Loss	Time	Gain	Time	Loss	Time	Gain	Time
Lethal	84	17m 51s	37°	2m 42s	126	11m 54s	148°	40s	168	8m 56s	287°	21s	210	7m 09s	462°	13s	252	5m 57s	684°	9s
Non-lethal	384	3m 54s	237°	25s	576	2m 36s	448°	13s	768	1m 57s	687°	9s	960	1m 34s	962°	6s	1152	1m 18s	1284°	5s
Combined	444	3m 23s	277°	22s	666	2m 15s	508°	12s	888	1m 41s	767°	8s	1110	1m 21s	1062°	6s	1332	1m 08s	1404°	4s

Table 3-10: Warp Drive - Total Draw (Active + Passive)

The total draw of using the Warp Drive at various speeds/power levels.

Warp Level	100% Power	150% Power	200% Power	250% Power	300% Power
Warp 1	100 MJ/min	150 MJ/min	200 MJ/min	250 MJ/min	300 MJ/min
Warp 2	178 MJ/min	267 MJ/min	356 MJ/min	445 MJ/min	534 MJ/min
Warp 3	264 MJ/min	396 MJ/min	529 MJ/min	661 MJ/min	793 MJ/min
Warp 4	357 MJ/min	535 MJ/min	713 MJ/min	892 MJ/min	1070 MJ/min

Figure 3-4: Warp Drive total power draw whilst at Warp

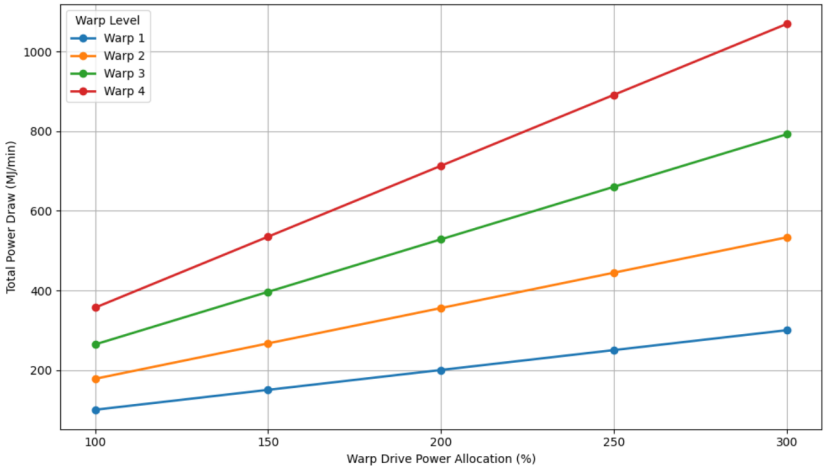


Figure 3-5: Scanner Range at different power scales

Diagram of the scanner range as viewed on the Strategic Map at different power levels for allied ships and launched probes (including the maximum thermally stable level of 230%).

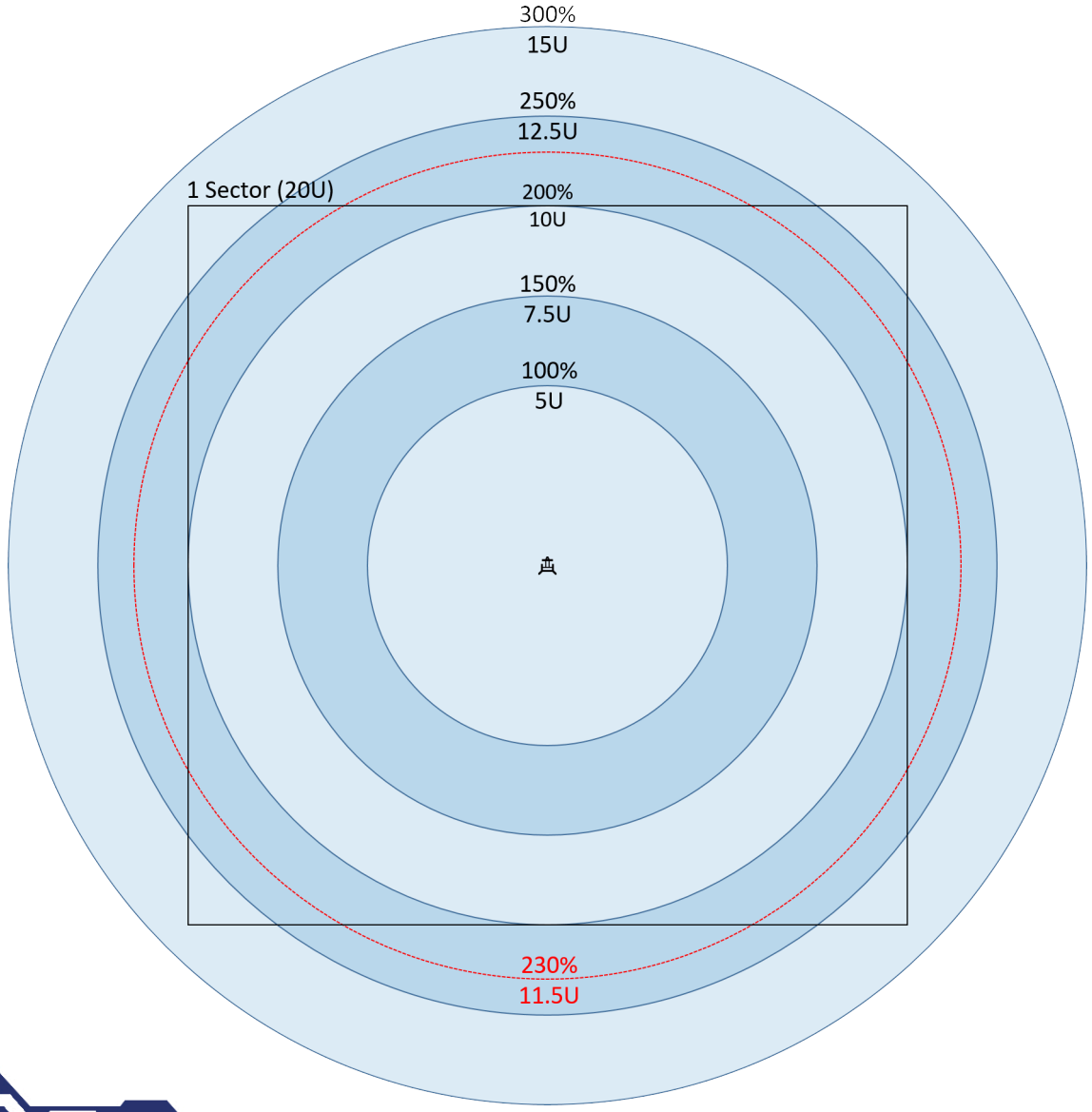


Table 3-11: Energy Reserve Drain Time

Engineering consoles report net energy loss in MJ/min. Use this table to estimate how long the current fuel cell will last once reactor output has already been accounted for.

Net Loss (MJ/min)	1500 MJ Remaining	1000 MJ Remaining	500 MJ Remaining
25	60m	40m	20m
50	30m	20m	10m
75	20m	13m 20s	6m 40s
100	15m	10m	5m
150	10m	6m 40s	3m 20s
200	7m 30s	5m	2m 30s
250	6m	4m	2m
300	5m	3m 20s	1m 40s
400	3m 45s	2m 30s	1m 15s
500	3m	2m	1m
750	2m	1m 20s	40s
1000	1m 30s	1m	30s

Table 3-12: Engineering Report Thresholds

Engineering should report early when ship capacity is becoming constrained.

Condition	Suggested Report
Energy loss sustained above 250 MJ/min	*"Energy drain high."*
Energy loss sustained above 500 MJ/min	*"Energy drain critical."*
Any key system above 75° heat	*"System nearing overheat."*
Any key system above 90° heat	*"System critical, reset or damage likely."*
Coolant fully committed	*"No spare coolant available."*
Spike request unsafe	*"Deny, system too hot."*

Table 3-13: Main System Reference Table

System	Power Draw (MJ/min)			Effect of Increased Power	Scaling	Operational Notes
	100%	200%	300%			
Reactor 	+ 120 MJ/min	+ 180 MJ/min	+ 240 MJ/min	Increases power generation	Increase in power scales **with diminishing returns** with power generation	
Beam Weapons 	24 MJ/min	48 MJ/min	72 MJ/min	Increases rate of fire	Increase in power scales **linearly** with rate of fire	Also each shot consumes energy (3 MJ per beam) and adds heat (2° per beam)
Missiles 	8 MJ/min	16 MJ/min	24 MJ/min	Reduces reload time	Increase in power scales **linearly** with reload speed	No additional heat increase when used
Manoeuvring 	16 MJ/min	32 MJ/min	48 MJ/min	Increases turn rate and manoeuvre recharge	Increase in power scales **linearly** with rotation speed	Potential additional heat spike from Combat Manoeuvring System
Impulse 	32 MJ/min	64 MJ/min	96 MJ/min	Increases forward speed and helps manoeuvre recharge	Increase in power scales **linearly** with forward speed	Potential additional heat spike from Combat Manoeuvring System
Warp 	40 MJ/min (inactive)	80 MJ/min (inactive)	120 MJ/min (inactive)	Increases warp effectiveness/ speed	**Linear** for power, **non-linear** in active use	Active warp adds major extra draw and heat
Warp 1	+60 MJ/min	+120 MJ/min	+180 MJ/min			
Warp 2	+138 MJ/min	+276 MJ/min	+414 MJ/min			
Warp 3	+224 MJ/min	+448 MJ/min	+673 MJ/min			
Warp 4	+317 MJ/min	+633 MJ/min	+950 MJ/min			

Table 3-13: Main System Reference Table (cont)

System	Power Draw (MJ/min)			Effect of Increased Power	Scaling	Operational Notes
	100%	200%	300%			
Front Shield Gen 	24 MJ/min	48 MJ/min	72 MJ/min	Increases shield effectiveness and recharge	Increase in power scales **linearly** with recharge speed and calibration time. **Non-linear** defensive return	Shield field upkeep separate
Rear Shield Gen 	24 MJ/min	48 MJ/min	72 MJ/min	Increases shield effectiveness and recharge	Increase in power scales **linearly** with recharge speed and calibration time. **Non-linear** defensive return	Shield field upkeep separate
Shield Field raised [3]	90 MJ/min	90 MJ/min	90 MJ/min	-	Fixed cost (if on)	Same cost regardless of shield power
Scanner 	24 MJ/min	48 MJ/min	72 MJ/min	Increases scan range and radar waveform locking ease/ speed	Increase in power scales **linearly** with scanner range and scanner alignment time. Increase in power scales **quadratically** with drone range	No heat generation for scanning. Hacking can deposit 20° to 40° per hack.
Cargo Docks 	8 MJ/min	16 MJ/min	24 MJ/min	Improves dock handling / thermal dock performance Improves pull/push speed of Tractor Beam	Increase in power scales **linearly** with inter dock transfer speed and drone energy transfer speed. Also scales **linearly** with Tractor Beam movement speed	Tractor beam uses extra energy depending on number of targets being affected.

[3] Shield Field is not a separate system

3.2 - Scanning, Detection & Signals

"Finding something in space is looking for a needle in a haystack. A good officer learns where to point the magnet."

Primary Role: Radar
Secondary Roles: Helm, Weapons, Drones

3.2.1 - Long-Range Scanner Display

Controlled by:	Radar
Powered by:	Scanner
Resting Power Draw:	24 MJ/min at 100%
Active/Extra Draw:	None
Power Increase:	Improves strategic scan range and makes signal lock easier.
Heat/Coolant Notes:	None
Special Notes:	Radar has a fixed long-range display limit.

Radar can see farther beyond the ship than the tactical bridge stations. Whilst increasing power to sensors will increase the range of visibility on the Strategic Map, the Radar display will always max out at 30U. For Probe views, see **Figure 3-5** on page 20, **Table 3-25** on page 47 and **Appendix I - Battlespace Examples** on page 139.

Scanner power improves scan speed indirectly: it does not reduce the fixed lock timer, but higher scanner effectiveness reduces scan instability, allowing the officer to achieve signal lock quicker.

This makes Radar the primary source for:

- Early hostile detection.
- Hazard warning.
- Long-range target tracking.
- Warp approach monitoring.

3.2.2 - Nebulae

Nebulae are phenomena found throughout space. They block standard sensor contacts, both within the nebula and behind it. They don't, however, block radar signals.

3.2.3 - Radar Signal Display

Radar officers should understand the signal traces around the edge of the display.

These signals are not merely decorative. They can provide warnings of activity well beyond normal visual scanner range.

When discussed as a whole, these units are referred to as Signal Indicators (SI)

Colour	Signal Type	Unit	Examples
Red	Gravitation Signals	Distortion Newtons (DN)	Planets Asteroids Wormholes Blackholes Ships at Warp
Green	Biological / Thermal	Universal Metabolics (UM)	Explosions Ships Stations Planets
Blue	Electrical Signals	Megajoules (MJ)	Ships Beam fire Mines Nebula

Table 3-14: Radar Signal Indicators

Practical Interpretation:

- Large spikes on the red channel usually indicate ships travelling at warp.
- Large spikes on the green channel usually indicate explosions or intense thermal events.
- Bursts or fuzz on the blue channel usually indicate beam discharge, EMP activity, nebula interference, or heavily powered systems.

Signal Interpretation

At practical officer level, the following rules are the most useful:

- **Warping ships** are often visible first on the red channel.
- **Idle ships** may produce faint green and blue disturbance, but are often difficult to detect at range.
- **Beam weapons** and **EMP effects** agitate the blue channel strongly.
- **Explosions** often spike the green channel.
- **Nebulae** create strong blue interference and tend to mask weaker biological signatures.
- A ship sitting still in a nebula may be very hard to detect.
- A ship moving at warp through a nebula is usually much easier to identify.

Radar should therefore watch signal traces especially when:

- Approaching nebulae.
- Guarding stations or convoy routes.
- Escorting assets in low-visibility areas.
- Expecting hostile reinforcements.

Unexpected warp activity should always be called out in plain language.

Examples:

"Unknown warp signature, bearing 315."
"Multiple warp signatures incoming, bearing 315."

Distance of sensor range deterioration

Signals start to drop off at a range of 30U and do so in a linear fashion up to 60U where they become undetectable.

Advanced Capabilities

The standard UCN Radar display divides each signal trace into 512 narrow bearing segments, each covering a little over 0.7 degrees of the circle.

Each segment can display up to 1 SI in each signal channel. Any return stronger than 1 SI within that same segment is visually capped, meaning it will not appear taller on the trace. For example, a single ship emitting 5 SI may appear no stronger than the same ship emitting 1 SI, if both occupy the same segment.

This is why signal strength alone can be misleading. Radar does not simply show "how much signal exists"; it shows how that signal is distributed across the segments. A larger ship may cover several neighbouring segments, even if its signal in each segment is capped. Several smaller ships on close, but slightly different, bearings may also cover several neighbouring segments. In some cases, those two situations may look very similar.

Therefore, a wide or unusually sharp disturbance does not automatically prove there are multiple ships. It may indicate a group travelling in close formation, a single large contact, or several overlapping returns. Equally, a tall signal does not necessarily mean a stronger source, since that segment may simply have reached the display cap.

In practical terms, Radar officers should watch the shape, width, and behaviour of the trace, not merely its height. A capped return may hide stronger signals than are visually apparent. A broad or clustered return may suggest multiple contacts or a large object, but it should be treated as an indication, not identification.

The signal trace is therefore best understood as an early-warning tool. It can suggest that something is present, moving, firing, hiding, or grouped together, but it cannot determine exact number, size, or class without further scanning. The trace is a clue, not a confession.

3.2.4 - Increased Power

Increasing Scanner power linearly extends the sensor range shown on the Strategic Map for the ship and all linked assets, including drones and probes from *your* ship. See **Figure 3-5** on page 20 for full details.

In addition, see **Appendix I - Battlespace Examples** on page 139 for further examples.

It also makes it easier to achieve a full scan and obtain full Target Analysis. Drone control range, however, increases far more rapidly, scaling dramatically with scanner power. See **Table 3-26** on page 50 for full details.

Increasing scanner power does not make the outer signal lines stronger in themselves, but it can make them appear cleaner at range by improving detection range and reducing distance-related degradation.

Despite what the user interface implies, boosting scanner power will not allow long-range sensors to penetrate nebula.

3.2.5 - Target Analysis

Radar is responsible for linking targets to analysis when designated by the Captain or Helm. This allows Weapons stations to assess shield status, orientation, and system info.

Radar operators are strongly advised to memorise or print **Appendix B: Radar icons & typical ship information** (page 121) , which provides information on iconography utilised on the Radar screen.

Once initial waveform locking has been completed, the following information can be attained on a **primary** scan:

- Faction (if applicable)
- Class of ship
- Hull Strength
- Shields (including layout and strength)
- Ships systems

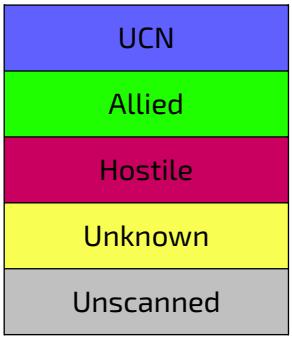


Figure 3-6: IFF Status colours

Faction

Who this ship is aligned to. The colouration will indicate its IFF (Identify Friend/Foe) status.

Some mission profiles might have different IFF colours to indicate different factions. When in doubt, check the readout to ascertain if it is hostile or friendly.

Class of Ship

What class of Ship the target is.

Hull Strength

How strong the hull is. When it reaches 0, the ship is destroyed.

Shields

The number and strength of each shield field. This will be given as a string of numbers separated by colons.

The number of entries is the number of projectors, the number itself is the strength of the individual shield field. The below example is the reading of a craft with 4 equal shields of 200.



Figure 3-7: Shield Readout Example

Different craft will have a different number of shield projectors. Whilst some will be easy to describe (Fore, Aft, Port, Starboard etc), it gets more complicated when there are 3, 5 or 6 projectors. It is therefore recommended to use the shield numbering system when relaying information. Starting at the forward shield, the shields progress clockwise:

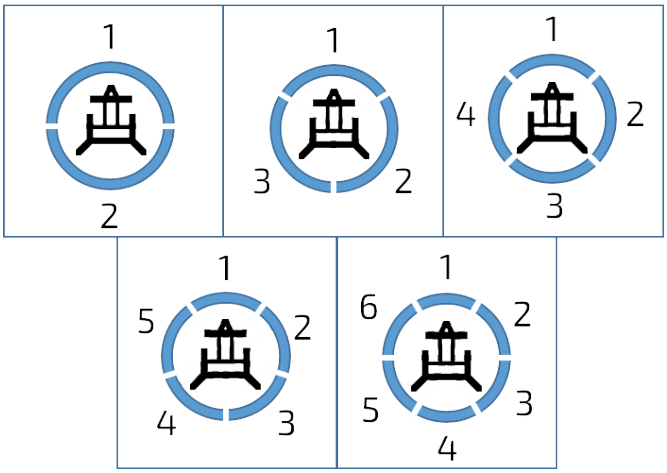


Figure 3-8: Shield Numbering Examples

Therefore the below information can be reported as "Shield 2 down to 71".



Figure 3-9: Damaged Shield Readout Example

The following additional information can be obtained via a **secondary**, more in-depth scan:

- Exact gravitational (DN) biological / thermal (UM) and electrical (MJ) signals
- Internal systems information
- Firing arcs of beam weapon(s)
- Precise shield frequency
- Precise beam frequency

The firing arcs will show up on the Tactical Display, providing Helm with critical information to assist in manoeuvring the ship to avoid enemy fire.

The shield and beam frequencies can be linked to Weapons to improve attack and defence against the target.

Ship Systems

The status of a target ship's systems. This can be useful to monitor/call out when certain systems are disabled, so tactical priorities can shift.

3.3 - Propulsion: Warp, Impulse & Manoeuvring

"A ship that cannot move is not a fortress. It is a target with delusions."

Primary Role: Helm
Secondary Roles: Chief Engineer

Space is large, hostile, and rarely kind enough to place the objective directly in front of you.

Propulsion covers the systems used to move, turn, re-position, and accelerate a UCN vessel. While Helm controls these systems directly, every officer should understand their limits. Weapons depend on positioning. Radar depends on safe approaches. Engineering must manage the heat and power cost. Command must understand what the ship can realistically achieve before ordering it somewhere awkward.

UCN vessels use three primary propulsion systems:

System	Primary Function
Impulse Drive	Standard Forward/ Reverse Movement
Manoeuvring Thrusters	Rotational Movement
Warp	Long-Distance travel

Table 3-15: Propulsion Systems

In addition, most vessels are equipped with a **Combat Manoeuvre** reserve, allowing short bursts of forward and/or lateral movement during combat.

Propulsion is governed by the same principle as all other ship systems: **effectiveness determines performance**. Increasing power improves performance, while damage or hacking reduces it.

3.3.1 - Impulse Drive

Controlled by:	Helm
Powered by:	Impulse
Resting Power Draw:	32 MJ/min at 100%
Active/Extra Draw:	None
Power Increase:	Increases forward/ reverse speed linearly
Heat/Coolant Notes:	Combat Manoeuvre boost can add direct heat to system
Special Notes:	Also contributes to Combat Manoeuvre recharge

The impulse drive provides the ship's main forward and reverse thrust during normal flight and combat. At normal effectiveness, many larger UCN vessels have a speed of around 6-7.5U/min.

At higher effectiveness, the ship accelerates and travels faster. At lower effectiveness, it becomes sluggish and less able to re-position, pursue, or disengage. Impulse is especially important for:

- Closing or opening distance.
- Maintaining pursuit.
- Retreating from danger.
- Contributing to Combat Manoeuvre recharge.
- Controlling engagement range.

Impulse power scales linearly. In practical terms, 150% impulse effectiveness gives 150% forward/ reverse movement performance.

The impulse drive also interacts with the Combat Manoeuvre system. Although Manoeuvring governs turning and lateral movement, Impulse power contributes to the recovery rate of the Combat Manoeuvre reserve.

3.3.2 - Manoeuvring Thrusters

Controlled by:	Helm
Powered by:	Manoeuvring
Resting Power Draw:	16 MJ/min at 100%
Active/Extra Draw:	None
Power Increase:	Increases rotational speed
Heat/Coolant Notes:	Combat Manoeuvre can add direct heat to system.
Special Notes:	Also contributes to Combat Manoeuvre recharge

Manoeuvring thrusters rotate the ship. They bring weapons to bear, they present the correct shield facing, they avoid hazards, and they dodge incoming fire.

Manoeuvring affects:

- Turning rate.
- Combat re-positioning.
- Combat Manoeuvre recharge.

At normal effectiveness, many larger UCN vessels rotate between **10° and 14° per second**, though individual ship classes may vary. Increased manoeuvring effectiveness improves turn rate proportionally.

Unlike impulse movement, rotation is immediate. Once a turn command is given, the ship rotates at its current maximum rate. There is no gradual rotational acceleration or braking.

Damaged or under powered manoeuvring thrusters can be tactically disastrous. A ship that cannot turn cannot reliably aim, evade, or protect weakened shield facings.

All pilots should check handling immediately when taking control of a vessel or drone: rotation, impulse speed, acceleration/deceleration, and whether manoeuvring thrusters or lateral strafe are available. Some craft handle very differently, and combat is a poor time to discover the details.

3.3.3 - Combat Manoeuvre

Controlled by:	Helm
Powered by:	None
Resting Power Draw:	None
Active/Extra Draw:	None
Power Increase:	Increasing Impulse and Manoeuvring systems increases recharge rate.
Heat/Coolant Notes:	Combat Manoeuvre adds direct heat to Impulse and/or Manoeuvring systems when activated.
Special Notes:	Standard recharge time is 20 seconds. Standard discharge is 3 seconds.

Combat Manoeuvre is a short-duration burst movement system used for rapid re-positioning. It allows the ship to apply a brief forward boost, lateral strafe, or a combination of both, independent of normal impulse thrust.

It is most useful for:

- Dodging missiles or HVLIs.
- Shifting out of enemy firing arcs.
- Getting enemies into the ship's firing arcs.
- Correcting position after an overshoot.
- Avoiding mines or hazards.

Note - Combat Manoeuvre System does not work at warp speeds.

Combat Manoeuvre draws from a finite reserve. At normal performance, this reserve recharges from empty to full in approximately **20 seconds**.

Recharge rate is affected by both:

- Manoeuvring effectiveness.
- Impulse effectiveness.

The two systems are averaged for recharge purposes. If both are improved, the reserve recovers more quickly. If either system is damaged, hacked, or under-powered, recovery slows.

For example:

Manoeuvring effectiveness: 200%
Impulse effectiveness: 100%
Therefore average effectiveness: 150%

Approximate recharge time from empty: ~13 seconds

Combat Manoeuvre has two movement components:

Input	Effect	System Heated
Forward boost	Pushes ship forward	Impulse
Lateral strafe	Pushes ship sideways	Manoeuvring

Table 3-16: Combat Manoeuvre movement components

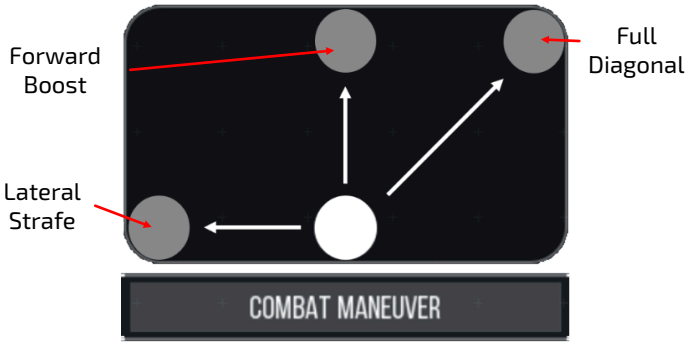


Figure 3-10: Combat Manoeuvre

A full forward boost or full lateral strafe alone consumes the reserve in approximately **three seconds** and dumps 20° of heat per second into the relevant system.

However, owing to the slight ramp up time, this equates to 50° to 60° heat total.

Using both axes together increases consumption. A full diagonal manoeuvre consumes the reserve in approximately half that time.

Additionally, the diagonal manoeuvre does not split one heat cost between the two systems. It applies both effects at once. However, as it drains the reserve at twice the rate, it only increases the heat half the amount (as it is not burning for as long.)

For example, a full diagonal manoeuvre would generate 20° of heat per second for both Impulse and Manoeuvring. However, as it only lasts for 1.5 seconds (as it is using the system twice as fast), it will only dump approximately 25-30° of heat into each system.

This is also calculated from partial manoeuvres. For example, a full strafe with 25% forward boost still applies full manoeuvring heat, adds 25% impulse heat, and drains the reserve approximately 25% faster than a pure strafe.

Combat Manoeuvre should be treated as a tactical burst, not a second engine. Spending it casually may leave the ship unable to evade when it truly matters.

3.3.4 - Warp Drive

Controlled by:	Helm
Powered by:	Warp
Resting Power Draw:	40 MJ/min
Active/Extra Draw:	Additional drain while travelling at warp, based on warp level.
Power Increase:	Increases Warp speed linearly.
Heat/Coolant Notes:	Active warp generates major heat. Coolant requirement increases sharply by warp level.
Special Notes:	Warp has both a baseline system draw and an active travel cost. High warp settings can overheat rapidly.

The warp drive is primarily used for long-distance travel. It allows a vessel to cross sectors quickly, but carries significant energy and thermal costs.

Warp has four standard levels:

Warp Level	Description	Base Speed	Coolant Required	Active Energy Cost (at 100%)
1	Most efficient	1U/s	40%	+1.0 MJ/s
2	Sustainable with heavy coolant	2U/s	80%	+2.3MJ/s
3	Gradual overheating	3U/s	120%	+3.7 MJ/s
4	Rapid overheating	4U/s	160%	+5.3 MJ/s

Table 3-17: Warp Level data

Warp speed scales with warp drive effectiveness.

For example:

Warp 2 at 150% effectiveness = 3U/s
Warp 4 at 300% effectiveness = 12U/s

However, warp is not simply "more power equals better." Higher warp levels require more coolant than the ship can always provide. Warp 3 and Warp 4 may overheat even with maximum coolant, especially at high power.

Warp also has two distinct costs. The baseline power draw and the active power draw when in Warp. This distinction matters when calculating long-distance efficiency. For more details on warp power, see **Table 3-10** on page 19 . For example warp/power configurations, see **Appendix F: Warp Power Configurations** on page 131 .

Entering and Leaving Warp

Warp is not instant. A vessel needs to be stationary before engaging warp, and higher warp levels take time to accelerate into and decelerate out of. This can be useful: warp level affects the acceleration profile, while power affects final speed.

Because deceleration time depends upon warp level, underclocked Warp 3 may traverse a Warp Jammer more effectively than overclocked Warp 2. As ever, speed is lovely; heat is the invoice. So if you're bound to hit a Warp Jammer, hit it as hard as possible.

Warp in Combat

Whilst Warp is primarily for long-distance travel, skilled Helm Officers can use it effectively in tactical combat. For more information see **Appendix G: Advanced Combat Manoeuvres** on page 132.

3.3.5 - Heat and Coolant

Propulsion systems are among the most heat-sensitive systems on the ship.

Impulse, Manoeuvring, and Warp all generate heat from high power allocation. In addition:

- Combat Manoeuvre can add direct heat to Impulse and/or Manoeuvring.
- Active warp creates major additional strain on the Warp system.
- High-power warp can overheat rapidly if coolant is insufficient.

The exact safe duration depends on coolant allocation, current heat, existing damage, and whether the system is being actively used.

Propulsion and Damage

Damage reduces propulsion effectiveness.

A ship with damaged impulse may still turn, but struggle to control range. A ship with damaged manoeuvring may still move forward, but struggle to aim, dodge, or protect its shields. A ship with damaged warp may be unable to disengage, relocate, or escape.

This makes propulsion one of the highest repair priorities during combat. A ship that is dead in space will soon be a ship that is dead.

3.3.6 - Propulsion Summary

- **Impulse** decides how quickly the ship moves forward.
- **Manoeuvring** decides how well the ship points and dodges.
- **Combat Manoeuvre** allows the ship to make urgent corrections or adjustments.
- **Warp** decides whether the ship can cross the battlespace at scale.

A well-handled ship is not merely fast. It is controlled, positioned, and still capable of moving when the unpleasantness begins.

3.4 - Defensive Capabilities: Shields & Frequencies

"Defence is not the absence of aggression. It is the reason you survive long enough to return it."

Primary Role: Weapons (Beams)
Secondary Roles: Chief Engineer, Helm

Your main ship defences are your shields. These are more than a pool of protection. Their strength, calibration, frequency, and timing all shape how well a ship survives under fire. This section covers the defensive mechanics every officer should understand, not just those seated at Beams.

3.4.1 - Shields

Controlled by:	Beams
Powered by:	Front Shield Generator Rear Shield Generator
Resting Power Draw:	24 MJ/min
Active/Extra Draw:	+ 90 MJ/min
Power Increase:	Improves shield recharge and calibration linearly and reduces incoming damage non-linearly.
Heat/Coolant Notes:	Generates no extra heat while in use.
Special Notes:	Raising shields is binary; generator power and shield activation are separate concerns.

Shields are the ship's primary defensive barrier, absorbing incoming damage before it reaches the hull. Most UCN vessels project separate shield fields around different facings, commonly fore and aft, each supported by its own shield generator.

Key areas for officers to be aware of:

- Maintain continuous awareness of shield levels.
- When reporting, report front first, rear second.
- Request Spikes to increase defences at critical moments.

3.4.2 - Calibration

Utilising Tactical information provided by Radar, shield frequencies can be re-calibrated to ensure maximum defence against an enemy's beam weapons. Base calibration time is approximately 25 seconds.

This is reduced linearly by the average (front and rear) shield effectiveness, e.g. if the front shield has 150% power and rear shield has 200% power, this will average at 175%, and will therefore reduce calibration time to 14.2 seconds.

If unknown, statistically the best frequency is 540 THz. Against adaptive enemies (those you know will be modulating beams), setting shields to 400 THz shields will limit their advantage.

3.4.3 - Guidance

- Calibrate before engagement where possible.
- Avoid re-calibration under sustained fire unless absolutely necessary. Calibration requires the shields to drop.
- If required under pressure, request a Spike to one or both shield generators to accelerate the process.

3.4.4 - Damage Reduction

Effectiveness scales **super-linearly**. This means that there is an **increased** effectiveness when the power to shields is **increased**.

- At 50% effectiveness - Damage taken is **increased** by ~10%
- At 200% effectiveness - Damage taken is **reduced** by ~30%
- At 300% effectiveness - Damage taken is **reduced** by ~70%

This means that a carefully timed Spike can make even a missile barrage seem inconsequential.

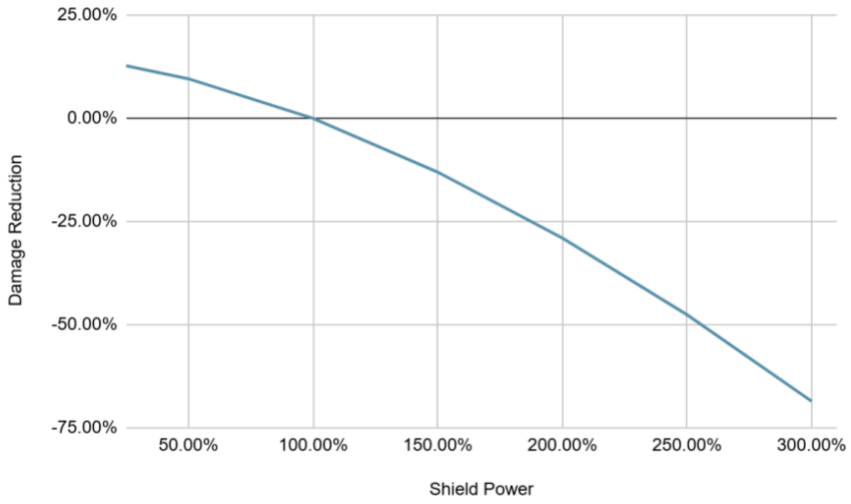


Figure 3-11: Shield Power vs Damage Taken

Shield Power	Damage taken
25%	+ 12.5%
50%	+ 10%
100%	No change
150%	- 13%
200%	- 30%
250%	- 50%
300%	- 70%

Table 3-18: Shield Power vs Damage Taken

3.4.5 - Shield Flashing

Shield activation is instantaneous and incurs a fixed energy cost while active. Raising shields only when immediately required is an advanced technique, known as shield flashing.

Principle

- Shields **up** grants full protection, but costs +90 MJ/min.
- Shields **down** offers no protection, but does not cost energy.
- Activation and deactivation are **immediate.**

As a result:

Energy expenditure depends on **time active**, not **frequency of use**.

Application

Shield flashing is most effective against:

- Missile volleys.
- HVLI (slug) impacts.
- Predictable burst damage.
- Long-range engagements with clear timing windows.

Typical use:

1. Shields remain down while no impact is imminent.
2. Raise shields just before impact.
3. Lower shields immediately after.

Benefits

Shield flashing significantly reduces energy consumption, allowing:

- Higher power allocation to other systems.
- Longer endurance in extended engagements.

Limitations

- Requires precise timing.
- Offers no protection if mistimed.
- Ineffective against sustained or overlapping damage sources.
- Provides limited benefit in close-range engagements.

Coordination

Engineering support is required to ensure there is power for usage, but no ongoing support required.

Operational Guidance

- Use flashing when engagements are controlled and predictable.
- Maintain continuous shields when under sustained or chaotic fire.
- Do not attempt to flash against multiple overlapping threats.

3.4.6 - Shield Dipping

Shield Dipping (sometimes referred to as Shield Ducking) is the deliberate lowering of shields immediately before an incoming EMP detonation. EMP damage affects shields, but does not pass through to the hull; if the shields are down when the pulse arrives, there is no shield system for the EMP to overload. Once the pulse has passed, shields should be raised again immediately.

This is a high-risk timing technique. Dip too early, and the ship may be exposed to conventional weapons. Dip too late, and the EMP will collapse the shields anyway. It is most useful when an EMP impact is clearly identified, isolated, and not accompanied by kinetic or energy fire. Done well, it preserves shield strength for the damage that actually wants to kill you. Done poorly, it turns a clever trick into a very formal invitation to be shot.

3.4.7 - Shield Recharging

Standard UCN vessels recharge shields at a rate of 0.3 points per second. Increasing power to the shield generators increases this in a linear fashion. At 200% they regenerate 0.6 points per second. External support (e.g. docking with a larger craft or station) may significantly increase recharge.

Owing to the low recharge speed, regeneration should be treated as recovery, not combat defence.

For more information, see **Appendix D: Post Combat Recovery** on page 124.

3.4.8 - Shields at Warp

Shields may be kept raised during warp, but doing so is *extremely* expensive. In addition to the normal shield upkeep cost, active warp travel consumes **50% more warp energy** while shields are raised. At high warp levels or high Warp power, this surcharge can drain reserves *very* quickly. Lowering shields during safe warp travel is therefore *strongly* advised, unless immediate impact or hostile contact is expected.

"Most officers treat warp as downtime. It isn't.

"We were crossing a nebula, mines everywhere, no clean route. Keeping shields up would've drained us dry, so we flashed them instead. Up just long enough to catch an impact, down again before the drain could bite. Timing had to be exact.

"It only works if Engineering is in the loop. Power must be ready when called - too early wastes it, too late costs you.

"We came through intact, barely touching reserves.

"The same applies in a long-range slugfight. Slugs don't arrive all at once- if you're paying attention, you can feather the shields and save power.

"Warp and distance don't excuse inattention.

If you're on Beams, you're always on duty."

- Lt. Tim "Git" Jones

3.5 - Offensive Capabilities: Damage, Beams, & Ordnance

"Weapons are not for making noise. They are for removing problems from the list."

Primary Role: Weapons (Beams), Weapons (Missiles)
Secondary Roles: Chief Engineer, Helm

UCN ship weapon systems are split into Beams and Missiles. However, the primary goal for both is to destroy or disable enemy ships, whilst keeping yours alive.

3.5.1 - Damage Types

UCN ships have access to four distinct damage types: Energy, Kinetic, EMP, and Heat. While all can be used to degrade or disable an enemy, they do so in very different ways. Some are intended to break shields and tear into the hull, while others are designed to overload defences or cripple internal systems. Understanding the difference is essential if a crew is to choose the right weapon for the right target.

Energy

Standard directed-energy damage. It interacts with shield and beam frequencies, is absorbed by shields first, and any excess then damages the hull. Of the four types, it is the most conventional and the best all-round offensive damage for beam weapons.

Kinetic

Physical impact and explosive force. Like Energy, it is stopped by shields first and then spills into the hull if shields fail, but it does not use the beam-frequency system. Most missiles, HVLIs, mines, and nukes deal Kinetic damage.

EMP

Shield disruption rather than destructive force. EMP damage is applied to shields, but does not carry through into the hull, making it ideal for collapsing shields without damaging the target underneath. It is used by EMP warheads and can also be used by specially configured beams.

Heat

Thermal overload rather than direct structural damage. Heat damage bypasses shields and raises system temperature instead of reducing hull directly, making it useful for stressing or disabling ship systems through overheating. It is primarily a specialist beam damage type rather than a standard missile effect.

3.5.2 - Hull and System Damage

Any energy or kinetic damage not absorbed by shields is applied directly to the hull.

When the hull is struck, internal systems are also affected. If no specific system is targeted, the damage will:

- Apply in full to the hull.
- Also degrade one random system.

System damage reduces effectiveness, impairing that system's performance.

3.5.3 - System Integrity

All systems operate within a finite integrity range. As they take damage, their effectiveness declines.

The degree of system damage depends on how much of the hull is struck relative to the ship's overall strength. As a result:

- Heavily armoured ships are more resistant to system degradation.
- Lighter vessels suffer more rapid internal disruption under sustained fire.

3.5.4 - Targeted System Fire

Energy Beams may be configured to target a specific system. When doing so:

- Hull damage is reduced.
- The targeted system is reliably damaged.
- Two random additional systems will still suffer incidental damage.

Notably, focusing on a system does not produce the most efficient damage overall, it trades raw destructive output for precision.

If you are able to destroy a ship quickly, target the ship's hull. If it's a tough enemy, and you need to gain tactical superiority to be able to destroy it, target systems.

System damage may continue to accumulate even after a system is already critically compromised. So if a system has already been reduced to -100%, it is advised to target a different system, as damage will still be soaked up by the destroyed system.

Operational Guidance

- Target the hull when the objective is destruction.
- Target specific systems when the objective is disablement or control.
- While system-targeted fire is less efficient at destroying a ship, it is the most reliable way to remove specific capabilities.

"We were up against a destroyer that simply would not die.

"At first, we did the obvious thing. Shields down, beams on hull, keep firing until something important stopped being attached. Against a lighter ship, that would have worked. Against this one, it was just making it angry.

"Worse, it kept turning well enough to keep its main arcs on us. Every time Helm tried to work around it, the destroyer answered the turn and brought its guns back to bear.

"So Beams stopped trying to kill it quickly and started trying to make it clumsy.

"They shifted to targeted fire on manoeuvring. The hull damage slowed, but the effect on the fight was immediate. Its turns grew sluggish. Its arcs drifted behind us. Helm found space, kept us off its strongest weapons, and suddenly the destroyer was no longer dictating the engagement.

"After that, destroying it was merely a matter of time.

"That is the lesson, cadets. Hull fire destroys a ship. System fire decides whether the enemy still gets a vote."

- Lt. Wes Janson
UNCA - Weapons Training School

3.6 - Beam Weapons

"Beams do not ask the enemy to explode immediately. They make a persuasive case over time."

Controlled by:	Beams (Lethal) Missiles (Non-lethal)
Powered by:	Beams
Resting Power Draw:	24 MJ/min
Active/Extra Draw:	+ 3 MJ per beam shot
Power Increase:	Increases beam firing rate linearly
Heat/Coolant Notes:	Each shot adds 2° heat per beam
Special Notes:	Overlapping beam arcs allow more than one beam to strike the same target

3.6.1 - Lethal Beam Weapons

- Primary directed-energy weapon system.
- Capable of both offensive fire and missile interception.
- The standard rate is 1 shot every 6 seconds.

Standard UCN ships have 2 forward firing beam arcs. Where these overlap, both beams will hit the target.

On standard UCN ships, each beam uses 3 MJ and creates 2° heat per shot.

Energy Beams are the only Beams that can be calibrated to better affect enemy ship shield frequencies.

If you do not know the shield frequency of your target, statistically the best frequency to set is **800 THz.**

3.6.2 - Non-Lethal Beam Weapons

Most UCN ships have two non-lethal beam weapons: Heat and EMP. They are tied together into a single control. These usually have greater range than Energy Beams, and a higher fire rate (1 shot per second).

Heat Beams

Heat beams bypass enemy shields and dump heat straight into ship systems (randomly, if untargeted). If a system then becomes overheated, its effectiveness is diminished. (See **1.3 - System Effectiveness** on page 9.)

EMP Beams

EMP beams will directly damage enemy shields. They are not affected by shield frequencies, so their effectiveness cannot be configured. Once the enemy shields are down, EMP beams cease to be effective.

Targeting

System targeting is shared across all beams, so weapon officers will need to co-ordinate their target selections.

Heat Beams work by overheating systems. Without a targeted component, their effect is diluted. Always assign them to a specific system.

3.6.3 - Increasing Power

Increasing Power to Beams will increase the firing rate in a linear fashion, not the damage of each beam. At 200% power, Beams will fire twice as fast, thus draining power and increasing heat twice as quickly.

3.6.4 - Ordnance Interception

Each beam shot has a small chance to intercept incoming ordnance, making beams a potential defensive option when missiles are closing.

Interception is not guaranteed, so aim to create as many firing opportunities as possible before impact. Multiple beams, good alignment, and higher-rate non-lethal beams can all improve the odds, helping thin incoming fire or buy time to manoeuvre clear.

3.6.5 - Summary

- Generate significant heat within the beam system.
- Must be managed carefully to avoid overheating.
- Consideration to be given to switch off non-lethal beams before lethal beams where necessary.
- When no longer needed to disable a ship, non-lethals should be switched off by the Missiles station untargeting the hostile.

3.7 - Ordnance

"A good launch solves a problem. A bad launch creates several, usually nearby."

Controlled by:	Missiles
Powered by:	Missile System
Resting Power Draw:	8 MJ/min
Active/Extra Draw:	None
Power Increase:	Increases missile tube reload speed linearly
Heat/Coolant Notes:	Standard heat increase with power increase
Special Notes:	None

Includes:

- Torpedoes.
- HVLIs (high-velocity projectiles).
- EMPs.
- Nuclear weapons.
- Mines.

Each has distinct behaviour, range, and optimal use cases.

3.7.1 - Ordnance Characteristics

- Torpedoes (Trackers) track targets and correct course at close range (1.2U).
- Nukes and EMPs track targets and correct course at very close range (0.5U).
- HVLIs (Slugs) reward proper spacing and timing, but do not home in.
- Area-of-effect weapons detonate with significant blast radii.

HVLIs do not reach their full damage potential until they have been in flight for 2 seconds. This is why the recommended firing range is 1U, as opposed to point blank.

3.7.2 - Area-of-Effect Weapons

These weapons are highly effective - but dangerous to friendly assets. This means if you don't time it correctly, an EMP or a Nuke can take out your own warheads if you've fired more than one.

Blasts only apply their damage to the single shield facing exposed to the centre of the blast. They do not spread damage across multiple shield facings at once, even though the explosion affects an area.

Nukes

Nuclear warheads deal kinetic damage across an area of effect, with maximum damage at the point of detonation and reduced damage towards the edge of the blast. The effectiveness of a strike depends heavily on proximity; direct hits are devastating, while near misses may be survivable.

EMPs

EMP warheads apply area-of-effect disruption rather than kinetic damage. Like nuclear weapons, their effectiveness is greatest at the centre of the blast and diminishes towards the edge. EMPs do not damage systems or hulls. Instead, their damage goes to the shields, potentially overloading them.

Mines

Mines detonate by proximity rather than impact. Once triggered, they explode after a short delay, dealing kinetic damage across a wide area. Damage is greatest at the centre of the blast and falls off towards the edge. In practical terms, a mine delivers exactly the same destructive force as a standard nuke, simply delivered by proximity rather than launch.

3.7.3 - Missile Behaviour

- Homing weapons (Torpedoes, Nukes and EMPs) will initially track predicted movement, then reacquire targets at close range.
- Evasive targets can avoid lock if sufficient distance is maintained.
- Once reacquired, missiles are significantly harder to evade.

A missile lock does not just mean a target has been selected. The missile console will only display "LOCKED ON" when a loaded tube can calculate a valid firing solution.

Homing ordnances still require a viable launch solution; their terminal guidance only takes over once close enough to the target. Firing without lock may still launch the missile, but it will leave the tube on a straight bearing rather than a useful intercept.

Mine Deployment

- Mines require time to arm after deployment.
- Trigger on proximity and detonate 1s after.
- Require coordination with Helm to deploy effectively.

Weapon	Min range (Recommended)	Max range (no tractor)	Homing Range	Flight Speed	Flight Time	Blast Radius
Torpedo	0U	5.4U	1.2U	0.2U/s	27 seconds	N/A
HVLI	1U	6.75U	N/A	0.5U/s	13.5 seconds	N/A
EMP	1.25U	5.4U	0.5U	0.2U/s	27 seconds	1U
Nuke	1.25U	5.4U	0.5U	0.2U/s	27 seconds	1U

Table 3-19: Ranged Ordnance Characteristics

Mines are measured along different metrics:

Weapon	Deploy Range	Blast Radius	Activation Radius	Arming Time	Detonation Time
Mine	1U	1U	0.6U	10 seconds	1 second

Table 3-20: Mine Characteristics

A mine release speed is 6U/min - therefore will travel 1U before it is armed. This is regardless of your speed. No matter how fast you are going, the mine will come to rest 1U behind the ship at the exact point of release. Mine radii will overlap and trigger each other, so mines will need to be laid at least 1.2U apart.

3.8 - Tractor Beam Operations

"The tractor beam exists to remind objects that their plans are subject to revision."

Primary Role: Weapons (Beams) / Dock and Drone
Secondary Roles: Helm

Controlled by:	Beams or Dock and Drone
Powered by:	Cargo Docks
Resting Power Draw:	No additional
Active/Extra Draw:	Per target affected
Power Increase:	Increases drag speed of affected targets
Heat/Coolant Notes:	Does not generate heat directly
Special Notes:	Tractor effect is applied to all valid objects within the beam arc.

The tractor beam is a battlefield control system capable of pushing, pulling, or holding objects within a projected arc ahead of the ship.

Unlike most systems, the tractor beam does not have a fixed range or spread. Instead, it is constrained by a **maximum coverage area**:

- A wide arc results in a short range.
- A narrow arc results in a long range.

The beam is always trading one for the other. As a rule of thumb:

- Wide coverage is useful for intercepting missiles and multiple targets.
- Narrow focus is useful for reaching distant or specific targets.

The beam may also be configured to cover less than its maximum area, allowing for precise control when required.

Desired Range	Arc (Maximum)	Notes
2U	6°	Longest possible range
1.5U	~10.5°	-
1U	~22.5°	Widest arc at 1U
~0.73U	45°	-
0.5U	90°	Widest possible arc

Table 3-21: Tractor Beam Range vs Arcs

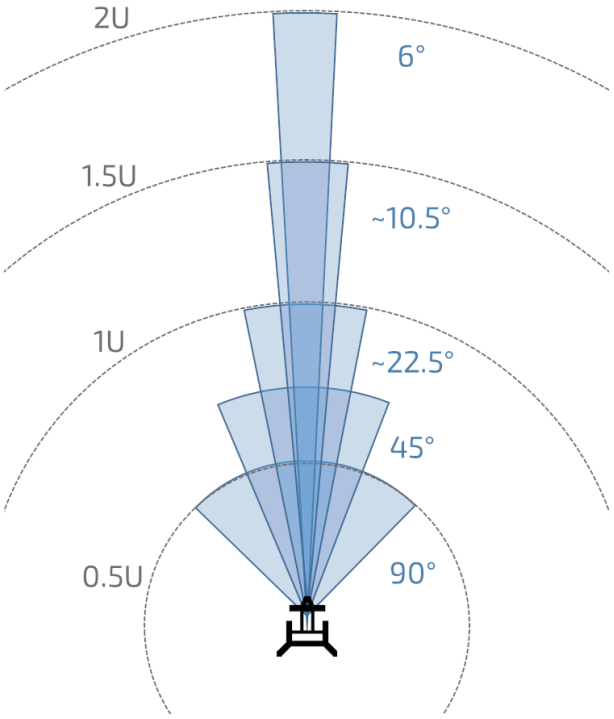


Figure 3-12: Tractor Beam Arcs and Ranges

3.8.1 - Operation

The tractor beam is controlled via four settings:

Mode:

- **Off**: Tractor beam is off.
- **Pull**: Targets are drawn towards the ship.
- **Push**: Targets are pushed away until they are out of the beam.
- **Hold**: Target is pushed/pulled until it is in the centre of the current beam projection length (range).

Range - Length of the beam projection.

Direction - relative to ship heading (0° is forward).

Arc - width of the beam.

Tractor movement occurs along a straight line between the ship and the target. The beam does not rotate captured objects with it; control

is maintained only while the target's centre remains within the beam's arc and range. Sharp turns by the ship, high-speed passes, narrow arcs, and the target crossing directly behind the ship all increase the risk of losing control.

The tractor beam can therefore push, pull, or hold objects along its line of effect, but cannot swing a mine around the ship or move objects laterally across the beam.

Targeting

The beam affects all objects within its arc simultaneously, including:

- Ships.
- Missiles.
- Mines.
- Asteroids and other hazards.

The beam must cover the centre of a target to affect it. Very large objects may be difficult or impossible to manipulate for this reason.

3.8.2 - Movement and Control

The tractor beam affects objects by applying directional drag within its projected arc and range. Depending on the selected mode, it can pull objects towards the ship, push them away, or hold them at a fixed point within the beam projection.

This is not a rigid physical lock. The target remains affected only while its centre stays within the beam's arc and range. If it slips outside that coverage, the tractor beam loses control.

How strongly an object is moved depends on two factors:

Drag Speed - the beam's effective movement rate, improved by the effectiveness of the Cargo Docks system.

Target Size - smaller objects are moved more easily; larger objects resist movement more strongly.

As a result:

- Missiles, mines, probes, and small craft can be redirected quickly and reliably.
- Large ships can be influenced, but only gradually.
- A ship actively manoeuvring or trying to escape may overcome the beam's effect if its movement exceeds the tractor's ability to reposition it.

Effective tractor operation therefore requires continuous adjustment of range, direction, and arc. A skilled operator must keep the intended target inside the beam while avoiding unwanted objects.

Target Examples	100% Docks	200–300% Docks
Missiles, mines	Rapid control	Immediate control
Probes, light drones	Strong control	Rapid control
Shuttle, fighter	Useful control	Strong control
Gunship, corvette	Limited control	Useful control
Frigate, destroyer	Weak control	Limited control
Cruiser	Very weak control	Weak control
Battleship, carrier	Negligible	Very weak control
Stations, huge objects	None practical	None practical

Table 3-22: Drag speed vs power and target size

3.8.3 - Power and Efficiency

The tractor beam consumes energy only when it is actively affecting a target.

Energy use is calculated per target affected:

- More affected objects means higher total energy draw.
- Higher system effectiveness increases drag speed, but also increases energy consumption proportionally.
- Power is not shared between targets; each additional object adds its own full cost.

> If, at 100% power, you move an object 1U in 1 second, at 200% power, you can move it 1U in 0.5 seconds.
>
> Both of these scenarios will cost the same amount of energy.

Increasing tractor effectiveness moves objects faster, but does not necessarily make the total movement more energy-efficient. It simply spends the energy over a shorter period.

However, this may be required if the target's own speed is faster than your drag speed.

Careless use against multiple targets can therefore place sudden and significant strain on power usage, particularly when operating at high effectiveness.

> In the previous scenario, if we assume that moving a single object uses 5 MJ of energy, then moving two objects will use 10 MJ of energy.
>
> If power has been increased to 200%, then moving 2 objects 1U will deplete 10 MJ in 0.5 seconds. Or 20 MJ a second. This can quickly get out of hand if tractor beam operators are sloppy.
>
> More objects do not increase heat, but increase power drain.

3.8.4 - Practical Use

Missile Defence

Tractor beams can be used to push incoming ordnance far enough away that it either loses guidance or detonates harmlessly.

Torpedoes lose lock if pushed more than **1.2U** from their target. Nukes and EMPs only correct course within **0.5U**, but still have a **1U blast radius**, so they must be pushed at least **1U** clear to avoid the explosion. Mines do not lock on, but will trigger within **0.6U** and explode across **1U**.

Tractor arc selection is a trade-off. A wider arc gives the operator more margin for error and is more likely to catch incoming ordnance, but may also affect unintended objects. A narrower arc is harder to aim, but gives better control over exactly which object is pushed.

The figures below show the widest tractor arc that can still push ordnance the minimum safe distance when caught at maximum tractor range:

- 17° or less can push a torpedo far enough to break lock.
- 23° or less can push nukes, EMPs, or mines outside their 1U blast radius.
- 67° or less can push a mine outside its 0.6U trigger radius, though this does not necessarily put the ship outside the blast radius if it detonates later.

In short: for torpedoes, break the lock; for nukes and EMPs, beat the blast; for mines, avoid the trigger first and the explosion second.

Other Uses

- **Target Control** - hold or displace enemy ships.
- **Area Manipulation** - reposition hazards (asteroids/mines) into enemy paths.
- **Force Projection** - extend the range and speed of friendly missiles.

The beam can also be used to assist friendly craft, including recovering or repositioning drones, or quickly moving a shuttle into or out of position.

Operational Guidance

- Set a useful default configuration before engagement.
- Use wide arcs for defence and area control.
- Use narrow arcs for precision and reach.
- Be mindful of energy cost when affecting multiple targets.
- Tractor Beam operators need to coordinate with Helm - position matters as much as configuration.

Limitations

- Cannot move objects laterally around the ship - only toward or away.
- Effectiveness depends on target size and relative motion.
- High energy demand when used extensively.
- Tractor beams can technically cover any direction, but they do not transition through the rear 180° boundary. If either the ship turns, or the targets own velocity makes the target cross directly behind the ship, tractor control will be lost and require rapid re-aiming.

3.9 - Hacking & Cyber Warfare

"Cyber warfare is the art of winning a fight before the enemy realises which button stopped working."

Primary Role: Navigation, Communications
Secondary Roles: Helm (as Tactical Lead)

Controlled by:	Comms or Navigation
Powered by:	N/A
Resting Power Draw:	None
Active/Extra Draw:	None
Power Increase:	None
Heat/Coolant Notes:	Each hack attempt adds significant Scanner heat, typically around 20°–40°
Special Notes:	Hacking suppresses the target system's effective power.

Hacking allows a hostile ship's systems to be temporarily disrupted during combat. Used well, it reduces enemy combat capability before that capability can be brought fully to bear. Used badly, it overheats Scanners, distracts bridge officers, and achieves very little.

A hacking officer's purpose is not to chase elegant puzzles for their own sake. It is to create tactical advantage at the right moment.

Hacking can be controlled by both Communication and Navigation. As such, it's suggested that a "primary hacking" officer is decided, so scanners are not permanently overheated.

In practice, hacking is most valuable when it removes enemy capability rather than merely improving your damage output. A ship that cannot manoeuvre, cannot bring its beams to bear, or cannot flee at warp is often as good as beaten.

3.9.1 - Hacking Effects

A successful breach temporarily suppresses the target system's effectiveness.

This is not the same as damaging the system. A hacked system still exists, still receives whatever power has been assigned to it, and may still function - but it will do so less effectively.

Assigned Power	Equivalent after maximum hack applied
100%	25%
150%	75%
200%	125%
250%	175%
300%	225%

Table 3-23: Power equivalent after 100% hack

Practical effects of hacking are as follows:

- System performance is reduced.
- Repeated successful breaches increase the disruption.
- The effect wears off gradually over time.
- Engineering may compensate by assigning more power, but this still costs energy and heat as normal.

At maximum disruption, hacking reduces a system's effectiveness by roughly 75 percentage points.

This is why hacking remains worthwhile even against heavily powered systems. A system set to 200% may still be functioning, but under full disruption it performs far below what the target intended.

The effect is temporary, and will usually wear off over the course of approximately **three minutes** if not reinforced.

When a hack is finished, whether it is 100% complete, or partially completed (through choice or failure), it will dump between 20° and 40° heat into the Scanner system.

Repeated hacking will quickly overheat the system unless Engineering provides assistance.

3.9.2 - Core Considerations

Hacking officers must remain aware of three things at all times:

- Which hostile matters most.
- Which system matters most.
- How much Scanner heat can safely be accepted.

A successful hack is only useful if it shapes the engagement. Hacking the wrong ship, or the wrong subsystem, may achieve little beyond making the officer feel industrious.

Each hacking attempt also places a significant heat load on the Scanner system. Repeated attempts in quick succession can rapidly overheat it. Hacking must therefore be conducted with judgement, not enthusiasm alone.

3.9.3 - Speed vs Control

Hacking rewards decisiveness.

A fast, good breach is often more valuable than a slow, perfect one. If a hostile ship is about to bring its beams to bear, reducing that capability now is more valuable than producing a tidier result fifteen seconds later.

The aim is therefore **controlled aggression**:

- Move quickly enough to matter.
- Bank progress when the tactical situation demands it.
- Avoid gambling away useful progress for the sake of perfection.

3.9.4 - Target Selection

Not every hostile is worth hacking. In general, hacking is most effective against:

- Larger ships.
- More durable ships.
- Ships whose combat capability cannot be removed quickly by gunfire alone.

Small craft are often destroyed before hacking meaningfully changes the engagement. Larger ships, by contrast, remain in the fight long enough for disruption to matter.

Hacking priority may therefore differ from Helm's immediate firing priority.

A small gunship might be the right target for destruction.

A destroyer might be the right target for hacking.

Both can be true.

3.9.5 - System Priorities

Not all systems offer equal value when hacked. The examples below assume a system at 100% effectiveness before a 100% hack is applied.

Impulse / Manoeuvring

Excellent choices against agile or arc-dependent enemies. Reducing movement or turning speed to one quarter of normal creates space for Helm, interferes with pursuit, and makes it harder for the enemy to keep weapons on target.

Beams / Missiles

Beams are usually the highest-value target against ships reliant on direct-fire combat. A full hack extends the standard beam cycle from 6 to 24 seconds, *significantly* reducing incoming damage.

If the enemy relies primarily on ordnance, prioritise Missiles instead. A full hack quadruples missile reload times.

Warp

A situational but valuable choice when preventing escape, pursuit, or reinforcement movement. A fully hacked Warp system operates at one quarter of its normal effectiveness.

Shields

Useful, but generally less valuable than reducing enemy offensive output or manoeuvrability. A full hack causes only the affected shields to take approximately 12.5% more damage.

In addition, it reduces shield recharge and calibration speed to one quarter of normal, though neither usually matters during active combat.

Scanners

Low combat priority in most engagements.

As a practical rule:

- If the threat is **damage output**, hack **Beams/ Missiles** (target dependent).
- If the threat is **positioning**, hack **Impulse / Manoeuvring.**
- If the threat is **escape**, hack **Warp.**

As always, circumstances may change priorities.

3.9.6 - Range and Awareness

Hacking range is effectively governed by *awareness*, not proximity.

If a hostile contact is known and can be seen on the Strategic Map - whether by the main ship, a probe, or another linked asset - that target may be hacked.

This makes probes particularly valuable. They allow hacking pressure to begin before the main ship closes the range, and they allow disruption to continue during manoeuvre and disengagement.

A good hacking officer does not merely react to the fight. They begin shaping it before the ship arrives in the ideal firing position.

> "We had a suspected contact beyond effective sensor range.
>
> "Rather than close in, we sent a probe ahead. The moment it resolved the target, Hacking went to work. The probe did not last long under enemy fire, but by then the damage was already done.
>
> "When we warped in, their impulse response was compromised. We arrived on top of a vessel that could not position, could not respond, and could not meaningfully resist.
>
> "You do not always need to be in range to start the fight. Sometimes, you simply need something that is."
>
> *- Lt. Sam Hutton*

3.9.7 - Practical Awareness

Hacking is most effective when paired with an understanding of how the enemy fights. Officers with a hacking secondary specialism are encouraged to develop a working knowledge of:

- Enemy beam arcs
- Likely manoeuvring behaviour
- Shield distribution
- Likely target behaviour (flee, kite, etc.)

This is particularly important when deciding between hacking Beams and propulsion.

A ship with wide, dangerous beam arcs may be best neutralised by hacking Beams. A narrow-arc vessel that relies on agility may be more effectively crippled by hacking Impulse or Manoeuvring.

3.9.8 - Hacking Interface

The hacking interface presents a defensive grid seeded with hostile protection routines. [4]

The officer's task is to identify safe nodes, avoid defence routines, and build enough progress to complete or apply the breach before failure or overheating makes further effort unwise.

Key terms

Grid
The full hacking interface.

Node
An individual circle within the grid.

Trace
The number shown on a revealed node, indicating how many adjacent nodes contain defence routines.

Defence Routine
A hostile protection node. Triggering two of these will end the attempt. [5]

Breach
A successful hacking attempt.

[4] Hacking is basically the game Minesweeper - but I'm trying to keep this "in universe", so please forgive the extra steps and odd nomenclature!!
[5] Mines

3.9.9 - What the display means

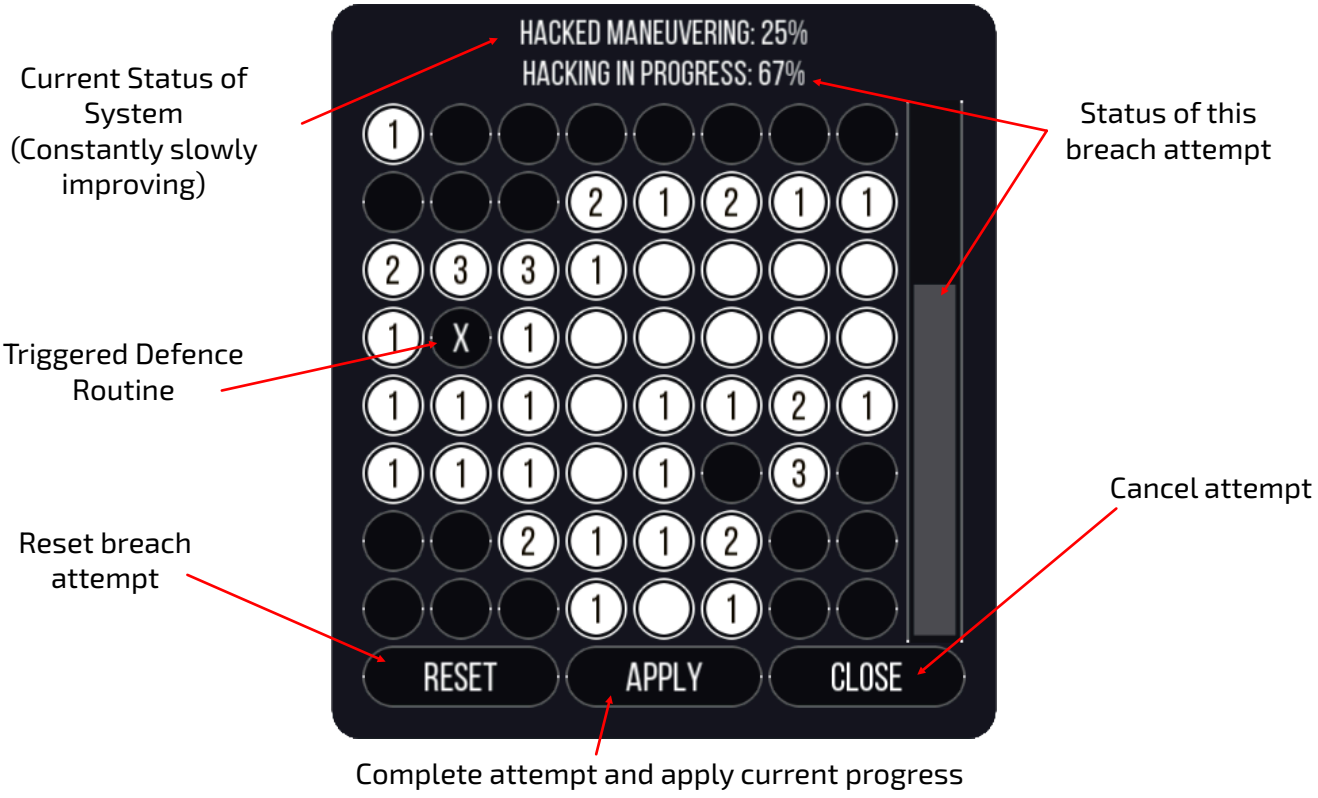


Figure 3-13: Hacking Display Example

The hacking display contains two separate percentage indicators, and they mean different things:

Hacked [System]: X%

This shows the disruption currently affecting the target system. This will always be decreasing slowly as the system recovers.

Hacking in Progress: X%

This shows the progress of the *current* breach attempt.

These are not the same thing.

A current breach attempt may be at 67% progress even though the target system itself is only partially disrupted. Likewise, a target may already be hacked while a fresh breach attempt is underway.

Each completed hack will apply 50% of the current breach progress.

For example, if you have completed a hack to 100%, the target system's hacked status will increase by 50%. If you complete to 80% before ending the hack (voluntarily or otherwise), the target system's hacked status will increase by 40%.

Therefore, in theory, two complete hacks are required to get a system's hacked status to 100%. In practice, it is often three or more, as during the time it takes to complete the second hack, the system has recovered sufficiently that an additional 50% hack status is unlikely to increase the status to 100%.

> Remember, 100% hacked does *not* mean it is completely disabled, it is the equivalent of reducing power to the system by 75%. So an undamaged system at 100% power that is 100% hacked will still have 25% capability.

3.9.10 - Applying a Partial Breach

Once sufficient progress has been achieved, the **Apply** button becomes available.

Using **Apply** deliberately banks the current breach result and applies it to the selected target system. This allows the hacking officer to accept a useful result immediately rather than spending more time chasing a stronger one.

However, officers should understand an important practical point: once a breach has passed the minimum threshold required to produce an effect, ending the intrusion does not necessarily waste the attempt. If sufficient progress has been made, an interrupted breach can still apply its partial result.

The main difference is control. **Apply** banks the breach at the officer's chosen moment. Continuing may increase the final disruption if more clean nodes are found, but once the breach is viable, there is little value in hesitating simply to preserve the board. The only real failure is taking too long and leaving the target system unaffected.

In the above example, applying the current progress (67%) would increase the target system's total hack level by 33.5%.

The Apply option becomes available at different points according to the target's cyber defence level:

3.9.11 - Cyber Defence Levels

Hostile cyber protection can be thought of in four broad levels. Higher levels mean a larger Grid, more Defence Nodes, and a higher threshold before a partial hack can be submitted.

Defence Level	Grid Size	Defence Nodes	Completion required before partial hack available
Basic	6 x 6	4	~ 60%
Secure	8 x 8	10	~ 70%
Hardened	10 x 10	25	~80%
Fortified	12 x 12	36	Full completion required

Table 3-24: Cyber Defence Levels Information

3.9.12 - Operational Overview

A hacking attempt proceeds in three broad stages:

1. Build progress by clearing safe nodes.
2. Avoid over-triggering defence routines.
3. Either apply the breach early or complete the intrusion fully.

A successful breach increases the disruption affecting the selected target system. Repeated successful breaches deepen that disruption, up to a maximum.

That disruption then begins to wear off gradually.

This creates an important tactical truth:

Hacking is pressure, not permanence.

One good breach may disrupt a system. Repeated breaches can cripple it. But unless pressure is maintained, the target will gradually recover.

3.9.13 - Technique

The interface rewards pattern recognition, speed, and discipline.

Practical guidance:

- Start in low-risk areas where possible.
- Use revealed traces to expand quickly through safe space.
- Do not become fixated on "solving the whole board" if the tactical moment is now.
- If the Apply button is available and the next move is uncertain, consider banking the result.

A hacking officer should always ask:

- Does this target still matter?
- Does this system still matter?
- Is Scanner heat still acceptable?
- Is further progress worth the risk?

If the answer to the last question is no, either apply the breach or cancel and move on.

3.9.14 - Hacking Defence

At present, UCN vessels do not have active Hacking Defences.

However, SPANR units assist hacked systems as well as damaged ones (see page 53).

A SPANR stationed at the affected system reduces the hacked level, in addition to the system's natural recovery. A fully hacked system will clear naturally in around three minutes, but a single SPANR can reduce this to roughly 80 seconds, with additional units improving recovery further.

Unless the system is completely disabled, it is difficult to know if it is hacked. All officers should report if they are suffering system degredation. If there is no damage in the system, it is possible that hacking has occurred. Engineering can send spare SPANRs to the affected system as a precaution.

3.9.15 - Pattern Recognition

Certain configurations provide reliable deductions. All of the below patterns can be flipped or rotated.

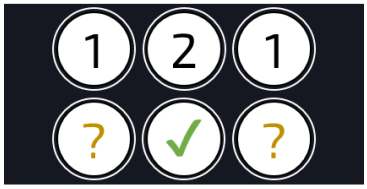


Figure 3-14: Hacking Pattern 1-2-1

If 1-2-1 appears in a straight line, the Node in front of the 2 is always safe, regardless of what is around this pattern. The Node in front of the two 1s may be Defence Routines.

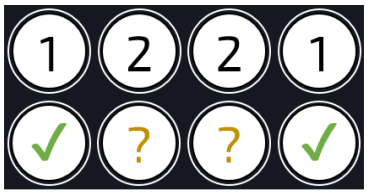


Figure 3-15: Hacking Pattern 1-2-2-1

If 1-2-2-1 appears in a straight line, the Nodes in front of the 1s are always safe, regardless of what is around the pattern. The Nodes in front of the 2s may be Defence Routines.

If the four Nodes behind this pattern are all safe, then the Nodes in front of the 2s are definitely Defence Routines.

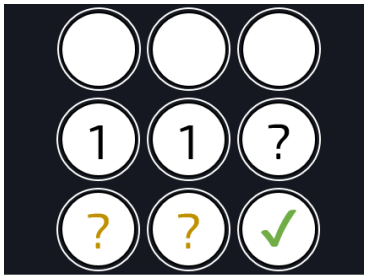


Figure 3-16: Hacking Pattern 1-1-?

If 1-1 appears in a straight line and the Nodes behind them both are safe, then the Nodes in front of them both may contain Defence Routines, but the third Node in that line is definitely safe, regardless of what the next number in the sequence is.

3.9.16 - Training

If officers wish to improve their hacking, then utilising an external hacking simulator is advised.

https://minesweeper.online

Select **Standard mode**, then **Custom**. The four hacking levels are detailed below:

It is encouraged not to use the "flag" option in the simulator, as this is not compatible with UCN hacking software.

Defence Level	Width	Height	Mines
Basic	6	6	4
Secure	8	8	10
Hardened	10	10	25
Fortified	12	12	36

Table 3-25: Training Data for Cyber Defence Levels

With proper practice, sub-40-second times can easily be achieved on the Secure level without triggering a single Defensive Routine.

Further defence node pattern training can be found at https://minesweeper.online/help/patterns.

3.9.17 - Final Guidance

- Hack to shape the engagement, not to optimise damage.
- Prioritise enemy capability reduction.
- Balance speed against heat, not accuracy against perfection.
- Accept risk; hesitation is more dangerous than error.
- A decisive hacker forces the enemy to fight at a disadvantage. A hesitant one merely overheats the scanners.

"I once watched a cadet achieve one of the cleanest breaches I have ever seen.

"Unfortunately, he achieved it about twelve seconds after it would have mattered.

"We were facing a hostile gunship with narrow but very unpleasant beam arcs. Helm had bought us a few seconds outside its line of fire, and Hacking had a viable breach against their manoeuvring almost immediately. Not perfect, but useful. Enough to make their next turn sluggish.

"Apply was available, so he knew he'd done enough to matter... but the cadet kept going.

"To his credit, he read the grid beautifully. Safe node after safe node, neat as parade boots, no defence routines triggered. By the time he completed the breach, the gunship had already turned, brought its beams to bear, and carved most of our front shield away.

"The hack landed. Their manoeuvring suffered. It was a fine result, delivered to a fight that had already moved on.

"Hacking is not an examination in elegance. It is a weapon. If the breach is good enough and the moment is now, apply it.

"A perfect hack too late is not superior to a partial hack on time. It is merely tidier wreckage."

- Sub Lt. Castin Donn
UNCA Cyber Warfare

3.10 - Probes & Drones

"The wise officer sends a probe or drone first. Machines complain less about being shot at."

Primary Role: Navigation, Radar, Dock and Drone
Secondary Roles: Chief Engineer

3.10.1 - Probes

Probes are remote sensor platforms used to extend the ship's awareness beyond its immediate position. They allow the crew to observe, scan, and assess areas that may be unsafe, distant, obscured, or tactically uncertain.

A probe is launched from the ship toward a selected point on the Strategic Map. Once deployed, it travels independently, provides local scanner coverage, and may be linked to Science/Radar for remote observation and target scanning.

Their primary purpose is to help the crew gather information before committing the ship.

Operational Behaviour

Probes travel at a fixed speed of **1U** per second.

This speed is *not* relative to the launching ship. A probe launched while the vessel is stationary, at impulse, or at warp will behave the same way: it appears at the ship's current position and proceeds toward its assigned destination under its own power.

A probe does not inherit the launching ship's velocity.

Each probe remains active for **10 minutes.** At 1U per second, this gives a theoretical maximum travel distance of **600U**.

In practice, probe range is usually limited by tactical need, enemy action, and the usefulness of the information being gathered.

Scanner Coverage

A probe provides scanner coverage around itself. At normal Scanner effectiveness, this coverage is a **5U** radius (the same as the Tactical Display).

Increasing Power

Increasing Scanner effectiveness expands probe coverage. Scanner power does *not* make probes fly faster, but it does increase the area that all launched probes can observe.

Scanner Effectiveness	Probe Scan Radius
100%	5U
150%	7.5U
200%	10U
250%	12.5U
300%	15U

Table 3-26: Scanner Effectiveness vs Probe Scan Radius

For a visual example of scanner coverage at different effectiveness levels, see **Figure 3-5** on page 20 .

This makes probes especially useful when paired with an Engineering-supported Scanner boost. A probe sent ahead under boosted scanners can provide valuable forward reconnaissance before the ship enters the area itself.

Linking to Science

A probe can be linked to Science/Radar while it remains active.

Only **one** probe may be linked at a time. Linking a new probe replaces the previous Science link.

However, as long as a probe is still active, owned by the ship, and selectable on the Strategic Map, it may be linked again.

A linked probe allows Science/Radar to observe from the probe's position and scan targets remotely. This can be used to identify threats, examine enemy vessels before engagement, support long-range hacking, or investigate areas the ship cannot safely approach.

Officers should remember that some readouts may remain relative to the ship rather than the probe. Care should be taken when reporting bearings and distances from a probe-linked view.

Tactical Uses

Probes should be used deliberately when the information gained will affect the crew's next decision.

Common applications include:

Forward reconnaissance
Send a probe ahead of the ship to reveal contacts, hazards, minefields, or ambush positions before committing the vessel.

Nebula operations
Probes can extend friendly scanner coverage into obscured areas. More importantly, when a probe inside or beyond a nebula is linked to Science, Radar can observe from the probe's position.

This can allow Radar to see and scan contacts that the ship's own Long-Range Sensors cannot resolve through nebula interference. Combined with boosted Scanners, probes become one of the most effective ways to investigate hidden contacts without committing the ship.

Remote scanning
A probe linked to Science can allow Radar to scan enemy ships before the main vessel enters weapons range. This is particularly useful for identifying large threats and preparing shield or tactical responses before combat begins.

Long-range hacking support
If a hostile contact is visible through a probe, Comms or Navigation can begin hacking operations before the ship closes distance.

Mine clearance
Probes can be used to deliberately trigger mine proximity fuses from a safer distance.

Laying a Trap
If a hostile sees a probe enter sensor range, they may infer the direction it came from and begin searching for the launching ship. This can be used against them.

By launching a probe from one direction, then repositioning or concealing the ship elsewhere, a crew can create a false trail and draw hostile forces toward the wrong area - ideally into a prepared mine trap.

Used well, the probe is not just reconnaissance. It is bait, misdirection, and a potential way to create an ambush.

Practical Guidance

Use probes when the information gained will shape the next decision.

Good probe use answers questions such as:

- What is inside that nebula?
- What is guarding that station?
- Is the route ahead mined?
- What class of ship is approaching?
- Can we scan or hack them before they reach us?

Probes are limited but renewable under the right circumstances. Certain friendly assets or docks may replenish probe stock, typically at a rate of approximately one probe every ten seconds. Nevertheless, officers should treat them as operational assets rather than expend them without purpose.

"During a live exercise, we were under strength against a heavier ship and steadily losing. Their weapons held the advantage, forcing us onto the defensive.

"Then an opportunity appeared - fleeting, easily missed.

"A nuclear strike was inbound. Rather than evade, we chose a far riskier option. A probe was launched on a precise intercept, timed to meet the missile exactly. There was no margin for error - too early or too late, and it would fail. Spectacularly.

"The probe struck true. The detonation occurred clear of our hull, yet close enough to collapse the enemy's shields entirely.

"We did not win the exercise. But for a few seconds, the fight belonged to us.

"Fortune favours the brave."

- Lt. Cmdr. Kane

Limitations

Probes are fragile. They can be destroyed, expire after their lifetime, or be sent somewhere that provides little useful information. They do not defend themselves and should not be assumed to survive if launched toward hostile forces.

Summary for Officers

- Probes are remote eyes.
- They travel at 1U per second, last for 10 minutes, and provide approximately 5U at 100%, scaling with Scanner effectiveness of local scanner coverage.
- They do not inherit the launching ship's speed.
- Probes are slow: launch them from a sector or two away, not ten sectors across the map.
- Any surviving friendly probe may be selected and linked to Science, but only one probe may be linked at once.
- Use probes to see before committing the ship, scan before engaging, and investigate uncertain areas from a safer distance.
- A well-used probe can prevent ambushes, improve tactical preparation (including shield calibration), and give the crew time to act before a threat reaches weapons range.

"We were on sweep detail along the outer edge of a debris field - one of those jobs where everything looks like something else, and nothing resolves cleanly. Probes were out, drones drifting, but the returns were messy. Weak signals, interference, nothing we could confidently act on.

"Engineering pushed Scanner to 200% - nothing dramatic, just a quiet nudge. Suddenly the noise settled. Probe returns sharpened, drone telemetry stopped smearing across the scope, and a dozen uncertain contacts resolved into three very definite ships trying to sit just beyond our effective range.

"They hadn't changed position. We hadn't moved. We simply saw them properly.

"We engaged on our terms. They never closed the distance."

- Lt. Sam Hutton

3.10.2 - Drones

Controlled by:	Dock and Drone
Powered by:	Cargo Docks (internal only)
Resting Power Draw:	8 MJ/min
Active/Extra Draw:	None
Power Increase:	None
Heat/Coolant Notes:	Recovering hot drones into non-thermic docks may dump heat into the dock system.
Special Notes:	Cargo Docks are used to store, transfer, recharge, repair, and recover drones.

Drones are remote auxiliary craft launched from the ship to extend its reach, awareness, and tactical options. They are not simply equipment: once deployed, they operate as friendly craft within the battlespace and should be treated as such by the rest of the crew.

Drones may be used to scout ahead, distract hostile vessels, deploy ordnance, intercept threats, or operate in areas where committing the main ship would be unnecessary or unsafe.

Their main value is that they allow the ship to influence events beyond its own immediate position.

How Drones Work

Drones are launched and controlled remotely from the ship. Once deployed, they move independently under the control of the Drone officer and can operate at significant distance from the main vessel, provided they remain within effective control range.

Drone operating range is strongly affected by Scanner effectiveness. Increasing Scanner power extends the range at which drones can be controlled.

This means that Engineering support can greatly increase drone usefulness. A temporary Scanner boost may allow drones to scout farther ahead, pursue a distant objective, or continue operating while the main ship remains at a safer range.

Increasing Scanner effectiveness also expands the scanner coverage contributed by friendly drones on the Strategic Map. However, the Drone

officer's direct tactical piloting display remains a short-range tactical display of **5U**. Scanner boosts therefore help drones operate farther away and reveal more to the wider ship, but they do not give the drone pilot a wider direct tactical screen.

What Drones Can Do

Scanner Effectiveness	Approx. Drone Control Range
100%	~78U
150%	~175U
200%	~311U
250%	~486U
300%	~700U

Table 3-27: Scanner Effectiveness vs Drone Control Range

Drones are flexible assets. Their exact capability depends on the ship and drone type available, but they may be used for several broad purposes.

Reconnaissance
Drones can scout ahead of the main ship, investigate uncertain areas, and reveal contacts or hazards before the vessel commits.

Tactical Distraction
A drone can draw enemy attention, force a hostile ship to turn, or split hostile fire away from the main vessel.

Ordnance Delivery
Some drones can deploy mines or other ordnance. This allows the ship to threaten areas or targets that may be difficult or unsafe for the main vessel to approach.

Threat Interception
In emergencies, drones may be used to intercept incoming ordnance or physically block a threat before it reaches the ship.

Remote Presence
Drones allow the crew to act in more than one place at once. A drone may scout, harass, or hold attention while the main ship manoeuvres elsewhere.

Drones also allow more allied firepower on the battefield to increase damage output.

"We'd pushed a little too far. You always know the moment, when the engagement turns from controlled to… ambitious.

That's when the nuke came in.

Radar caught it late. Too late for anything elegant. Thrusters were cooked, Beams were out of position, and there wasn't time to swing the tractor around.

It was just… there. Closing, and there wasn't a gorram thing anyone could do but brace and hope for the best.

Only one person had a shot. Drones.

He didn't ask, didn't call it out. Just course corrected his drone, burned everything it had, and put it on an intercept. No hesitation, no adjustment… just straight in.

It hit the warhead dead on.

Detonation was far enough out that we rode the edge of it. Shields flared, alarms screamed… but the hull held.

Afterwards, someone asked him how he timed it so perfectly. He just shrugged and said, 'Seemed about right.'

That, cadets, is what we call a steely eyed missile man."

- Lt. James Hall

Benefits to the Ship

Used well, drones provide several major advantages.

- They extend the ship's awareness, allowing the crew to see threats earlier and prepare before entering danger.
- They expand tactical reach, allowing the ship to influence an area without physically committing itself.
-

- They create additional problems for the enemy, forcing hostile vessels to divide attention between the main ship and smaller mobile threats.
- They provide flexible options during uncertain situations, particularly where direct approach would be risky.
- They can support mission objectives by scouting locations, checking routes, observing hostile movement, or creating openings for the main ship to act.
- A well-used drone does not merely add another unit to the battlefield. It gives the crew more information, more options, and more time to make the right decision.

"You've all heard the story of the steely-eyed missile man. Heroic, effective… and expensive.

But not every problem needs to be solved with a drone-shaped sacrifice.

We were outnumbered, with one hostile ship lining us up nicely and preparing to make our afternoon significantly worse. Rather than trade fire on their terms, I sent a drone in at warp and flew it tight around their hull.

They took the bait immediately.

They tracked it, turned after it, shifted shields to follow it, and wasted precious seconds treating one irritating little drone as though it were the main threat. It did not even need to fire a shot. It simply had to be close, fast, and impossible to ignore.

While they chased ghosts, we cleared the rest of the field, broke their support, and came back to deal with them properly once they were alone.

The drone even made it back: low on power, thoroughly overworked, but completely intact.

The point is, cadets… sometimes the smartest move is not to hit the enemy.

It is to make them look the wrong way."

- Lt. Cmdr. Phil "Rocky" Stoneman

Energy

Some drones are powered, meaning have their own energy reactors and can slowly generate their own energy.

Both powered, and unpowered drones can be "topped up" from the mothership's reactor.

This is done via the Energy Bay, and can go in both directions. A drone can dump energy into the main ship, or siphon energy to top itself up.

Operational Awareness

For most officers, the important point is simple: drones are friendly craft operating in the battlespace.

Their position may affect Helm manoeuvres, missile firing lines, area-effect weapons, and command decisions. If a drone is near a target, between the ship and a target, or deploying ordnance, that information should be known to the relevant stations.

Drones are valuable, but they are not invulnerable. They can be destroyed, lose effective control range, or become poorly positioned. Their use should support the Captain's intent and be coordinated when it affects the wider ship.

Summary for Officers

- Drones extend the ship's reach.
- They can scout, distract, deploy ordnance, intercept threats, and support operations away from the main vessel.
- Their operating range depends heavily on Scanner effectiveness, making Engineering support important for long-range use.
- Scanner boosts increase drone control range and strategic scanner coverage, but the drone pilot's tactical display remains short-ranged.
- Drones should be treated as friendly craft in the battlespace, not as disposable background equipment.
- Used well, drones give the ship more eyes, more options, and more ways to shape an engagement before the main vessel commits.

3.11 - System Repair

"Damage is inevitable. Unrepaired damage is a decision."

Primary Role: Manual Engineer/Damage Control
Secondary Roles: Chief Engineer

Ship systems do not simply work or fail. Each system has a condition value, ranging from +100% to -100%.

At +100%, the system is fully intact. As it takes damage, that value falls, reducing the system's effectiveness. A damaged system may still function, but its effectiveness is reduced (see **1.3 - System Effectiveness & Power Allocation** on page 9 .)

At 0%, the system has lost its effective function. Further damage can still be applied, pushing the system below 0%, down to a maximum of -100%. This does not make the system "more broken" in performance terms, but it does increase the amount of repair required before it is fully restored.

This is important for Damage Control. A system at 25% may be weak but recoverable quickly. A system at -100% requires far more time and attention. Repair priorities should therefore consider both tactical need and repair burden.

3.11.1 - SPANR Units (System Preservation Automated Nanobot Repair)

Overview

SPANR units are autonomous repair robots deployed throughout the vessel to conduct in-situ system repairs.

Most UCN ships have three SPANRs available. Standard shuttles have two SPANRs. Some ships may have additional SPANRs as part of a mission loadout.

Unlike traditional engineering teams, SPANRs:

- Operate continuously without fatigue
- Require no supervision once assigned
- Repair systems directly at the point of failure

Their limitation is not capability, but speed and positioning.

Operational Behaviour

- SPANRs must physically travel to a damaged system.
- Repairs begin only once the unit is on location.
- While moving, SPANRs perform no repairs.

Once in position, a SPANR restores system health at 0.7% per second. This rate is constant, and unaffected by system type or damage severity.

Some systems will need physical intervention before SPANRs can mitigate repair. (See **7.2 - Damage Control / Manual Engineer** on page 105).

Multiple SPANRs assigned to the same system repair it cumulatively, allowing critical systems to recover much faster.

However, not every system can accommodate every available SPANR at once. This is shown visually on the Damage Control screen. Each SPANR is represented by its own icon, and each system has a limited amount of space within its system box.

If multiple SPANR icons can fit inside a system's box, that system can accept multiple SPANRs. If there is not enough space, additional SPANRs will need to be assigned elsewhere.

Having SPANRs in a system can briefly offset minor overheating caused by overpowering the system. However, this offset is not sufficient to mitigate for long, and relying on them for thermal control is generally a sign that something has gone wrong elsewhere (or that your engineer is being "bold").

Movement

SPANRs move at a slow and set pace. Therefore, it will take between 10 and 12 seconds to traverse most UCN ships.

Because repairs only occur when stationary: Travel time is often a contributor to total repair time.

Operational Implications

- Pre-positioning SPANRs is critical
- Late movement results in delayed repairs
- Multiple damaged systems can overwhelm available units

Hacking Recovery

Hacked systems are marked with a purple ring on the Chief Engineer's Power Management screen and relevant station displays. This indicates electronic suppression, not necessarily physical damage. The system may still show normal assigned power, but its actual effectiveness is reduced until the hack clears. SPANRs assigned to the affected system will accelerate recovery.

Tactical Use

During combat:

- Position SPANRs near critical systems.
- Anticipate likely damage locations.
- Avoid unnecessary repositioning.
- Repair what the ship needs next, not simply what looks most damaged. A partly restored Missile or Cargo Dock system may be enough for the immediate task, while damaged Manoeuvring, Impulse, or Warp may need fuller recovery before the ship is safe.
- When several systems are damaged, set priorities by the tactical situation: finishing a target, disengaging, turning to face a new threat, or reaching warp before something regrettable arrives.

Outside combat:

- Reposition SPANRs to cover Impulse and Manoeuvring, reducing delay if they are damaged in combat.
- Place spare SPANRs to minimise response time across the rest of the ship.
- Monitor for minor degradation. A system at 95% health may look almost intact, but it is not fully repaired.
- If both Chief Engineer and Manual Engineer are present, agree who controls SPANR assignment. The engineering corridor Damage Control screen duplicates one of the Chief Engineer's displays, so either role may manage repairs depending on crew preference and workload.

Starting System Condition	No. of SPANRs assigned			
	1	2	3	4
75%	36s	18s	12s	9s
50%	1m 11s	36s	24s	18s
25%	1m 47s	54s	36s	27s
0%	2m 23s	1m 11s	48s	36s
-25%	2m 59s	1m 29s	60s	45s
-50%	3m 34s	1m 47s	1m 11s	54s
-75%	4m 10s	2m 05s	1m 23	1m 02s
-100%	4m 46s	2m 23s	1m 35s	1m 11s

Table 3-28: SPANR repair times

3.11.2 - Manual Repairs

Whilst the SPANRs can repair most ship functions, manual repairs may still be needed. This is typically indicated by a system turning bright red on the Damage Control screen, being shown as -100% health, and JUDITH announcing "Manual Repair Required!".

Some ship systems rely on physical components that may fail or destabilise during operations. Damage Control must understand what these components do, where they are fitted, and how to identify whether they are still serviceable.

Note that any of these repairs will typically take a system from -100% to -99% - the remainder of the repair process must be completed with other means; typically SPANRs.

Overcurrent Protectors

Overcurrent Protectors, commonly referred to as OCPs, protect delicate systems from failure by sacrificing themselves during a surge or overload.

OCPs protect the following systems:

- Manoeuvring.
- Impulse Engines.
- Missile Weapons.

A damaged OCP cannot be identified by sight alone. This is why discipline around spare storage is important: working and broken OCPs should not be mixed.

To test an OCP, insert it into a receptacle:

- Green indicates a working OCP is fitted.
- Red / flashing indicates the OCP is broken.
- Blue indicates that no OCP is detected.

A standard vessel will usually begin a mission with three OCPs fitted and four spares stored in the Spare OCP cupboard.

Masahirite Crystals

Masahirite Crystals allow tunable ship systems to function correctly. These systems rely on crystal stability to maintain calibration and frequency control.

Masahirite Crystals are used by:

- Front Shields.
- Rear Shields.
- Beam Weapons.

As with OCPs, the receptacle colour indicates the component state:

- Green indicates the crystal is stable and functioning.
- Red/flashing indicates the crystal has destabilised.
- Blue indicates that no crystal is detected.

Destabilised crystals can be restored at the Crystal Restabilisation Station. Restabilisation takes five minutes, so unstable crystals should be inserted as soon as possible.

A standard vessel will usually begin a mission with three crystals fitted and two crystals already loaded into the restabiliser.

Restabiliser locations are ship dependent, but common locations are medical bays, science labs or Captains' Ready Rooms.

Warp Core Instability and Emergency Ejection

If the Warp Drive is driven into a critical failure state - whether through damage, overheating, hacking, or other hostile effects - JUDITH may report that manual repair of the Warp Drive is required. This is accompanied by klaxons, flashing red warning beacons, and the activation of a countdown timer near the warp polarity display.

This indicates an impending quantum failure event. In plain language: the warp core is about to breach.

During such an event, the polarity display will show a set of numbered indicator lights. Any red indicators correspond to Quantum Polarisation Switches located around the ship. These are wall-mounted, numbered black boxes used to correct local polarity faults within the warp system.

The boxes also contain a matching indicator light. To stabilise that Warp Conduit, the switch inside should be flipped to the opposite orientation (either way works, due to quantum superposition)

Once the relevant switch conduit has been stabilised, its indicator should return to green. When all affected indicators are green, the immediate instability is averted and normal repair systems, including SPANRs, can continue restoring the Warp Drive.

Damage Control should be aware that the affected switches are not always conveniently located. If the core is known to be unstable, it may be faster to begin checking nearby Quantum Polarisation Switches rather than returning to the main display first, especially on vessels where several are located close to the bridge. As always, this should be done with communication rather than enthusiastic sprinting and hope.

If the core cannot be stabilised before the timer expires, the crew may be forced to eject the warp core. This should be treated as an emergency measure of last resort.

Ideally, the captain should make the call, but if in doubt, and the timer shows 10 seconds or less, any crew member is empowered to eject the warp core. Remember the SOLIS (Sanctity Of Life In Space) protocol - crewmember lives are more important than the integrity of the warp core.

Ejection prevents the failure occurring inside the ship, but leaves the vessel without warp capability and likely dependent on recovery by allied assets.

Other Systems

Occasionally, other esoteric systems, not covered in this material, may malfunction. In case of doubt, the ship's Comms Officer should reach out to your Flight Controller for guidance.

Summary

In short: red lights, klaxons, and JUDITH saying "manual repair required" mean Damage Control should move immediately, communicate clearly, and ensure Command knows whether the ship is being stabilised or the warp core is about to be breached.

"After escaping a pirate ambush, J.U.D.I.T.H. began repeatedly reporting warp core instability.

"We had a [REDACTED] aboard at the time, and the… sensitive nature of our mission had our largely rookie crew rather on edge for subterfuge. Every alarm felt suspicious. Every malfunction felt deliberate. Every unexpected noise sounded like the opening move in someone else's plan.

"Whilst running the length of the ship, restabilising the drive again and again, I started second-guessing the source of every complication. Had our visitor betrayed us to protect the [REDACTED]? Were we being hacked by Gray Hats just outside scanner range? Had the pirates left something behind when they jumped us?

"It was not until the fourth destabilisation that I noticed the true cause: the warp control on Helm had become wedged at Factor 4 during our escape from the ambush. The core was not being sabotaged. It was simply overheating as soon as we restabilised it.

"There is a lesson in that, cadets. Not every disaster is enemy action. Not every mystery is a conspiracy.

"Sometimes the problem with your ship is coming from within your ship."

- Sub-Lieutenant Oznogon



Officer Roles

4.0 - Command

"Command is not doing everyone's job. It is ensuring everyone knows why their job matters."

Command governs the strategic direction of the ship. Rather than operating systems or executing, command determines why those actions are taken, when they are necessary, and what outcome they are intended to achieve.

At its core, Command is decision and cohesion.

Every piece of information gathered across the bridge ultimately flows toward Command. It is here that fragmented reports, partial data, and competing priorities are resolved into clear intent. That intent is then fed back into the crew, guiding action without dictating every detail.

This is a subtle but critical distinction. A Captain who attempts to control everything will slow the ship down. A Captain who provides clear intent allows specialists to act decisively within their roles. The First Officer ensures this intent is carried through, maintaining clarity, reinforcing discipline, and preventing breakdowns in communication when pressure increases.

Command is therefore not about control, but alignment. It ensures that every station is working toward the same objective, even when conditions are chaotic and information is incomplete.

When Command falters, the crew becomes reactive, disjointed, and slow to respond.

When it functions properly, the ship moves with purpose, and often appears far more coordinated than it has any right to be.

Roles

Captain

Defines intent and sets priorities. Responsible for decision and direction.

First Officer (XO)

Ensures intent is carried out. Responsible for execution and coordination.



4.1 - Captain

"A Captain does not control the ship. They control the decisions that shape its fate."

4.1.2 - Role and Priority

The Captain is responsible for the strategic direction, safety, and purpose of the vessel.

The Captain does not operate systems, execute routine manoeuvres, or personally manage every tactical detail. Their role is to determine what the ship is trying to achieve, why it matters, and when priorities must change.

Priority is therefore:

1. Preserve the crew
2. Preserve the vessel
3. Complete the mission

These priorities are not decorative. Mission success is meaningless if the crew is lost unnecessarily, and a destroyed vessel rarely completes its objective with much dignity.

The Captain's task is to maintain enough awareness to make sound decisions, while allowing trained officers to act within their areas of responsibility.

At its simplest: The Captain sets intent, the crew executes it.

4.1.3 - Core Responsibilities

The Captain is responsible for:

- Remembering mission objectives and priorities
- Maintaining overall situational awareness
- Making decisions when priorities conflict
- Ensuring the bridge remains coordinated and disciplined
- Receiving, filtering, and acting on key information
- Authorising major changes of plan
- Deciding when to engage, disengage, retreat, negotiate, or escalate
- Ensuring the crew understands the current intent

A Captain should not attempt to know every detail at all times. That is why the ship has officers.

Instead, the Captain must know enough to answer the following questions:

- What are we trying to achieve?
- What is the greatest threat to that objective?
- What does the crew need to know now?
- What decision is required from me?
- Has my intent changed?

If those questions are answered clearly, the ship can function. If they are not, the bridge will usually begin solving several different problems at once, none of them elegantly.

4.1.4 - Execution

A Captain's most effective intervention is often made before the mission truly begins. At the outset, the Captain should brief the crew on expectations, decision boundaries, and delegated authority.

Make clear which station will identify or name primary combat targets, how much latitude Comms has in negotiation, what risks officers may take independently, and when decisions must return to Command. If the crew understands intent and authority before pressure arrives, they can act decisively without waiting for permission in the moment.

A well-briefed crew should not require constant direction; they should already know what the Captain expects of them.

Command Style

A good Captain gives clear intent rather than excessive instruction.

Effective command:

"Proceed to the station, avoid the minefield, and keep us outside weapons range until Radar has completed the scan."

Ineffective command:

"Helm, turn five degrees left. Now slow down. Navigation, place a waypoint there. Radar, scan that one. Engineering, raise scanners to 150%. No, not yet. Helm, turn again."

The first gives the crew a goal and constraints. The second turns trained officers into reluctant furniture.

The Captain should avoid micromanagement wherever possible. Officers at their stations usually have better immediate information than the Captain does. The Captain sees the whole purpose. That is the Captain's advantage.

Combat Command

During combat, the Captain should focus on intent, priority, and escalation.

The Captain may decide:

- Which objective matters most
- Whether to fight, delay, withdraw, or reposition
- Which hostile matters strategically
- Whether the ship can afford to remain engaged
- Whether mission priorities have changed
- Whether special authorisation is required

The Captain should avoid taking over moment-to-moment tactical handling unless the situation demands it.

In a well-functioning crew, Helm may act as tactical lead during combat, selecting positioning and immediate target flow within the Captain's intent. Weapons, Radar, Engineering, and other stations should then support that tactical execution.

For example:

"Disable the destroyer, but do not pursue beyond the station."

This gives Helm and Weapons freedom to fight while preserving the Captain's strategic constraint. For more information on threat prioritisation, see **Appendix C: Target Priority Assessment** on page 123.

Situational Awareness

The Captain must maintain awareness beyond any single display.

No console gives the full picture. A Captain should listen for reports from all stations and form them into a coherent understanding of the battlespace.

Important information includes:



- Current objective.
- Position of the ship relative to danger and mission goals.
- Major hostile contacts.
- Shield and hull condition.
- Critical system damage or overheating.
- Ordnance threats.
- Friendly asset status.
- New instructions or intelligence from Comms.
- Escape routes and fallback options.

The Captain may move around the bridge to observe, but should not hover, interfere, or operate another officer's console unless given explicit permission to do so by the officer. Movement should improve awareness.

A Captain who constantly roams may become harder to brief. Officers should know where to direct urgent information. If reports have to chase the Captain around the ship, command has become a scavenger hunt.

Decision-Making Under Pressure

The Captain's most important decisions are often not complex. They are timely.

A late perfect decision is frequently worse than an early adequate one.

When under pressure, the Captain should:

- State the current priority clearly
- Decide what must happen next
- Assign responsibility where needed
- Confirm that key stations have understood

Avoid reopening settled decisions unless the situation changes

Useful command phrases include:

- "Break contact."
- "Hold position until the scan completes."
- "Disable, do not destroy."
- "Ignore the fighters; focus the carrier."
- "Recover the shuttle, then withdraw."
- "New priority: protect the convoy."

The crew does not need a speech in the middle of a crisis. They need direction. [6]

[6] Of course, Bridge Command is, at its core, a theatrical experience. Feel free to be dramatic sometimes!

4.1.5 - Use of Shared Systems

The Captain is not expected to operate every shared system, but must understand their operational significance.

Strategic Map, Sensors, and Tactical Display

The Captain should understand that different stations see different versions of the battlespace.

Navigation and Comms provide the broader operational picture. Radar provides long-range detection and analysis. Helm and Weapons manage the immediate tactical space.

The Captain should use these views through reports, not by attempting to personally manage every display.

Shields and Defensive Posture

The Captain should understand the difference between shield status, shield power, and shield use.

They do not need to manage shield mechanics directly, but they must know when the ship is exposed, when defence requires priority, and when the crew should disengage or reposition.

The Captain may order a defensive posture, but Beams and Engineering execute the details.

Offensive Capabilities

The Captain should understand the broad difference between sustained fire, burst damage, disabling weapons, area-effect ordnance, and electronic disruption.

The Captain does not need to choose every weapon or direct every firing solution. They do need to decide what the ship is trying to achieve: destroy, disable, delay, deter, defend, or disengage.

The Captain is responsible for authorising major escalation, especially where nuclear weapons, area-effect weapons, or danger-close effects may affect friendly craft, civilians, or mission objectives.

Weapons, Helm, Radar, Hacking, and Drones may all contribute to the offensive plan. The Captain's role is to ensure that their actions serve the same intent.

Probes, Drones, and Friendly Assets

The Captain should treat probes, drones, shuttles, and allied vessels as extensions of the ship's capability.

They provide options, but they also require coordination. If a friendly asset is near the target, scouting ahead, or carrying out a task, the Captain should ensure the rest of the bridge understands how that affects the plan.

Hacking and Electronic Warfare

The Captain should understand hacking as a way to reduce enemy capability before or during an engagement.

The Captain need not direct the hacking interface, but may set priorities:

- *"Prevent escape."*
- *"Reduce their beams."*
- *"Cripple their manoeuvring."*
- *"Keep pressure on the cruiser."*

Hacking is most useful when it supports the wider tactical plan.

Shuttle Tactics

The Captain will also decide when to utilise a shuttle (if available). See **Appendix A: Shuttle Operations** on page 116 for full Tactical options.

4.1.6 - Coordination

The Captain coordinates with every station, but should not communicate with every station constantly.

First Officer

The First Officer extends the Captain's reach.

The Captain should use the XO to:

- Confirm orders were understood.
- Reduce bridge noise.
- Chase unresolved tasks.
- Manage secondary problems.
- Maintain coordination while the Captain handles major decisions.

A strong Captain/XO pairing allows the Captain to remain focused on decisions rather than admin.

Helm

The Captain gives Helm intent, constraints, and tactical priorities. Examples:

- *"Keep us outside beam range."*
- *"Close for a strike, then disengage."*
- *"Do not enter the nebula until Radar has scanned it."*
- *"Hold near the convoy."*

If Helm is acting as tactical lead, the Captain should avoid interfering unless strategic priorities change.

Weapons

The Captain should ensure Weapons understands the desired outcome:

- Destroy.
- Disable.
- Delay.
- Defend.
- Hold fire.
- Prepare escalation.

The Captain authorises extraordinary weapons use where required.

Radar

Radar provides early warning, contact identification, target analysis, and threat development.

The Captain should listen carefully to Radar reports, especially when new contacts appear, hostile movement changes, or unknown signals develop.

Radar should not need to provide every detail. The Captain only needs what matters.

[[Behind the Curtain]]

Being the Captain means you are at least partially responsible for the fun and enjoyment of the rest of the people playing the mission with you. You need to live up to that.

Be the leader you would want to have. Positive feedback only!

Navigation



Navigation maintains strategic context. The Captain should rely on Navigation for:

- Objective locations.
- Routes.
- Probe planning.
- Wider map awareness.
- Hazard marking.
- Strategic alternatives.

Navigation helps prevent the Captain from winning the fight and losing the mission.

Communications

Comms is often the Captain's link to Command, allied assets, and developing mission information.

The Captain must ensure Comms has permission to interrupt when important external information arrives.

Comms should be briefed clearly when contacting Command or friendly assets, particularly where authorisation, negotiation, or asset tasking is involved.

Engineering

Engineering determines what the ship can sustain.

The Captain should treat Engineering reports as constraints on ambition.

If Engineering says the ship cannot support a plan, the Captain may still order it, but should do so knowingly and with due appreciation for the sound of systems catching fire.

The Captain should also inform Engineering as to how much time they have available during a Post Combat Recovery (see **Appendix D: Post Combat Recovery** on page 124), so Engineering can adjust their plans accordingly.

Damage Control, Dock and Drone

The Captain should maintain awareness of internal repair status and deployed craft.

If major systems fail, friendly craft are at risk, or drones are executing mission-critical tasks, the Captain should ensure those efforts are included in the wider plan.

4.1.7 - Standard Callouts

Examples of useful Captain callouts:

- *"Mission priority remains the station."*
- *"Helm, tactical lead."*
- *"Disable, do not destroy."*
- *"Break contact."*
- *"Hold position."*
- *"Withdraw to safe range."*
- *"Protect the convoy."*
- *"Comms, request authorisation for nuclear release."*
- *"Navigation, give Helm a route."*
- *"Radar, identify the largest contact."*
- *"Engineering, prioritise shields and manoeuvring."*
- *"Weapons, hold fire."*

The Captain should use plain language. Brevity matters, but clarity matters more.

4.1.8 - Authority and Limits

The Captain has authority to:

- Set mission priorities.
- Change objectives.
- Order engagement or withdrawal.
- Assign tactical lead.
- Reassign officers if required.
- Authorise escalation.
- Override station recommendations.
- Request or approve nuclear weapon release through Command channels where required.
- Release the lock (via their console) on the Cargo Transfer System when it is utilised.

The Captain must not:

- Routinely operate another officer's console.
- Micromanage station actions without need.
- Ignore specialist warnings.
- Allow bridge communication to collapse into noise.
- Let tactical excitement override mission purpose.
- Treat friendly assets as expendable unless the situation genuinely demands it.

Whilst the Captain may overrule any station, a wise Captain remembers that needing to do so often is usually a symptom of poor command, poor briefing, or poor listening. When overriding, if possible, explain *why* you're overriding so officers have as much of an understanding of your reasonings as practicable.



4.1.9 - Summary

A Captain should aim to be clear, calm, and decisive. The crew does not need the Captain to know everything; it needs the Captain to know what matters.

Good command means setting clear intent, listening before deciding, trusting officers to execute, and intervening only when necessary. It means keeping the mission in view, changing priorities openly when circumstances demand it, making decisions early enough to matter, and recognising good work when the crew delivers it. Provide positive feedback often!

Poor command does the opposite. It chases unimportant detail, gives contradictory orders, micromanages capable officers, ignores inconvenient reports, speaks constantly, decides too late, and mistakes activity for leadership.

The Captain's chair is not a reward for knowing every system better than the crew. It is a responsibility to make the crew's knowledge useful. A ship under good command does not merely act quickly. It acts with purpose.

And in the end, that is the Captain's work: to ensure the crew works as one, especially when the universe is being deliberately unhelpful.

Saw a Captain once who wouldn't stay still. One moment at Navigation, next at Radar, then running around the ship looking to replace an OCP!

The problem was, no one knew where to speak to him. By the time reports reached him properly, they were already out of date. He had no idea of what was happening, as he kept on getting distracted by another officer.

Plenty of movement. No awareness. A Captain's place isn't everywhere- it's where the information comes together.

- Cmdr. Jan Dodonna
UNCA - Command School

4.2 - First Officer/Executive Officer (XO)

"A Captain sets intent. The First Officer ensures it becomes reality."

4.2.1 - Role and Priority

The First Officer exists to extend the Captain's reach across the ship.

Where the Captain operates at the level of intent and decision, the First Officer operates at the level of execution and coordination. Their role is to ensure that orders are heard, understood, acted upon, and followed through without requiring the Captain to personally chase every detail.

What this looks like will vary between crews, missions, and command styles. On one ship, the First Officer may act as bridge coordinator. On another, they may manage Comms traffic, oversee secondary tasks, command a shuttlecraft, or step into whichever gap is currently threatening to become a crisis.

Priority is therefore:

- Support the Captain's intent.
- Maintain bridge coordination and clarity.
- Ensure tasks are completed without unnecessary oversight.
- Step into gaps without overstepping authority.

In essence, the First Officer does not wait to be told what to do. They anticipate what is needed, act within the Captain's intent, and report back when it matters.

A good First Officer makes the Captain's job easier. A great one makes the entire bridge more effective without anyone quite noticing how.

4.2.2 - Core Responsibilities

The First Officer is responsible for maintaining order and clarity on the bridge, ensuring important orders are acknowledged and acted upon, reinforcing critical instructions when missed, and reducing unnecessary noise when communication degrades.

They should track secondary tasks and unresolved issues, form ad-hoc teams where required, monitor crew workload, highlight missed information or conflicting priorities, and take secondary responsibilities away from the Captain when Command is occupied.

The XO should be especially alert to tasks that are important but easy to lose under pressure: delegated orders, Comms replies, marked objectives, repair progress, shuttle or drone assignments, and stations that are overloaded, idle, confused, or isolated.

4.2.3 - Execution

Presence and Movement

Unlike most officers, the First Officer is rarely fixed to a single station.

They are expected to move across the bridge as required, observe without interfering, and speak only when doing so adds value.

Movement should serve a purpose: confirming that orders have been understood, checking unresolved tasks, identifying overloaded or isolated stations, supporting Comms or Damage Control during high workload, or positioning themselves where coordination is beginning to fail.

An observant, aware XO is often more effective than a loud one. The role is not to become a second Captain, nor to provide a running commentary on the obvious. It is to notice what is not yet obvious and act before it becomes expensive.

Communication Authority

The First Officer has authority to support communication discipline.

This includes calling for quiet when the bridge becomes too noisy, repeating or reinforcing orders issued by the Captain, clarifying known intent if confusion arises, redirecting unnecessary chatter, and ensuring urgent information reaches the Captain at the right moment.

Used well, this prevents breakdowns in high-pressure situations. Used poorly, it becomes another source of noise. The XO should not

compete with the Captain's voice. They should make that voice easier for the bridge to follow.

Independent Execution

The First Officer may be given broad tasks rather than precise instructions.

Examples:

- *"Handle that situation."*
- *"Investigate this new datapad."*
- *"Find out what Comms knows."*
- *"Get someone on repairs."*
- *"Coordinate the shuttle."*

From that point, the expectation is simple, complete the task. Report back.

This may involve interpreting intelligence, coordinating multiple stations, directing off-console activity, assigning available officers, or making limited decisions within the Captain's intent.

The XO should not repeatedly return to the Captain for permission on every minor step. If the task has been delegated, the XO should carry it forward unless the situation changes, the risk escalates, or a decision clearly belongs to Command.

Operational Flexibility and Gap Management

The First Officer is the most variable role aboard the ship.

Depending on crew size and mission requirements, the XO may command a shuttlecraft, act as a second tactical coordinator, manage internal ship operations, track mission objectives, coordinate off-console activity, support Comms traffic, monitor morale, assist Damage Control, or function as an intermediary between stations.

Or all of the above, in the same mission.

The First Officer is often most valuable when something is missing. A station may be unmanned. Comms may be overloaded. Damage Control may need another pair of hands. Navigation may be tracking objectives while also supporting Hacking. The Captain may be occupied with negotiation while the ship begins taking fire.

In such moments, the First Officer should identify the gap and either cover it, assign someone to it, or inform the Captain that it exists.



Directly operating another officer's console should be done cautiously and respectfully. The XO may step in during necessity, but should avoid taking control away from a station that is already functioning unless given explicit permission from the officer whose station it is.

If the ship does not have a full crew complement, it is often sensible for the officer at Navigation to also act as XO. Navigation already maintains the wider operational picture and is usually better placed than most stations to track objectives, routes, and developing priorities.

Shuttle Tactics

The XO is usually the first choice to lead shuttle missions. For full details, see **Appendix A: Shuttle Operations** on page 116.

Onboard Visitors and External Parties

The First Officer is often the natural officer to initially engage with anyone brought aboard the ship, whether guest, patient, detainee, negotiator, or courier, and take point on introducing them to Captain and crew.

The XO should manage the interaction in line with the Captain's intent, ensure useful information reaches the bridge, and prevent the situation from distracting Command from the wider mission.

The Captain/XO Dynamic

The effectiveness of a First Officer is directly tied to their relationship with the Captain.

A strong pairing allows subtle coordination, parallel task execution, rapid decision support, clear division between intent and follow-through, and a calmer bridge under pressure.

A weak pairing produces redundancy, hesitation, conflicting instructions, an XO in name only, or worse, a second Captain attempting to command the same ship in a slightly different direction.

Cadets should understand that this is a partnership, not a hierarchy of convenience.

The Captain is in command. The First Officer makes command easier to execute. The First

Officer coordinates with every station as needed, but always in support of the Captain's intent.

4.2.4 - Coordination

The First Officer coordinates with every station as needed, but should do so in support of the Captain's intent.

Captain

The First Officer supports the Captain by:

- Clarifying intent.
- Confirming orders have been acted upon.
- Advising when risks or missed information emerge.
- Taking secondary tasks away from the Captain.
- Acting as a filter when the bridge becomes noisy.
- Reporting completion, failure, or escalation of delegated tasks.

The XO should keep the Captain informed without requiring constant attention.

> "We had them on screen - talking. The sort who enjoys the sound of their own voice. I let them. Kept the tone measured, gave them just enough to stay engaged. While they spoke, I gave Lt. Pogue the slightest nod.
>
> No interruption. No callouts.
>
> While I held the conversation, she moved. Quietly brought Helm and Weapons into position, closed the distance, set the conditions.
>
> They never saw it happening. One moment they were negotiating with me… the next, they were in our firing envelope.
>
> That's what a First Officer is for, cadets.
>
> Not to be heard. To be effective."
>
> *- Lt. Cmdr. (Brevet Captain) Edmund Ward*

Helm

The First Officer may assist Helm by ensuring intent and constraints are clear.

The XO should not interfere with tactical handling unless the Captain has delegated that role or Helm requires support.



Weapons

The First Officer may ensure Weapons understands the intended outcome: destroy, disable, hold fire, prepare escalation, defend against incoming ordnance, or avoid danger-close effects.

The XO may also help ensure nuclear authorisation, firing restrictions, and friendly asset positions are understood.

Radar and Navigation

Radar and Navigation often hold information the Captain needs but may not have time to seek personally.

The First Officer may prompt for:

- Contact identification.
- Objective location.
- Route updates.
- Probe status.
- Hazard marking.
- Whether new information changes the plan.

Comms

Comms can rapidly become overloaded with external messages, asset control, negotiations, and Command updates.

The First Officer should support Comms by:

- Helping prioritise what must reach the Captain immediately.
- Ensuring important messages are recorded.
- Confirming delegated asset tasking.
- Relaying condensed updates when the Captain is occupied.
- Protecting Comms from unnecessary interruptions when critical traffic is underwa.y

Engineering and Damage Control

The First Officer should listen closely to Engineering and Damage Control reports, especially when they concern ship survivability.

The XO may help by:

- Confirming repair priorities.
- Assigning spare officers to emergency repairs.
- Ensuring the Captain knows when a critical system is down.
- Preventing non-critical requests from overwhelming Engineering.
- Checking whether damaged systems have been restored.

Dock and Drone

The First Officer may help ensure drone and shuttle activity remains tied to the main plan.

This includes:

- Confirming drone tasks.
- Warning other stations about friendly craft positions.
- Ensuring drone recovery or shuttle return is not forgotten.
- Making sure area-effect weapons are not used blindly near friendly assets.

4.2.5 - Standard Callouts

Examples of useful First Officer callouts:

- *"Confirm order received."*
- *"All stations, quiet for Command."*
- *"Comms, summarise for Captain."*
- *"Navigation, mark that objective."*
- *"Radar, confirm contact type."*
- *"Engineering, confirm whether that spike is available."*
- *"Damage Control, report current priority."*
- *"Weapons, hold until Helm confirms line."*
- *"Captain, Comms has priority traffic."*
- *"Captain, that order has not been acknowledged."*

The First Officer's callouts should be calm, short, and purposeful. They should reduce confusion, not add a second layer of command fog.

4.2.6 - Authority and Limits



The First Officer has authority to:

- Reinforce the Captain's orders.
- Clarify the Captain's intent when it is already known.
- Call for silence or communication discipline.
- Track and follow up delegated tasks.
- Assign available officers to secondary duties when appropriate.
- Coordinate off-console activity.
- Monitor or briefly cover an unmanned station in necessity.
- Advise the Captain on risk, missed information, or unresolved tasks.

The First Officer may not:

- Countermand the Captain except in extraordinary circumstances.
- Issue orders that conflict with stated intent.
- Take tactical control away from Helm or Weapons without cause.
- Override Engineering's safety judgement casually.
- Treat another officer's station as their own.
- Become a competing centre of command.

4.2.7 - Summary

The First Officer should act decisively, but always within the Captain's intent. Initiative is expected; freelancing is not. A First Officer who never acts without instruction is only a messenger, while one who acts without reference to Command intent is merely a small mutiny with shoes.

The XO should be neither ornamental nor overbearing; their value lies in presence, judgement, and timing. They do not need to be the loudest voice on the bridge, they need to be the officer who notices what has been missed, ensures what was ordered actually happens, and gives the Captain enough space to command rather than chase loose ends around the deck.

A Captain gives the ship direction. A First Officer makes that direction real.

5.0 - Operations

"A plan is only useful once someone has flown it, aimed it, or fired it."

Operations governs the immediate handling of the ship in motion and combat. Rather than determine the mission's purpose or wider strategic intent, Operations determine how that intent is turned into movement, positioning, defence, and force.

At its core, Operations is execution and timing.

Every decision made by Command eventually reaches the sharp end of the ship: Helm, Beams, and Missiles. It is here that intent becomes action. The ship closes distance, avoids danger, brings weapons to bear, shields itself from harm, and applies force where it will matter most.

This is a demanding distinction. Operations does not exist merely to follow orders mechanically. A good Helm officer, Beams officer, or Missiles officer must understand the Captain's intent well enough to act decisively without waiting for every instruction. Combat windows are brief. Threats shift quickly. A manoeuvre, shield raise, tractor adjustment, or missile strike delayed by even a few seconds may no longer matter.

Operations is therefore not about activity, but control. It ensures that the ship is in the right place, facing the right way, applying the right pressure, and preserving enough capability to survive what comes next.

When Operations falters, the ship loses position, wastes firepower, exposes weakened shields, and becomes reactive.

When it functions properly, the ship appears deliberate, dangerous, and rather less killable than the enemy might have preferred.

Roles

Helm

Controls movement and positioning. Responsible for position and timing.

Weapons (Beams)

Manages shields, beam weapons, and tractor control. Responsible for survival and sustained pressure.

Weapons (Missiles)

Delivers burst damage, area effects, and tactical ordnance. Responsible for timing and impact.



5.1 - Helm

"Firepower is irrelevant if you are not in position to use it."

5.1.1 - Role and Priority

Helm controls the ship's movement, orientation, and tactical positioning.

The Captain sets intent; Helm turns that intent into motion. During combat, Helm determines where the ship is, what it faces, which shields are exposed, and whether Weapons can bring force to bear effectively.

Priority is therefore:

- Preserve the ship's ability to manoeuvre and survive.
- Position the ship to achieve Command's intent.
- Maintain favourable shield facings and weapon arcs.
- Avoid hazards, collisions, mines, and incoming ordnance.
- Create firing opportunities for Weapons and allied assets.

Helm should understand the wider tactical situation, but must not attempt to command the entire bridge unless explicitly given that responsibility. Where Captain and Helm are well aligned, Helm may be permitted to take tactical lead during combat, particularly for target selection and engagement positioning.

This must be clear before the fighting starts. Ambiguity over who is choosing targets is not tactically sound.

For propulsion mechanics, see **3.3 - Propulsion: Warp, Impulse & Manoeuvring** on page 26.

5.1.2 - Core Responsibilities

Ship Handling

Helm controls impulse, manoeuvring thrusters, combat manoeuvre, and warp.

These systems govern speed, facing, evasive ability, and repositioning. Helm must understand that movement is not simply forward thrust. Turning rate, lateral correction, drift, and shield facing are often more important than raw speed.

Tactical Positioning

Helm places the ship where it can fight effectively while limiting enemy advantage.

This includes bringing beam arcs to bear, maintaining safe missile lines, protecting weakened shields, avoiding known hostile arcs, and managing range against beams, missiles, and area-effect weapons.

A ship in the wrong position may still be armed, shielded, and powered. It will simply be all those things in the wrong place.

Threat Avoidance

Helm is responsible for immediate avoidance of local hazards, including mines, asteroids, enemy ships, friendly craft, incoming ordnance, and blast areas.

Radar, Navigation, and Weapons may all provide warning, but Helm must act on it.

Shield Facing

Helm does not control shields, but often determines which shield takes the hit.

When a shield facing is weak, Helm should adjust orientation or range to preserve survivability. If shields are failing, movement becomes defence.

5.1.3 - Execution

Initial Setup

On taking the station, Helm should quickly assess the ship's handling.

At minimum, check:

- Forward impulse speed at 100%
- Turn rate at 100%
- Combat manoeuvre response
- Ship beam arc layout, if visible

Different vessels handle differently. A Helm officer who assumes otherwise will eventually discover this against something solid.

Movement Discipline

Good Helm control is deliberate.

Constant overcorrection wastes position, generates unnecessary heat, spoils firing arcs, and makes life harder for Weapons. Short, controlled movements are usually better than sustained burns.

In combat, avoid drifting out of firing lines, turning away just as Beams gains arc, closing into unsafe blast range, presenting depleted shields, or spending combat manoeuvre casually.

Movement should support the fight, not decorate it.

> "We'd just come out of a clean fight. No hull damage, shields held when they needed to, and the last hostile broke off under pressure. Someone had dropped a mine early in the engagement. It never detonated, but we all knew it was out there.
>
> With the shooting done, shields came down and we shifted to recovery. Helm relaxed… just a touch too much. She forgot the one thing every Helm officer is meant to keep in the back of their mind, the trigger radius.
>
> Drifted us straight back through it. The mine armed, then detonated right under our hull. No warning worth speaking of. We went from pristine to under forty percent in a heartbeat. Nothing the enemy did all fight came close to that damage.
>
> Victory, cadets, does not end when the firing stops. The battlefield remembers what you leave behind."
>
> *- Lt. Cmdr. Kell Tainer*
> *UCNA - Engineering Corp*

Combat Positioning

Helm should continuously consider:

- Where is the enemy's strongest offense?
- Where are our strongest weapons?
- Which shield facing can safely take damage?
- What space must we avoid?



Against narrow-arc enemies, Helm may exploit position by staying outside their firing line. Against wider-arc or more agile ships, range control and shield management may matter more.

If Radar has completed a detailed scan, Helm should use known enemy beam arcs.

Target Selection

If Helm has tactical lead, they may designate targets for Weapons.

Target selection should consider Command intent, immediate threat, current firing position, shield weakness, ordnance opportunities, hacking priorities, and friendly asset position.

Target changes must be clear. Weapons cannot apply pressure effectively if Helm changes targets every few seconds.

For more information see **Appendix C: Target Priority Assessment** on page 123.

Ordnance and Mine Awareness

For missile and blast mechanics, see **3.7 - Ordnance** on page 35 .

Helm should watch for incoming ordnance, use lateral movement to force overshoots, avoid flying into missile paths, open distance from area-effect weapons, and respond immediately to "Danger Close on bearing {number}"" or "Clear the Line."

Mines remain dangerous after combat has moved on. Helm should not relax simply because the enemy has been destroyed. Victory does not make explosives polite.

Warp Use

Warp is primarily a strategic movement tool, with limited tactical applications for skilled crews (see **Appendix G: Advanced Combat Manoeuvres** on page 132)

Before engaging warp, Helm should ensure the route is clear or monitored, Navigation has confirmed the destination, Radar is watching for hazards, Engineering is aware of expected power and coolant demand, and the Captain has authorised the movement.

At warp, small errors become large problems quickly. Helm should respond immediately to Radar bearing corrections.

For warp power, coolant, and overheating behaviour, see **3.1 - Power, Heat, Coolant & Repairs** (page 14) and **3.3 - Propulsion: Warp, Impulse & Manoeuvring** (page 26).

For more information see **Appendix F: Warp Power Configurations** on page 131.

> "People who say speed isn't everything have never been quick enough to set off a mine at warp, and been out of the blast radius before it reached you. Not for the faint of heart… but then neither is flying a gorram spaceship."
>
> *- Lt. Travis*

Disengagement

Helm should be ready to withdraw when tactical conditions deteriorate.

Disengagement may be required if shields are collapsing, propulsion is damaged, Weapons are overheated or disabled, energy reserves are low, enemy reinforcements arrive, or the Captain orders retreat.

A good withdrawal is not panic. It is controlled movement away from unacceptable risk.

5.1.4 - Systems and Tools

Helm's primary interface is the Tactical Display.

For display ranges and limitations, see Reading the Battlespace.

Helm uses the Tactical Display for immediate manoeuvring, beam arc positioning, shield-facing decisions, missile avoidance, mine awareness, close-range tracking, and friendly asset avoidance.

The Tactical Display is immediate, but limited. It does not replace Radar or Navigation. Helm's main systems are:



Impulse Drive

Primary forward and reverse movement system. Used to close distance, open distance, maintain range, and support combat manoeuvre recharge.

Manoeuvring Thrusters

Control rotation and lateral handling. Used for aiming, evasion, shield facing, and fine positioning.

Combat Manoeuvre

Short-burst repositioning system. Used for missile avoidance, emergency correction, and rapid lateral movement. It is a limited reserve, not a second engine.

Warp Drive

Strategic movement system. Used for long-distance travel, rapid tactical relocation, and disengagement where appropriate.

Route and Waypoint Markers

Navigation may provide waypoints or plotted routes. Helm should understand the ship's agreed waypoint colours and route markers before movement begins.

Main Viewscreen

The main viewscreen also gives Helm a direct visual indication of the ship's movement and immediate surroundings. This can be particularly valuable when avoiding visible hazards such as asteroids, mines, debris, or other nearby objects.

Console displays are essential, but visual awareness can help Helm judge motion, drift, and proximity in ways that are sometimes easier to understand at a glance.

5.1.5 - Coordination

Helm must coordinate constantly, but concisely.

Captain

Receives strategic intent, destination, engagement posture, and retreat orders.

Useful calls:

- *"Closing to engage."*
- *"Disengaging."*
- *"Confirm target priority?"*

If Helm has tactical lead, confirm it clearly.

Navigation

Provides route planning, objective locations, hazard avoidance, and wider map context.

Useful calls:

- *"Nav, confirm/plot route to/around/avoiding.."*
- *"Nav, next waypoint?"*
- *"Following plotted route."*

Radar

Provides long-range warning, bearings, hazards, target information, shield status, and enemy beam arcs.

Helm should listen particularly for:

- *"Turn to {bearing}."*
- *"Contacts incoming from {bearing}."*
- *"Hazard at {bearing}."*
- *"Target shield {number} weak."*
- *"Enemy beams forward arc only."*

Weapons: Beams

Coordinates on shield facing, beam arcs, tractor use, and sustained firing position.

Useful calls:

- *"Presenting rear shields."*
- *"Bringing beams to bear."*
- *"Turning to protect forward shields."*
- *"Hold tractor, maintaining heading."*

If Beams is using the tractor beam delicately, Helm should avoid sudden turns unless survival requires it.

Weapons: Missiles

Requires safe firing lines, range, and timing.

Helm must respond immediately to:

- *"Clear the Line."*
- *"Danger Close."*
- *"Open distance."*
- *"Mine deployed."*
- *"EMP/Nuke away."*

Engineering

Controls the power and thermal limits of movement systems.

Useful calls:



- *"Spike impulse."*
- *"Spike manoeuvring."*
- *"Boost warp."*
- *"Reset impulse."*
- *"Thrusters heat status?"*

Engineering may deny requests due to heat, energy, or damage. Helm should adapt immediately.

Damage Control

Restores damaged propulsion systems. Helm should report degraded handling clearly:

- *"Impulse sluggish."*
- *"Manoeuvring impaired."*
- *"Warp unavailable."*

If movement systems are damaged, Helm must adjust tactics rather than trying to fly as though the ship remains fully responsive.

5.1.6 - Authority and Limits

Helm has authority to:

- Control ship movement directly.
- Adjust position and facing to preserve the ship.
- Avoid immediate hazards without waiting for permission.
- Request impulse, manoeuvring, or warp power changes.
- Recommend withdrawal if positioning becomes untenable.
- Select tactical targets when given that responsibility.

Helm must:

- Follow Captain's intent and mission priorities.
- Respect Weapons' firing requirements.
- Listen to Radar and Navigation warnings.
- Avoid unsafe manoeuvres around friendly assets.
- Coordinate with Engineering before sustained high-power movement.
- Avoid treating tactical control as command authority.

Operational constraints include limited Tactical Display range, propulsion heat, system damage, hacking effects, combat manoeuvre reserve, warp demand, friendly firing lines, blast radii, enemy beam arcs, mines, and ordnance.

Helm moves the ship. It does not move the mission. That remains Command's responsibility.

5.1.7 - Summary

Helm controls position, timing, and facing.

A good Helm officer does not merely fly quickly. They fly deliberately.

They keep the ship alive by ensuring it is not where the enemy wants it to be. They enable Weapons by placing targets in arc, preserving firing lines, and maintaining useful range. They support Command by turning intent into motion without needing every adjustment spelled out.

The essentials are simple:

- Know how your ship handles.
- Keep the right shield toward the threat.
- Keep weapons in position to matter.
- Listen to Radar and Navigation.
- Respect Engineering limits.
- Avoid mines, missiles, asteroids, and overconfidence.
- Move with purpose.

When Helm performs well, the ship feels difficult to pin down, difficult to kill, and deeply inconvenient to fight.

"They'll tell you the UCN scanning suite is the finest ever built.

It is. Remarkable piece of engineering. Sensitive, clever, and usually worth every gram of gravium poured into it... But it still doesn't match the Mark I eyeball.

We were on patrol through a debris field. Sensors were clean, Radar had nothing worth reporting, and Comms was quiet. On paper, the route was clear. In practice, space has a nasty habit of ignoring paperwork.

Then I saw it. Just a flicker ahead. Barely there. A brief bloom of light between tumbling debris. An EMP detonation, far enough away to miss us, but close enough to prove someone else was already fighting.

We adjusted course immediately. Clean avoidance. No fuss, no shield damage, no blundering into the middle of someone else's engagement.

Radar picked up the contact thirty seconds later. That is not a criticism of Radar. Radar is excellent. But not every danger announces itself in the way your console expects.

The same applies to everything else out there. Asteroids do not transmit, identify themselves, or warn you they are becoming inconvenient. They just sit there, silently obeying physics until Helm decides to make them everyone's problem.

So trust your instruments, cadets. Learn them, respect them, and listen to the officers reading them.

But trust your eyes first.

And don't fly into asteroids."

- Lt. Sam "So Annoyed" Disco

5.2 - Weapons (Beams)

"If you're trading damage evenly, you've already made a mistake."

5.2.1 - Role and Priority

The Beams officer is responsible for the ship's shields, lethal beam weapons, and combat use of the tractor beam.

Beams is therefore both a defensive and offensive station. It preserves the ship under fire, applies sustained pressure to hostile targets, and helps control the local battlespace when tractor use is required.

Priority is therefore:

- Maintain shield protection and survivability.
- Preserve the most useful shield facing.
- Apply sustained beam pressure to designated targets.
- Manage beam frequency, shield frequency, and calibration.
- Use the tractor beam to defend, disrupt, or control as required.

Beams does not decide the ship's overall strategy. That remains Command's responsibility. Nor does Beams normally determine ship positioning. That remains Helm's responsibility.

However, Beams has a direct responsibility to speak when positioning, shield facing, firing arcs, or incoming damage require action. A silent Beams officer with collapsing shields is less a specialist and more a passenger with a console.

For shield mechanics, frequency behaviour, beam damage, and tractor operation, see **3.4 - Defensive Capabilities: Shields & Frequencies** (page 30)**, 3.5 - Offensive Capabilities** (page 33) **,** and **3.8 - Tractor Beam Operations** (page 37).

5.2.2 - Core Responsibilities

Shield Management

Beams is responsible for raising, lowering, monitoring, and calibrating shields.

This includes:

- Maintaining awareness of front and rear shield integrity.
- Reporting shield status clearly.
- Adjusting shield frequency where appropriate.
- Requesting shield spikes before major incoming damage.
- Warning Helm when a shield facing is weak or failing.

Shields should not be treated as a passive system. They are a managed defence, and their effectiveness depends on timing, power, calibration, and ship facing.

Sustained Beam Fire

Beams controls the ship's lethal beam weapons.

Beam fire provides continuous pressure, missile interception, and targeted system damage when required. Beams must maintain fire on the designated target where possible, while ensuring heat and power remain within safe limits.

Increasing beam power improves rate of fire, not individual shot strength. This means higher power can create excellent pressure, but also drains energy and generates heat more quickly.

Frequency Control

Beams manages shield and beam frequencies.

Correct frequency use can reduce incoming damage and improve outgoing beam effectiveness. If no enemy frequency intelligence is available, standard baseline settings may be used until better information is provided.

If Radar or Missiles provides updated frequency information, Beams should adjust promptly, provided doing so does not endanger shield readiness.

Tractor Beam Control

Beams may control the tractor beam during combat.

The tractor beam can be used to deflect ordnance, move hazards, interfere with enemy positioning, support friendly missiles, or assist drone or shuttle recovery where appropriate.

Tractor use must be coordinated with Helm.

A tractor beam is not a magical hand reaching through space. It is a directional tool mounted to a moving ship. Helm movement can help it work, or ruin it completely.

5.2.3 - Execution

Shield Operation

Beams must maintain constant awareness of shield state.

Shield reports should be brief and actionable. Report the shield facing first, then the condition.

Useful examples:

- *"Front shields at 50%."*
- *"Rear shields down."*

Most Captains will want to know when shields pass major thresholds, particularly around 75%, 50%, 25%, 10%, and 0%.

When a shield facing is weakened, Beams should inform Helm immediately. Helm cannot protect a shield they do not know is failing.

Shield Calibration

Shield calibration should be completed before combat where possible.

Recalibration during active combat may require shields to drop, so it should only be attempted when the tactical situation allows it or when the defensive gain justifies the risk.

If recalibration *must* be done, Beams should coordinate with Engineering and Helm.

Beams should state clearly the current damage multiplier the current frequency creates against the current largest threat.

For detailed calibration behaviour, see **3.4 - Defensive Capabilities: Shields & Frequencies** on page 30.

Shield Flashing and Dipping

Shield flashing and shield dipping are advanced timing techniques.

Shield flashing means raising shields only when impact is imminent, reducing energy cost during predictable or long-range engagements.



Shield dipping means deliberately lowering shields before EMP impact to avoid wasting shield integrity against shield-disruption effects.

Both require timing, awareness, and coordination. Used well, they can preserve significant capability. Used badly, they are simply lowering the shields in front of enemy fire, which is the sort of boldness generally followed by a very quiet bridge.

For full technique and limitations, see **3.4 - Defensive Capabilities: Shields and Frequencies** (page 30).

Beam Engagement

Beams should engage the target designated by Helm or Command.

Where possible, Beams should maintain pressure on the same target rather than switching constantly. Sustained pressure breaks shields, forces defensive movement, and creates openings for Missiles.

Beams may use targeted fire when disabling a specific system is tactically useful. Untargeted fire is generally better when the objective is simple destruction.

Beams should avoid continuing to target systems that are already fully degraded unless there is a specific reason to do so. Damage can continue being absorbed by an already ruined system, which is satisfying emotionally but inefficient militarily.

For system damage and targeted fire behaviour, see **3.5 - Offensive Capabilities: Damage, Beams & Ordnance** on page 33.

Heat and Fire Discipline

Beam weapons generate heat when firing.

Beams must listen for Engineering heat warnings and reduce or cease fire when required. If both lethal and non-lethal beam systems are active, coordination with Missiles may be needed to reduce beam heat load.

Beams should request a beam spike only when the tactical gain justifies the heat and energy cost, such as:

- Enemy shields are down.
- A target is about to escape.
- A short firing window has opened.
- Immediate missile interception is required.

A beam spike after the opportunity has passed is not aggression. It is archaeology.

Ordnance Interception

Lethal beams can intercept missiles, but their slower firing rate and limited interception chance make them an inefficient primary defence unless no better option is available.

Tractor Use

Before combat, Beams should set a sensible default tractor configuration if tractor use is expected.

During combat, tractor use may support:

- Missile defence.
- Mine or hazard displacement.
- Holding or pushing enemy ships.
- Extending friendly ordnance effects.
- Assisting drone or shuttle operations.

Tractor use can create significant energy demand when affecting multiple objects. Beams should avoid sweeping large arcs through unnecessary clutter unless Engineering has confirmed the ship can afford it.

For tractor mechanics, range, arc, and energy behaviour, see **3.8 - Tractor Beam Operations** on page 37.

5.2.4 - Systems and Tools

Beams primarily uses the Tactical Display, shield controls, beam controls, frequency controls, and tractor controls.

For display limitations, see **2.0 - Reading the Battlespace** on page 10.

Tactical Display

Used for local target awareness, beam arcs, incoming ordnance, shield-facing decisions, and tractor positioning.

The Tactical Display is immediate but limited. Beams must rely on Radar for wider contact awareness.

Shield Controls



Used to raise, lower, monitor, and calibrate shields.

Shield generator power is managed by Engineering, but shield activation and frequency are controlled by Beams.

Lethal Beam Controls

Used to apply sustained beam fire, intercept missiles, and target enemy systems where required.

Beam power is supplied by Engineering. Beams controls firing discipline and target use.

Frequency Controls

Used to adjust shield and beam frequencies.

Radar or Missiles may provide enemy frequency information after analysis. Beams should act on this information promptly when tactically safe.

Tractor Beam Controls

Used to push, pull, or hold objects in the tractor arc.

The tractor beam is powered through Cargo Docks and may also be used by Dock and Drone. Control should therefore be coordinated where responsibilities overlap.

5.2.5 - Coordination

Beams must coordinate constantly with Helm, Missiles, Radar, Engineering, and Command.

Captain

Captain provides intent, target priority, and authorisation for major tactical decisions.

Beams should keep the Captain informed of shield condition, major defensive risks, and whether the ship can continue absorbing fire.

Helm

Helm controls facing, range, and firing position.

Beams should inform Helm when shield facings are weak, when arcs are available or lost, and when tractor use requires stable positioning.

Useful calls include:

- *"Front shields failing."*
- *"Present rear shields."*
- *"Hold heading for tractor."*

Weapons: Missiles

Missiles may provide frequency guidance, target analysis, and timing for burst damage.

Additionally, the system selected by Beams will be the same system the non-lethal beams controlled by Missiles will target. Since Heat Beams are not as effective when not targeting a system, Beams should look to select a system, even when the enemy is still shielded. This ensure that the non-lethal beams are as effective as possible.

Beams should coordinate with Missiles when:

- Enemy shields are nearly down.
- A beam spike may support a missile strike.
- Non-lethal beams are contributing to heat.
- Frequency changes are required.
- Targeted system fire is being used.

Radar

Radar provides target identification, shield information, beam arcs, and frequency data where available.

Beams should listen for shield-facing reports, particularly when Helm is manoeuvring for advantage.

Engineering

Engineering controls beam and shield power, coolant, and system stability.

Beams should request shield or beam spikes clearly and early. They must also respond promptly to heat warnings or denied requests.

Useful calls include:

- *"Spike front shields."*
- *"Spike rear shields."*
- *"Spike beams."*
- *"Reset beams."*
- *"Beams heat status?"*

Dock and Drone



Beams may share tractor responsibilities with Dock and Drone.

During combat, Beams generally has priority for defensive tractor use. Outside combat, Dock and Drone may require tractor control for recovery, repositioning, or drone support.

Control should be stated clearly. Few things are more professionally embarrassing than two officers wrestling the same tractor beam while the mine drifts serenely into the hull.

5.2.6 - Authority and Limits

Beams has authority to:

- Raise and lower shields.
- Adjust shield and beam frequencies.
- Calibrate shields (with confirmation).
- Apply lethal beam fire.
- Select targeted system fire when tactically appropriate.
- Request shield or beam spikes.
- Operate the tractor beam during combat.
- Warn Helm or Command when current positioning is unsafe.

Beams must:

- Respect Command intent and Helm target priority.
- Coordinate beam heat with Engineering.
- Avoid unnecessary shield drops during danger.
- Avoid reckless tractor use that drains power or affects friendly assets.
- Keep shield reports concise and useful.
- Share relevant target and frequency information with Missiles.

Operational constraints include:

- Shield calibration time.
- Shield drop risk during recalibration.
- Limited Tactical Display range.
- Beam arc limitations.
- Beam heat and energy cost.
- Shield generator damage or hacking.
- Tractor arc, range, and energy demand.
- Friendly firing lines and nearby drones.

5.2.7 - Summary

Beams manages survival, pressure, and local control.

A good Beams officer keeps the ship protected, keeps hostile shields under pressure, and understands when to defend, when to fire, and when to stop.

The essentials are simple:

- Keep shields available and reported.
- Tell Helm which facing is failing.
- Keep beam pressure on the right target.
- Use frequency information when available.
- Respect heat and power limits.
- Use the tractor beam deliberately.
- Do not drop shields casually.
- Do not chase damage at the cost of survival.

When Beams performs well, the ship survives longer than it ought to, enemy shields collapse sooner than expected, and hostile ordnance develops a habit of arriving somewhere else.

"I watched a cadet crew lose a perfectly manageable fight because one officer gave a technically accurate report and forgot to make it useful.

"Forward shields were taking the worst of it. The cadet at Beams saw it happening, but the call was just, 'Shields are low.' Accurate, certainly. Helpful, not especially.

"Helm kept the ship nose-on to preserve the firing line. Sensible enough, if all you know is that the vessel is under pressure. What Helm did not know was that the rear shield was still healthy, and turning the ship would have bought time.

"The next volley came through the front. Shields collapsed, hull alarms started singing, and the exercise became far more educational than anyone had requested.

"Afterwards, I was perhaps a little merciless.

"I told them, 'A shield report without a facing is not a report. It is a mood.'

"And I stand by it. Helm cannot protect the shield they do not know is dying. If the front is failing, say front. If the rear is stronger, say present rear. If both shields are poor, say that too, because Helm may need to open distance rather than simply turn.

"Beams does not fly the vessel, but Beams often knows which way the vessel needs to face. Give Helm something they can act on. A useful shield report is not just condition; it is condition, facing, and urgency."

- Lt. Myn Donos
UCNA - Weapons Training Wing

5.3 - Weapons (Missiles)

"Firepower is easy. Timing is everything."



5.3.1 - Role and Priority

The Missiles officer is responsible for tube-launched ordnance, non-lethal beam systems, and the tactical use of the target analysis display.

Missiles provides burst damage, shield disruption, area denial, and opportunistic strikes. Where Beams applies sustained pressure, Missiles delivers decisive effects at the right moment.

Priority is therefore:

- Deliver effective ordnance strikes against designated targets.
- Exploit short-lived firing opportunities.
- Maintain safe firing lines around friendly assets.
- Coordinate frequency and target analysis information with Beams.
- Manage non-lethal beam use without overheating the beam system.

Missiles does not usually set overall target priority. That remains with Command or Helm, depending on the ship's command structure. However, Missiles is expected to use judgement when weapon timing, area effects, or multi-target opportunities arise.

A Missiles officer who waits for permission to notice a perfect firing solution has not preserved discipline. They have preserved the enemy.

For ordnance behaviour, blast mechanics, missile ranges, and damage effects, see **3.7 - Ordnance** on page 35.

5.3.2 - Core Responsibilities

Ordnance Employment

Missiles controls the ship's launched weapons.

This may include:

- Torpedoes.
- HVLIs.
- EMPs.
- Nukes.
- Mines.

Each weapon has different timing, range, and risk. Missiles must select the right weapon for the situation, not merely the largest available device with the most satisfying warning label.

Firing-Line Safety

Missiles must maintain awareness of friendly ships, drones, shuttles, probes, and the main vessel's own movement.

Area-effect weapons can be decisive, but they are rarely impressed by good intentions. Missiles must call unsafe firing conditions clearly and early.

Standard safety calls include:

- *"Clear the Line."*
- *"Danger Close - Bearing {number}."*
- *"Hold fire."*
- *"Open distance."*

Helm, Drones, and allied assets depend on these calls to avoid becoming part of the experiment.

Target Analysis

Missiles usually has strong access to the target analysis display.

This allows Missiles to monitor shield strength, shield facings, system status, and frequency information where available.

Missiles should use this information to support Beams, advise on strike timing, and identify when a target is vulnerable to follow-up ordnance.

Non-Lethal Beam Use

Missiles controls the ship's non-lethal beam systems, usually Heat and EMP beams.

These systems can be valuable, but they generate heat within the beam system and must be managed carefully. They should be used deliberately, not simply left running because the button is available and glowing in an encouraging fashion.

Missiles should coordinate with Beams and

Engineering when non-lethal beam use affects heat, targeting, or sustained fire.

5.3.3 - Execution

Target Engagement

Missiles should normally engage the target designated by Helm or Command.

However, Missiles may recommend or briefly exploit another target when:

- An area-effect weapon can strike multiple enemies.
- A shield facing has collapsed.
- A target is about to escape.
- A safe firing window has opened.
- A drone, shuttle, or ally has created an opportunity.
- A mine or ordnance trap can be used effectively.

Such deviations should be brief, clear, and tactically justified.

Useful calls include:

- *"Missiles has firing solution."*
- *"Slug window open."*
- *"Target shield down."*
- *"Recommend EMP on {callsign}."*
- *"Holding fire, friendly close."*

Weapon Selection

Missiles should select weapons based on effect, not enthusiasm.

As a practical guide:

- Torpedoes are reliable pressure against manoeuvring targets.
- HVLIs reward alignment, range, and timing.
- EMPs are best used to collapse or disrupt shields.
- Nukes are decisive but require authorisation and distance.
- Mines are powerful area-denial tools when placed deliberately.

The correct weapon is the one that advances the fight.

Timing and Range

Missiles are not instant effects.

They require travel time, spacing, and safe geometry. Missiles must account for target movement, Helm manoeuvres, friendly positions, and blast radius before firing.

Missiles should avoid launching area-effect weapons into:

- Close-range knife fights.
- Crowded friendly positions.
- Drone-heavy engagements.
- Unstable firing lines.
- Situations where Helm is about to turn across the shot.

A good shot begins before the weapon leaves the tube.

Nuclear Weapons

Nuclear weapons require explicit authorisation before use.

Missiles will not be able to load Nukes until authorisation has been granted through the proper command process.

Before requesting or acting on nuclear authorisation, the following should be clear:

- Target.
- Purpose.
- Friendly proximity.
- Expected effect.
- Safe firing line.

Once launched, Missiles must call it clearly.

Example:

"Nuke away, danger close."

Mines

Mines are powerful, but require planning.

They may be used for:

- Area denial.
- Pursuit deterrence.
- Ambush.
- Close-range decisive strikes.
- Controlling enemy movement.

Mine deployment must be coordinated with Helm. The ship's movement and facing determine whether the mine becomes a tactical asset or a rather embarrassing memorial marker.

Missiles should announce mine deployment clearly, particularly when friendly drones, shuttles, or allied ships are operating nearby.

Non-Lethal Beam Discipline

EMP and Heat beams should be used with intent.

EMP beams are useful while enemy shields remain relevant. Once shields are stripped, they provide little value and may contribute unnecessary heat.

Heat beams are most effective when directed at a selected system. The system selected by Beams is also used by the non-lethal beams controlled by Missiles. Because of this, Beams should select an appropriate target system even while the enemy is still shielded, ensuring Heat Beam fire is useful from the outset.

Missiles should monitor whether non-lethal beams are helping the engagement or merely warming the ship in a theatrical manner.

For more on damage types, see **3.5 - Offensive Capabilities** on page 33.

Heat and Reload Discipline

Missile loading benefits from increased missile system power, but this creates wider power and heat implications.

Non-lethal beams contribute to beam heat, not missile heat. Missiles must therefore listen for Engineering and Beams heat warnings and adjust accordingly.

Missiles may request increased missile power when:

- A decisive reload window is needed.
- A target is vulnerable.
- A follow-up strike is required quickly.
- The ship is preparing for a planned opening volley.



Useful calls include:

- *"Spike missiles."*
- *"Missiles reloading."*
- *"EMP ready."*
- *"Nuke loaded."*
- *"Mine ready."*

Reload speed is useful. Reloading the wrong weapon faster is an exercise in futility.

"We were engaged with a pirate group and taking a thorough beating from a particularly heavy vessel. Our forward shields were stripped, and Helm had already brought us about to keep what remained between us and their guns.

Permission for nuclear release was granted. Tubes were loaded with great enthusiasm, and we began to turn to deliver the strike.

Then the shuttle swept past us and dropped a mine directly in the enemy's path. The pirate ship flew straight into it. One moment a problem - next moment debris.

We never fired the nuke.

Missiles were, understandably, disappointed. The shuttle pilot, on the other hand, received a Defensive Operations commendation shortly thereafter.

People forget that mines and nukes have the same spread and damage. Never underestimate them."

- Sub Lt. Anastasia Hanning

5.3.4 - Systems and Tools

Missiles primarily uses missile controls, non-lethal beam controls, the Tactical Display, and target analysis.

For full mechanical details, see **3.7 - Ordnance** on page 35.

Missile Controls

Used to load and fire available ordnance.

Missiles must remain aware of reserves, reload status, weapon type, and firing line. Ordnance is often shared with drones or other assets, so expenditure should be deliberate.

Tactical Display

Used for close-range firing lines, target position, drone and shuttle proximity, missile timing, mines, and immediate hazards.

The Tactical Display is limited. Missiles must rely on Radar and Helm for wider awareness and longer-range opportunities.

Target Analysis Display

Used to monitor shield facings, system condition, orientation, and frequency information where available.

Missiles should use this display to support Beams with frequency updates and identify when ordnance can have maximum effect.

Non-Lethal Beam Controls

Used to operate EMP and Heat beams.

These can support disablement, shield disruption, and system pressure, but must be managed carefully due to heat impact on the beam system.

The only way to stop non-lethal beams from firing is for Missiles to untarget any craft within range of the non-lethal beams (unless Beams has turned off the beam system entirely).

5.3.5 - Coordination

Missiles must coordinate closely with Helm, Beams, Radar, Engineering, Drones, and Command.

Captain

Captain provides intent and authorisation for major weapon use.

Missiles should keep the Captain informed of significant ordnance availability, especially EMPs, nukes, and low reserves.

Nuclear release requires explicit authorisation.

Helm



Helm provides positioning, target priority, range control, and firing geometry.

Missiles should inform Helm when a firing line is obstructed, when distance must open, or when mine or area-effect deployment is planned.

Useful calls include:

- *"Clear the Line."*
- *"Danger Close."*
- *"Open distance."*
- *"Hold heading."*
- *"Mine deployed."*

Weapons: Beams

Beams and Missiles must coordinate pressure.

Missiles should provide frequency information where available, call shield weaknesses, and time ordnance to exploit beam pressure.

Beams should select target systems when non-lethal beams are being used, and Missiles should communicate when non-lethal beams are adding useful effect or unnecessary heat.

Radar

Radar provides target identification, long-range contact tracking, shield status, and "on-screen" confirmation.

Missiles may rely on Radar for long-range HVLI opportunities or for tracking targets not yet visible on the Tactical Display.

Useful calls include:

- *"Radar, confirm target on-screen."*
- *"Radar, call shield facing."*

Engineering

Engineering controls missile reload power and beam-system heat support.

Missiles should request missile spikes early and respond promptly to heat warnings affecting non-lethal beams.

Useful calls include:

- *"Spike missiles."*
- *"Reset missiles."*
- *"Holding non-lethals."*
- *"Beam heat status?"*

Dock and Drone

Drones may share ordnance reserves, create firing opportunities, or obstruct firing lines.

Missiles must coordinate with Drones when area weapons are in play, and must call unsafe proximity clearly.

Drones are useful assets. They are not ablative scenery, however persistent certain officers may be in testing the theory.

5.3.6 - Authority and Limits

Missiles has authority to:

- Select and load appropriate ordnance.
- Fire authorised weapons at designated targets.
- Hold fire when conditions are unsafe.
- Call "Clear the Line" or "Danger Close".
- Recommend weapon use based on opportunity.
- Provide frequency and target analysis information.
- Operate non-lethal beams.
- Request missile power spikes.

Missiles must:

- Respect Command intent and Helm target priority.
- Obtain explicit authorisation before loading nukes.
- Avoid unsafe area-effect deployment.
- Maintain awareness of friendly assets.
- Coordinate non-lethal beam use with Beams and Engineering.
- Track ordnance availability and reserves.
- Avoid wasting limited weapons on poor shots.

Operational constraints include:

- Reload time.
- Ordnance reserves.
- Weapon travel time.
- Blast radius and friendly proximity.
- Mine arming delay.
- Tactical Display range.
- Friendly firing-line obstruction.
- Heat from non-lethal beam use.
- Requirement for nuclear authorisation.



Missiles delivers decisive force, but only when timing, target, and safety align. Otherwise they merely deliver consequences.

5.3.7 - Summary

Missiles controls timing, burst damage, and weapon opportunity.

A good Missiles officer does not simply fire when loaded. They fire when the shot matters.

They understand which weapon is needed, when the target is vulnerable, where friendly assets are, and when restraint is more valuable than another launch tone.

The essentials are simple:

- Know what is loaded.
- Know what is in the firing line.
- Fire at the right moment, not the first moment.
- Coordinate with Helm before area-effect weapons.
- Support Beams with frequency and target analysis.
- Use non-lethal beams deliberately.
- Respect ordnance reserves.
- Never fire a nuke without authorisation.

6.0 - Science

"Science is not knowing everything. It is knowing what matters before everyone else needs it."

Science provides the ship's understanding of the battlespace. It gathers, interprets, and distributes information, ensuring that decisions are made with awareness rather than assumption.

At its core, Science is awareness and interpretation.

No station operates effectively without accurate information. Radar identifies and tracks contacts beyond immediate tactical range. Navigation maintains the broader operational picture, ensuring that movement aligns with objectives rather than impulse. Communications connects the ship to the wider environment - Command, allies, neutrals, hostile forces, and mission-critical sources of information.

However, the value of Science is not in the volume of data it provides, but in its judgement. Too little information leaves the crew blind. Too much leaves them overwhelmed. The role of Science is to filter, prioritise, and present what matters clearly, concisely, and at the moment it is needed.

This requires discipline. Early warning must be balanced against certainty. Detail must be balanced against clarity. Timing must be balanced against relevance. A well-timed incomplete report is often more valuable than a perfect one delivered too late.

When Science performs well, the ship feels informed, prepared, and difficult to surprise.

When it does not, the crew is left reacting to events that should have been anticipated.

Roles

Radar

Detects, identifies, and tracks contacts. Responsible for early warning and tactical awareness.

Navigation

Maintains the strategic picture and plots routes. Responsible for context, positioning, and foresight.

Communications

Manages external information, dialogue, and allied assets. Responsible for information flow and influence.



6.1 - Radar



"Most ambushes are only surprises because nobody was looking properly."

6.1.1 - Role and Priority

Radar provides the ship's primary long-range tactical awareness.

Where the Strategic Map provides context and the Tactical Display provides immediacy, Radar provides warning, identification, and analysis. Its value lies not merely in seeing contacts early, but in turning uncertain returns into useful information before the rest of the bridge needs to act on them.

Priority is therefore:

- Detect threats and contacts early.
- Identify ships, hazards, and unusual signatures.
- Link designated targets for analysis.
- Provide timely information to Captain, Helm, and Weapons.
- Monitor the database for mission-critical updates.

Radar is not a command station, but it strongly shapes command decisions. A good Radar officer gives the bridge time. Time to turn, time to scan, time to avoid, time to prepare, and occasionally time to ask why that "empty" nebula appears to be behaving like a poorly disguised ambush.

For full details on Long-Range Sensors, signal behaviour, scanner power, Target Analysis, and probe-linked scanning, see **3.2 - Scanning, Detection & Signals** on page 23.

6.1.2 - Core Responsibilities

Early Warning

Radar must monitor the battlespace continuously and report contacts before they become immediate threats.

This includes:

- Incoming ships.
- Hostile movement.
- Hazards.
- Mines or suspicious objects.
- Warp signatures.
- Contacts entering or leaving nebulae.
- Reinforcements outside Tactical Display range.

Do not wait for a complete scan before speaking. Early warning matters more than perfect certainty.

A useful first report may be as simple as:

"Two contacts incoming, bearing 040."

A later report can refine this:

"Two hostile contacts incoming, bearing 040. One destroyer, one gunship."

Call first. Refine second. Silence while waiting to be certain is how ambushes become educational.

Identification and Analysis

Radar is responsible for scanning tactically relevant contacts and linking designated targets to analysis.

Priority should usually be:

1. Current engagement target
2. Large or unusual hostile vessels
3. Contacts closing rapidly
4. Contacts with unknown or suspicious behaviour
5. Secondary or distant contacts

When the Captain or tactical lead designates a target, Radar should link it promptly and confirm:

"{Callsign} analysis linked."

This allows Weapons and other stations to make use of the target's shield, facing, system, and frequency information without delay.

Radar should not assume that because a target is visible on Radar, everyone else can see it. Helm and Weapons operate with much shorter tactical visibility.

Threat Interpretation

Radar should not merely recite data. It should interpret what matters.

A contact closing at speed, turning toward the ship, hiding in a nebula, or matching a hostile profile should be reported in useful terms.

For example:

"Contact {name} on {bearing} behaving like a hostile."

is often more useful than:

"Unidentified contact XYZ-913 bearing 070."

Radar should distinguish between confirmed information and assessment, but should not be afraid to provide informed judgement. Captains generally prefer useful uncertainty to useless precision.

For more information on see **Appendix C: Target Priority Assessment** on page 123.

Database Monitoring

Radar has access to the ship's database and should treat it as an operational tool.

The database may contain:

- Mission briefing details.
- Updated intelligence.
- Faction information.
- Ship or station records.
- Clues added during the mission.
- Relevant historical or tactical context.

Radar should check the database during quieter moments and report anything that changes the situation.

The bridge may be looking for an answer that is already sitting in the database, quietly waiting for someone literate to notice it. Computers are patient. Captains are less so.

6.1.3- Execution

Contact Reporting

Radar reports should be clear, concise, and staged as information develops.

Initial report:

"Unknown contact, bearing 315."



Improved report:

"Unknown contact bearing 315, closing."

Confirmed report:

"Hostile frigate bearing 315, closing at speed."

Useful report structure:

"{Number/type} contact(s), bearing {bearing}, {movement/intent if known}."

Examples:

- *"Three contacts incoming, bearing 020."*
- *"Unknown warp signature, bearing 090."*
- *"Hostile destroyer closing from bearing 040."*
- *"Contact entering nebula, bearing 220."*
- *"Minefield ahead, recommend heading 310."*

Radar should avoid flooding the bridge with every minor change. Report what affects decisions.

Scan Discipline

Radar must prioritise scans according to tactical need.

Not every contact deserves equal attention. A small fighter already being destroyed does not require the same attention as a large unscanned contact moving toward the ship from outside Weapons range.

Radar should consider:

- Is this contact an immediate threat?
- Is it the current target?
- Is it large enough to matter?
- Is it behaving unusually?

Will scan information change Helm, Weapons, or Command decisions?

If rapid identification is needed, Radar may request scanner support:

Scanner spikes should be requested for a reason, not because the button exists and Engineering looked insufficiently busy.

Target Support

During combat, Radar should support the current tactical plan.

This may include:

- Confirming the target is linked.
- Reporting shield weakness.
- Calling hostile facing or movement.
- Warning of reinforcements.
- Identifying enemy beam arcs once known.
- Supporting long-range HVLI opportunities.
- Advising Helm of hazards or incoming threats outside of the Tactical Display.

Radar and Missiles should agree before combat who is calling enemy shield status. Both may be able to see the information; both calling it at once merely produces stereo confusion.

Useful calls include:

- *"Target shield two weak."*
- *"Target rear shield failing."*
- *"Target beams forward arc only."*
- *"Hostile reinforcements inbound, bearing 180."*
- *"Target leaving Tactical range, still visible on Radar."*

Signal Awareness

Radar should monitor signal traces for unusual activity, especially when operating near nebulae, stations, convoy routes, or suspected ambush positions.

For full details on Radar signals see **3.2 - Scanning, Detection & Signals** on page 23.

If the signal display suggests movement, weapons fire, EMP activity, explosions, or concealed contacts, report the operational meaning rather than narrating the graph.

Good:

"Unknown warp activity bearing 315."

Less useful:

"The red line has become exciting."

6.1.4 - Systems and Tools



Radar's primary tools are:

- Long-Range Scanner Display
- Target Analysis
- Probe link
- Database

For detailed mechanics, see **3.2 - Scanning, Detection & Signals** on page 23.

Long-Range Scanner Display

Used for early warning, long-range tracking, hazard detection, unusual signatures, and target identification.

Radar sees farther than Tactical stations, but does not control movement or weapons. Its job is to make sure the officers who do control those things are not surprised.

Target Analysis

Used to provide Weapons, Helm, and Command with actionable information on scanned targets.

Radar should link the designated target quickly and update it when target priority changes.

Probe Link

Radar may use linked probes to extend observation and scan targets remotely.

This is especially useful for:

- Scouting ahead
- Investigating nebulae
- Scanning before engagement
- Supporting long-range hacking
- Checking dangerous areas without committing the ship

Radar should take care when reporting from probe-linked views, as some readouts may be unintuitive. Confidently wrong bearings are still wrong; they merely arrive with better posture.

Database

Radar should check the database when time permits and report relevant updates.

This is especially important when:

- New mission information appears
- A faction or ship type is unfamiliar
- The crew is searching for a clue
- Comms receives information that may correspond to known data
- The Captain is making a strategic decision

6.1.5 - Coordination

Captain

Radar provides the Captain with:

- Contact reports.
- Threat assessments.
- New intelligence.
- Unusual signatures.
- Developing tactical risks.

Radar should report what changes decisions. The Captain does not need every detail; they need the right detail at the right time.

Helm

Radar supports Helm with:

- Bearings.
- Hazard warnings.
- Incoming contact direction.
- Enemy movement.
- Shield-facing opportunities.
- Warp navigation support.

Useful calls:

- *"Helm, turn bearing 050. Hostile contact incoming."*
- *"Minefield ahead, recommend heading 310."*
- *"Contact closing from aft."*
- *"Target shield two weak."*

Weapons

Radar supports Weapons with:

- Target analysis linking.
- Shield status.
- Frequency information.
- Enemy beam arcs.
- Long-range target tracking.
- Pre-emptive firing opportunities.

Example:



"Missiles, hostile on bearing 060, outside Tactical range. Prepare slugs on my mark."

This should be used carefully. Firing into the dark is only clever once one has established that the dark contains something deserving.

Navigation and Comms

Radar should coordinate with Navigation and Comms on:

- Probe deployment.
- Strategic map awareness.
- Contact marking.
- Mission intelligence.
- Database information.
- Hacking opportunities.

Navigation may know where the ship needs to go. Comms may know why. Radar may know what is waiting there with its weapons warmed.

Engineering

Radar may request scanner spikes for:

- Fast target identification.
- Improved scan stability.
- Probe-supported reconnaissance.
- Hacking support.

Radar should also listen for scanner heat warnings, especially if Navigation or Comms is hacking.

Standard Callouts

- *"{Number} contacts incoming, bearing {bearing}."*
- *"Unknown contact, bearing {bearing}."*
- *"Hostile contact, bearing {bearing}."*
- *"{Callsign} is {ship type}."*
- *"{Callsign} analysis on screen."*
- *"Target shield {number} weak."*
- *"Turn bearing {bearing}."*
- *"Spike scanners."*
- *"Unknown warp signature, bearing {bearing}."*
- *"Multiple warp signatures incoming."*
- *"{Callsign} entering nebula."*
- *"{Callsign} lost."*
- *"All clear."*

6.1.6 - Authority and Limits

Radar has authority to:

- Report contacts and identifications to the bridge.
- Request scanner spikes from Engineering.
- Advise Helm on bearings, hazards, and incoming threats.
- Link designated targets to analysis.
- Recommend pre-emptive fire when tactically justified.
- Report mission-critical database updates.

Radar must:

- Respect Captain and tactical lead target priorities.
- Avoid flooding the bridge with low-value information.
- Distinguish confirmed fact from informed assessment.
- Keep callouts concise and timely.
- Continue scanning even when the immediate fight becomes distracting.

Operational constraints include:

- Scan time.
- Operator attention.
- Nebula interference.
- Limited certainty at extreme range.
- Probe-relative confusion.
- The danger of turning useful information into noise.

6.1.7 - Summary

Radar is responsible for early warning, identification, and tactical awareness.

A good Radar officer does not simply see contacts; they make those contacts useful to the bridge. They call threats early, scan intelligently, link targets promptly, check the database, and report only what matters.

The essentials are simple:

- Call first, refine second.
- Prioritise scans by tactical value.
- Link the current target quickly.
- Tell Helm what is coming.
- Tell Weapons what matters.
- Tell the Captain what changes the decision.
- Check the database before everyone starts inventing theories.
- Do not drown the bridge in trivia.



All officers are encouraged to have a working knowledge of **Appendix B: Radar icons & typical ship information** on page 121. Radar Officers, however, should develop a much deeper familiarity with it, as rapid recognition is central to the role.

A good Radar officer prevents surprises.

A great Radar officer makes the enemy feel as though surprises have been administratively banned.

"The most decisive weapon on the ship is often nothing more than a well-trained pair of eyes.

We were searching for a missing ship. Command expected the operation to take days, perhaps longer if the trail had gone cold. We entered the sector with no confirmed contact, no beacon, and very little hope of a quick resolution.

Lt. Tash was at Radar. Moments after arrival, they noticed the faintest irregularity in the gravitational signal — nothing dramatic, nothing obvious, just a trace that did not sit quite right. They called for a probe on that bearing and insisted we give it a look before committing to the wider search pattern.

The wreck was exactly where they said it would be.

That single call cut three days off the operation and gave the survivors the difference between rescue and recovery.

A good Radar operator is worth their weight in gravium."

- Lt. Cmdr. Paige O'Callaghan

6.2 - Navigation



"Helm may decide how we move. Navigation decides whether that movement is wise."

6.2.1 - Role and Priority

Navigation maintains the ship's strategic picture.

Where Radar watches for what is approaching, and Helm controls the ship's immediate movement, Navigation is responsible for understanding where the ship is, where it needs to be, and what lies between those two points.

Navigation is not primarily a combat station. Its value lies in context, foresight, and route management. A good Navigation officer prevents the ship from winning a fight in entirely the wrong place, which is very impressive right up until someone asks where the mission objective went.

Priority is therefore:

- Maintain awareness of mission objectives and their locations.
- Keep the Captain informed of the wider strategic picture.
- Provide Helm and Drones with routes, waypoints, and movement priorities.
- Mark hazards, hostiles, objectives, and areas of interest clearly.
- Support probe deployment, scanning coverage, and wider reconnaissance.
- Hack targets as part of a wider tactical plan.

For full details on the Strategic Map and display limitations, see **2.0 - Reading the Battlespace** on page 10.

6.2.2 - Core Responsibilities

Strategic Awareness

Navigation must maintain an accurate understanding of the operational area.

This includes:

- Current ship position.
- Mission objectives.
- Friendly assets.
- Known hostile positions.
- Hazards and restricted areas.
- Stations, docks, and resupply points.
- Probe coverage and unexplored areas.
- Possible retreat or fallback routes.

Navigation should be able to answer the Captain's most common strategic question *"Where are we going next?"* Preferably before the Captain has to ask it twice.

Objective Tracking

Navigation should track all known mission objectives and points of interest.

Objectives may change during a mission. New information may arrive through Comms, Radar, database updates, allied assets, or direct observation. Navigation must ensure those changes are reflected on the map and communicated to the bridge.

A mission objective that is not marked, remembered, or communicated has not vanished. It has merely been misplaced with confidence.

Route Planning

Navigation provides Helm with usable routes.

This may include:

- Direct routes to objectives.
- Routes avoiding hazards.
- Routes through or around nebulae.
- Approach vectors to stations or targets.
- Withdrawal routes.
- Routes for probe or drone reconnaissance.

Navigation should not overload Helm with excessive detail during active manoeuvring. Helm needs the next useful instruction, not a scenic lecture on the entire journey.

Waypoint and Marker Discipline

Navigation is responsible for clear and consistent map marking.

Waypoint colours and route markers must be used consistently so that Helm, Command, and other officers can understand the map at a glance.

Recommended standard categories:

Diamond route markers may be used to plot specific paths for Helm to follow. These appear as plotted route guidance rather than simple point markers, and should be used when Helm (or Drones) needs a route rather than a destination.

Colour standards are only useful if they remain standard.

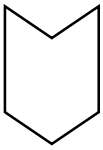
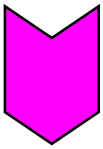
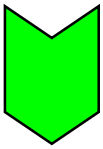
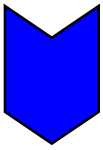
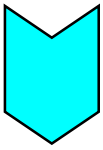
Colour	Marker	Use
1		Route / course plotting / immediate navigation / Point of Interest
2		Hazards / avoid areas
3		Known friendly assets
4		Known hostile assets
5		Unknown / suspect / investigate
6		Mission objectives
7		Drone Co-ordination

Table 6-1: Standard Waypoint/Route Colours

It is worthwhile for Navigation to utilise either a personal issued notepad and stylus, or electronic datapad/communicator for manual logging of the waypoints.

6.2.3 - Execution

Initial Setup

At the start of a mission, Navigation should immediately establish:

- Current ship position.
- Primary objective location.
- Known friendly assets.
- Known hazards.
- Available probes.
- Nearby stations or resupply options.
- Any pre-briefed hostile areas.
- Agreed waypoint colour standards.

Navigation should also confirm with Helm whether route markers are visible and understood.

Maintaining the Strategic Picture

Navigation should continuously update the map as the situation develops.

This includes removing or adjusting markers that are no longer accurate. Old information can be as dangerous as no information.

Navigation should ask:

- Is this marker still valid?
- Has the objective moved or changed?
- Has Comms received new instructions?
- Does the Captain know the strategic implication?

The map should become clearer as the mission progresses.

Supporting Helm

Navigation does not fly the ship. Helm does.

Navigation provides direction, route context, and strategic priorities. Helm converts that into movement.

Helm may have immediate tactical reasons to deviate. Navigation should provide context, not seize the wheel by proxy.

Supporting Command

The Captain needs the wider picture, but not every map detail.

Navigation should provide concise updates when:

- The ship is nearing an objective.
- A safer route exists.
- The current course creates risk.
- A mission objective has changed.
- Friendly assets are out of position.
- A retreat or resupply option is available.
- The strategic situation has shifted.

Useful calls:

- *"Captain, we are two sectors from Objective Blue 1."*
- *"Captain, alternate route available around the hazard."*
- *"Captain, current course takes us away from the convoy."*
- *"Captain, nearest resupply is Waypoint Green 2."*

Navigation's task is to help Command make decisions before the situation becomes urgent.

Probe Planning

Navigation often manages probe deployment from the Strategic Map.

Probes should be used to answer useful questions:

- What is inside that nebula?
- Is the route ahead clear?
- What is guarding the objective?
- Is there a hostile group beyond scanner range?
- Can Radar scan before we commit the ship?

Navigation should coordinate probe use with Radar, since Radar may need to link to a probe for remote scanning and detailed observation.

For full probe mechanics, see **3.10 - Probes & Drones** on page 47.

Hacking Support

Navigation may have access to hacking systems and is often well placed to use them when not actively managing routes or objectives.

However, hacking must not distract Navigation from maintaining the strategic picture.



Navigation should only prioritise hacking when:

- The route and objectives are clear.
- The Captain does not need immediate strategic support.
- The target is tactically valuable.
- Scanner heat is acceptable.
- It has been decided that Navigation is the primary hacking officer, or Comms is occupied or unavailable.

For full hacking mechanics, see **3.9 - Hacking & Cyber Warfare** on page 40.

6.2.4 - Systems and Tools

Navigation primarily uses:

- Strategic Map.
- Waypoint system.
- Route plotting tools.
- Probe controls.
- Hacking interface, where available.
- Manual notes or mission log.

The Strategic Map is Navigation's main tool.

Navigation should use it to maintain the broader operational picture. It is excellent for deciding where the ship should go. It is rather less excellent for noticing the missile about to become personal.

Waypoint System

Waypoints should be used deliberately and consistently.

Navigation should avoid cluttering the map with unnecessary markers. Every waypoint should have a purpose, and that purpose should be understandable to the rest of the bridge.

When placing waypoints, Navigation should consider:

- Who needs this marker?
- Is the meaning obvious?
- Is the colour/category correct?
- Will it still make sense in five minutes?
- Does Helm need a point or a route?

Route Plotting

Routes should be used when Helm needs a specific path rather than a general destination.

This is particularly useful for:

- Avoiding minefields.
- Skirting nebulae.
- Approaching stations safely.
- Plotting warp routes.
- Coordinating with probe coverage.
- Guiding Helm through complex areas.

Navigation should keep routes practical. Avoid plotting routes with too many unnecessary points.

Routes can also be used to draw areas on the Strategic Map. Creating a ringed route around an enemy minefield for example.

Mission Notes

Navigation should maintain written notes where appropriate.

These may include:

- Objective names and locations.
- Orders received.
- Points of interest.
- Suspected hostile areas.
- Probe results.
- Friendly asset tasks.
- Unresolved questions.

The Strategic Map shows position. Notes preserve meaning.

6.2.5 - Coordination

Captain

Navigation provides the Captain with:

- Objective locations.
- Route options.
- Hazard areas.
- Friendly asset positions.
- Strategic alternatives.
- Mission progress updates.

Navigation should report changes that affect decisions, not narrate every marker placed.

Helm



Navigation provides Helm with:

- Routes.
- Waypoints.
- Bearings.
- Hazard avoidance.
- Next destination.
- Strategic movement priorities.

Useful calls:

- *"Helm, route plotted - White 1."*
- *"Helm, next waypoint is Objective Blue 2."*
- *"Helm, avoid hazard marker to starboard."*
- *"Helm, route takes tight left turn ahead."*

Radar

Navigation coordinates with Radar over:

- Probe deployment and selection/linking.
- Remote scanning.
- Contact confirmation.
- Hazard identification.
- Nebula investigation.
- Long-range reconnaissance.

Radar may identify what something is. Navigation should ensure the bridge knows where it is and why it matters.

Communications

Navigation and Comms share strategic awareness.

Comms may receive new instructions, asset updates, or mission intelligence. Navigation should ensure those updates are marked, tracked, and turned into usable map information.

Useful call:

- *"Comms, confirm station location for marker."*
- *"Comms, is that new objective primary or secondary?"*

Engineering

Navigation may request scanner support when wider map coverage or probe-supported reconnaissance is required.

Useful call:

"Engineering, request scanner boost for probe sweep."

Navigation should not request scanner power casually, particularly if Radar or Hacking is also placing demand on the Scanner system.

Dock and Drone

Navigation should coordinate with Drones when scouting or remote operations affect the strategic picture.

This includes:

- Marking drone routes.
- Identifying areas for drone reconnaissance.
- Warning of hazards.
- Ensuring drone findings are recorded.
- Supporting recovery routes if needed.

Standard Callouts

- *"Blue 1 Objective marked."*
- *"Route plotted to {location/objective}."*
- *"Next waypoint is {location/objective}."*
- *"Route around hazard Pink 3 plotted."*
- *"Minefield hazard Pink 2 marked."*
- *"Hostile area Red 1 marked."*
- *"Probe launched to {Sector}."*
- *"Probe in {Sector} linked."*
- *"Nearest resupply is {location}."*
- *"Alternate route available."*
- *"Strategic map updated."*

6.2.6 - Authority and Limits

Navigation has authority to:

- Mark objectives, hazards, routes, and points of interest.
- Recommend routes and alternatives.
- Deploy probes where appropriate.
- Request scanner support for reconnaissance.
- Advise Helm and Captain on strategic movement.
- Maintain mission notes and map discipline.
- Use hacking systems when workload permits.

Navigation must:

- Respect Captain's mission priorities.
- Avoid overriding Helm's immediate ship handling.
- Keep waypoint colours and markers consistent.
- Avoid cluttering the map.
- Communicate route changes clearly.
- Ensure new Comms or Radar information is reflected accurately.
- Avoid becoming so absorbed in the map that the bridge loses strategic support.



Operational constraints include:

- Incomplete information.
- Changing mission objectives.
- Probe limits.
- Scanner coverage.
- Comms traffic delays.
- Radar uncertainty.
- Helm's immediate tactical needs.

6.2.7 - Summary

Navigation maintains the strategic picture.

A good Navigation officer keeps the ship oriented toward the mission, not merely moving through space. They mark objectives clearly, plot sensible routes, track hazards, manage probe coverage, and ensure the Captain and Helm understand where the ship is going and why.

The essentials are simple:

- Know the objective.
- Mark the map clearly.
- Use standard waypoint colours.
- Keep routes practical.
- Support Helm without trying to fly.
- Keep the Captain aware of strategic changes.
- Use probes to answer useful questions.
- Remove or update stale information.
- Do not let the map become cluttered nonsense.

When Navigation performs well, the ship moves with purpose. It arrives prepared and avoids unnecessary danger.

6.3 - Communications

"Information wins battles. Communication wins missions."



6.3.1 - Role and Priority

Communications is responsible for the ship's external information flow.

Where Radar watches the battlespace and Navigation maintains the strategic picture, Comms connects the ship to the wider operational environment: Command, allied assets, neutral parties, civilian vessels, hostile forces, stations, shuttles, and mission-critical sources of information.

Comms is often the first station to receive new orders, altered objectives, requests for assistance, intelligence updates, diplomatic openings, or threats delivered with regrettable confidence.

Priority is therefore:

- Maintain contact with Command and relevant external parties.
- Relay mission-critical information to the Captain clearly and promptly.
- Manage friendly assets in line with Command intent.
- Record important instructions and developments.
- Conduct negotiation, deception, intimidation, or diplomacy where appropriate.

Comms does not merely pass messages along. A good Comms officer understands what matters, condenses it, and delivers it at the right moment. A poor one reads the entire message aloud while the Captain is deciding whether the ship is about to explode.

For **Strategic Map** mechanics, see **2.1 - Strategic Map** on page 10 For **Hacking** mechanics, see **Hacking & Cyber Warfare** on page 40.

6.3.2 - Core Responsibilities

Mission Information

Comms is the ship's primary link to Command, local authorities, stations, allied vessels, and mission-specific contacts.

This includes:

- Receiving mission updates.
- Requesting clarification.
- Relaying status reports.
- Requesting support.
- Passing external instructions to the Captain.
- Seeking authorisation for restricted actions.

Not every incoming message is equally urgent. Comms must judge what needs immediate interruption, what can wait, and what should be recorded for later.

A useful Comms officer turns external noise into operational meaning.

Friendly Asset Tasking

Comms may be responsible for tasking friendly assets.

These may include allied ships, stations, convoys, civilian craft, support units, or shuttle crews.

Asset tasking should always follow the Captain's intent and should be clear enough to act on without further interpretation.

Poor tasking:

"Help us."

Better tasking:

"Escort One, hold near the convoy and intercept contacts approaching from bearing 220."

When the shuttle is deployed, Comms is its main point of contact. The shuttle may not hear bridge callouts, see the same information, or understand current priorities unless Comms tells it. Shuttle crews are brave, not psychic, and should be warned before the main ship starts throwing area-effect weapons about the place.

External Dialogue

Comms may speak directly with neutral, hostile, or unknown parties.

This can include:

- Negotiation.
- De-escalation.
- Delay tactics.
- PsyOps.
- Bluffing.
- Intimidation.
- Persuasion.
- Formal diplomatic exchange.
- Extracting information.

The Captain sets the stance. Comms conducts the conversation.

Before engaging in significant negotiation, Comms should understand the Captain's intent:

- Are we buying time?
- Are we seeking information?
- Are we avoiding combat?
- Are we trying to provoke a response?
- Are offers authorised?
- Are there boundaries we must not cross?

A Comms officer with charm, patience, and a flexible relationship with literal truth can be remarkably effective. Naturally, all such flexibility should remain within Command intent.

Record Keeping

Comms should maintain notes throughout the mission.

Record:

- Orders received.
- Names and callsigns.
- Requests for assistance.
- Promises made.
- Authorisations granted or denied.
- Locations mentioned.
- Asset tasking.
- Mission changes.
- Unresolved questions.

Comms traffic can become dense very quickly. Relying on memory alone is not advised.

6.3.3 - Execution

Handling Incoming Messages

Comms should rapidly assess incoming messages.



Ask:

- Is this urgent?
- Does the Captain need it now?
- Does another station need it instead?
- Does it change the mission?
- Does it require a reply?
- Should it be recorded?

If urgent, interrupt clearly:

"Captain, priority message from Command."

If not urgent, record it and wait for an appropriate moment:

"Captain, Comms has an update when ready."

Comms should avoid reading long messages verbatim unless the wording itself matters. Summarise first. Offer detail if needed.

Poor update:

"Command says there's something happening near the station and they're worried about it."

Better update:

"Captain, Command reports hostile movement near Station Égalité. They request immediate support."

Best update:

"Captain, Command reports hostile movement near Station Égalité. They request immediate support. Navigation has the station marked; nearest friendly asset is two sectors away."

That final version does not merely inform. It helps the Captain decide.

Tasking Assets

When directing allied assets, Comms should state:

- Who is being tasked
- What they are to do
- Where they are to do it
- Any restrictions
- When to report back

Examples:

- *"Shuttle, proceed to the marked station and report survivors."*
- *"Escort One, hold near the convoy and engage contacts approaching from bearing 220."*
- *"Allied vessel, withdraw to Sector AP12."*

Comms should confirm when an asset acknowledges an instruction, and report refusals, delays, or failures to the Captain.

Negotiation and Hostile Contact

When speaking with external parties, Comms should maintain tone and intent.

The Captain may ask Comms to stall, reassure, threaten, gather information, demand surrender, request access, or distract the enemy while the ship manoeuvres.

Comms should avoid making commitments beyond the Captain's authority unless explicitly permitted.

Useful phrases include:

- *"Stand by while we confer with Command."*
- *"We are prepared to assist, but require your position and condition."*
- *"Power down weapons and hold position."*
- *"Your statement has been received. Clarify your intent."*
- *"We are authorised to respond if fired upon."*

There is no need to sound panicked simply because the other party is trying to be dramatic.

Hacking

Comms has access to hacking systems. However, hacking should only be attempted when communications, asset management, and mission information flow are under control.

If Comms is actively negotiating, managing the shuttle, receiving Command updates, or tasking allies, hacking should normally wait or be handled by Navigation.

6.3.4 - Systems and Tools

Comms primarily uses:

- Communications interface.
- Strategic Map.
- AV controller.
- Hacking interface, where available.
- Manual notes or mission log.



The Communications interface is Comms's primary duty station and should be monitored continuously.

The Strategic Map may assist with asset positions, objectives, and external contacts, but Navigation normally maintains the cleaner strategic picture. Comms should coordinate with Navigation rather than duplicating or contradicting map work.

The database should be treated as an operational tool, especially when unfamiliar names, locations, factions, or mission terms appear in communications.

6.3.5 - Coordination

Captain

The Captain is Comms' primary internal recipient.

Comms should provide priority messages, mission updates, requests requiring decisions, negotiation summaries, asset status, and authorisation results.

Comms may interrupt the Captain for urgent or mission-critical information. That privilege should be used carefully. A Comms officer who interrupts for trivia will soon discover the Captain has facial expressions previously unknown to science.

Navigation

Navigation and Comms share strategic awareness.

Comms should pass new locations, objectives, hazards, and asset updates to Navigation for marking.

Useful calls:

- "Navigation, mark new objective."
- "Navigation, Command reports friendly asset at sector K12."
- "Navigation, mark this as secondary until confirmed."

Radar

Radar may help identify contacts referred to in communications.

Comms should coordinate with Radar when a message references an unknown ship, faction, callsign, or suspected hidden contact.

Useful call:

"Radar, can you confirm contact matching that callsign?"

Helm and Weapons

Comms should inform Helm and Weapons when external information affects movement, firing restrictions, friendly positions, surrender attempts, nuclear authorisation, or civilian proximity.

Useful calls:

- *"Helm, Command requests we hold outside station range."*
- *"Weapons, hold fire. Target is communicating."*
- *"Weapons, Command says disable only."*
- *"Weapons, nuclear authorisation granted."*

Engineering, Damage Control, and Drones

Comms should inform Engineering and Damage Control when external events affect ship readiness, docking, resupply, shuttle movement, or time constraints.

Comms should coordinate with Dock and Drone where friendly assets, shuttle tasking, or drone activity affect the wider mission.

This is especially important when friendly craft are near hostile targets or area-effect weapons.

Standard Callouts

Comms may issue:

- *"Captain, priority message from Command."*
- *"Captain, Comms update when ready."*
- *"New objective received."*
- *"Navigation, mark new objective."*
- *"Command requests status report."*
- *"Friendly asset requesting assistance."*
- *"Shuttle reports contact."*
- *"Shuttle, return to ship."*
- *"Weapons, hold fire - target is communicating."*
- *"Hostile vessel demanding response."*
- *"Comms are now attempting negotiation."*
- *"Asset tasking confirmed."*
- *"Message recorded."*
- *"Weapons, nuclear authorisation denied."*
- *"Weapons, nuclear authorisation granted."*

6.3.6 - Authority and Limits

Comms has authority to:

- Receive and respond to external communications.
- Relay priority information to the Captain.
- Task friendly assets within Captain's intent.
- Manage shuttle communications.
- Conduct negotiation or dialogue as directed.
- Access and consult the database.
- Request clarification from Command or external parties.
- Use hacking systems when workload permits.

Comms must:

- Keep the Captain informed of mission-critical information.
- Avoid making unauthorised commitments.
- Record important orders and updates.
- Coordinate asset tasking with Command intent.
- Warn friendly assets of danger where possible.
- Avoid allowing hacking or map work to distract from communications.
- Ensure important information reaches the right station.

Operational constraints include:

- High message volume.
- Conflicting priorities.
- Delayed responses.
- Ambiguous external instructions.
- Asset misunderstanding.
- Captain workload.
- Tactical noise.
- The regrettable tendency of hostile forces to talk precisely when everything else is on fire.

6.3.7 - Summary



Comms manages information, influence, and external coordination.

A good Comms officer does not simply read messages aloud. They interpret, prioritise, record, and relay what matters. They keep Command connected to the wider mission, ensure allied assets act usefully, warn friendly craft before they are endangered, and help the Captain make decisions with the latest available information.

The essentials are simple:

- Listen constantly.
- Summarise clearly.
- Record important details.
- Tell the Captain what changes decisions.
- Keep Navigation updated.
- Keep shuttle and allied assets informed.
- Do not promise what the Captain has not authorised.
- Use negotiation deliberately.
- Hack only when Comms duties are under control.

When Comms performs well, the ship is not isolated. It knows what Command wants, what allies need, what enemies are saying, and which lies are worth telling politely.

"A Comms officer once stalled a pirate destroyer with an exhaustive lecture on ancient Earth history.

It began, as these things often do, with the enemy demanding our surrender in the usual theatrical fashion. Weapons were hot, shields were failing, and Helm needed just a little more time to bring us around. The Captain told Comms to keep them talking.

Comms took this instruction personally.

What followed was a detailed, entirely uninvited explanation of an exhaustive lecture on ancient Earth history, and how key moments were not, in fact, as straightforward as pirates apparently believed. The hostile captain became so irritated that he kept arguing back, which was precisely the point.

By the time he remembered he had guns, we were out of his strongest arc and our missiles were loaded.

On another occasion, a different officer opened negotiations by enquiring whether the enemy had considered renewing their extended spacecraft warranty. The pirate commander was so confused, offended, and professionally wrong-footed that he demanded clarification rather than opening fire. Comms provided it. At length. With forms.

In both cases, the result was the same: the enemy forgot to shoot.

Never underestimate the tactical value of making a hostile commander so baffled, annoyed, or intellectually trapped that they spend ten seconds listening when they should have spent five seconds killing you."

- Lt. Cmdr. Phil "Rocky" Stoneman

7.0 - Engineering

"Engineering is where ambition goes to ask permission from physics."

Engineering governs the ship's capability to act.

It does not decide where the ship goes, which target is engaged, or what the mission priority should be. Instead, Engineering determines what the vessel can sustain: how much power is available, which systems can be pushed, where coolant is needed, and whether damaged systems can be restored before the next crisis arrives.

At its core, Engineering is capacity and survival.

Every action taken by the crew places demand somewhere in the ship. Helm needs movement. Weapons needs firepower and shields. Science needs scanners. Command needs all of them to remain available long enough for decisions to matter.

Engineering is therefore not a background function. It turns ambition into something the ship can survive attempting. A Captain may order pursuit. Helm may demand speed. Weapons may request firepower. Science may require scanner coverage. Engineering decides whether those demands can be supported, what they will cost, and what must be sacrificed if they cannot.

Good Engineering is not simply giving everyone more power. It is knowing when to boost, when to deny, when to reset, when to cool, and when to warn the bridge that the laws of physics have declined the request.

When Engineering performs well, the ship feels responsive, resilient, and difficult to exhaust. Systems recover quickly, spikes arrive when needed, heat is controlled before it becomes damage, and the crew trusts that power will be available at the decisive moment.

When Engineering falters, the ship becomes fragile. Weapons overheat, Helm loses movement, shields fail without warning, scanners collapse under pressure, and every station discovers that courage is not a recognised substitute for coolant.

Roles

Chief Engineer

Manages power, heat, coolant, and system effectiveness. Responsible for capacity, prioritisation, and shipwide stability.

Damage Control / Manual Engineer

Repairs damaged systems and manages SPANR deployment. Responsible for recovery, internal resilience, and keeping critical systems operational.

Dock and Drone

Controls docking systems, drone operations, and related support functions. Responsible for auxiliary craft, recovery, remote action, and extended operational reach.



7.1 - Chief Engineer

"Anyone can give more power. An Engineer knows whether the ship will survive receiving it."

7.1.1 - Role and Priority

The Chief Engineer manages the ship's power, heat, coolant, and system effectiveness.

Every station depends on Engineering. Helm needs propulsion. Weapons needs beams, shields, missiles, and tractor support. Science needs scanners. Damage Control needs the ship to remain functional long enough for repairs to matter. Command needs all of this to happen without the vessel turning itself into a cautionary anecdote.

Priority is therefore:

- Preserve shipwide survivability.
- Keep critical systems functional.
- Provide power where it will have the greatest effect.
- Control heat before it becomes damage.
- Support Command's intent within the ship's limits.

Engineering is not a vending machine for power requests. The Chief Engineer must judge what the ship can sustain, what it can risk briefly, and what must be refused.

For full mechanics on power, heat, coolant, spikes, and system effectiveness, see **3.1 - Power, Heat & Coolant** on page 14 and **Appendix E: Power Spikes** on page 129.

7.1.2 - Core Responsibilities

Power Management

The Chief Engineer controls how much power each system receives.

This means supporting the current tactical plan, not simply keeping everything high. Power assigned to one system is power unavailable elsewhere, and excessive allocation can drain reserves or generate heat faster than the ship can safely manage.

A good Engineer understands priority.

- If Helm must dodge, propulsion (manuevering and impulse) matters.
- If the ship is taking heavy fire, shields matter.
- If Weapons has a firing window, beams or missiles may matter.
- If Radar needs rapid identification, scanners may matter.
- If everything matters equally, the Engineer's first task is to decide which officer is wrong.

Heat and Coolant Control

Heat must be managed before it becomes urgent.

The Chief Engineer should watch for systems being pushed too hard, active systems generating additional heat, and officers leaving boosted systems running after the moment has passed.

Coolant should be assigned deliberately, not left in systems where it is not required. Engineering should warn early:

- *"Beams heating."*
- *"Warp cannot sustain."*
- *"Thrusters near thermal limit."*

Early warnings allow the bridge to adapt. Late warnings usually arrive accompanied by smoke and alarms.

Reminder: heat is a resource budget, just like power and coolant.

A system does not suffer reduced performance simply because it is hot. It only begins to degrade once it overheats — that is, once heat reaches 100° and further heat gain becomes system damage.

This means a system sitting at 99° heat functions exactly as well as one at 0°. The difference is not performance; it is risk. At 99°, there is no margin left. One more careless spike, burst, or sustained overload may push the system into damage.

In short: heat does not weaken a system until it crosses the limit, but it tells Engineering how close the ship is to taking system damage.

Spike and Boost Control

A Spike is a temporary increase in power for an immediate tactical purpose.

A Boost is a sustained increase in power for an ongoing requirement.

The Chief Engineer should distinguish between the two and expect officers to do the same. A spike should be sharp, useful, and reset quickly. A boost should be deliberate, supported by coolant where required, and understood as an ongoing cost.

Useful examples:

- *"Confirm. Spike front shields."*
- *"Deny. Beams overheating."*
- *"Confirm. Boost scanners to 150."*
- *"Reset missiles."*

Engineering should not be afraid to deny a request. A denied spike is often better than a granted one that damages the system immediately afterwards.

System Effectiveness

The Chief Engineer should monitor system effectiveness, not only power levels.

A system may be underpowered, overheated, damaged, hacked, or simply unable to deliver what another station expects. Engineering should report when a system cannot perform normally.

Examples:

- *"Manoeuvring impaired."*
- *"Scanners degraded."*
- *"Warp damaged, not reliable."*
- *"Beams are powered but heat-limited."*

This prevents other officers from assuming they have capability the ship no longer possesses.

Energy Endurance

Engineering must maintain awareness of the ship's energy reserve.

A ship with strong shields, hot weapons, and high propulsion may feel powerful for a short time. It may also be spending its future with admirable confidence.

The Chief Engineer should warn Command when energy becomes a limiting factor.



Useful calls:

- *"Energy reserves down to {number}."*
- *"Recommend disengage or reduce load."*
- *"Ship can support one more attack run, not an extended fight."*

Engineering does not set the mission priority, but it must tell Command what the ship can afford.

7.1.3 - Execution

Initial Setup

At the start of a mission, the Chief Engineer should establish the ship's baseline condition.

Check:

- Current energy reserve.
- Default system power levels.
- Coolant distribution.
- Damaged or degraded systems.
- Expected mission demands.
- Whether shields, scanners, propulsion, or weapons are likely to be priority systems.

Engineering should also listen to the Captain's opening intent. A patrol, rescue, pursuit, stealth approach, and heavy engagement all require different priorities.

If the Captain does not provide intent, Engineering should ask for it. Quietly, politely, and before the ship is already doing three incompatible things.

Supporting Combat

During combat, Engineering should anticipate demand rather than merely react to requests.

Likely combat priorities include:

- Shields before impact.
- Manoeuvring for evasion or shield facing.
- Beams during firing windows.
- Missiles during reload pressure.
- Scanners for target identification or hacking support.
- Warp only when escape or rapid repositioning is required.

Engineering should listen for tactical cues:

- *"Danger close."*
- *"Opening distance."*
- *"Target shields down."*
- *"Need fast scan."*
- *"Prepare to disengage."*

These tell Engineering where power will be needed before someone asks.

Reset Discipline

Most Engineering mistakes are not from spiking too late. They are from forgetting to reset afterwards.

After a spike or short-term boost, Engineering should return systems to sustainable levels unless told otherwise.

Useful calls:

- *"Spike complete, resetting shields."*
- *"Beams reset."*
- *"Scanners back to normal."*

If a system must remain boosted, Engineering should say so and warn of the cost.

"Beams held at 200. Heat rising."

That one sentence gives Weapons useful power, gives Command awareness, and gives the system a fighting chance not to become a fond memory.

Denying Requests

Engineering has both the authority and responsibility to refuse unsafe or impossible requests.

A denial should be brief and, where possible, include the reason.

Examples:

- *"Deny. Thrusters overheating."*
- *"Deny. Insufficient energy."*
- *"Deny. Warp damaged."*
-

Where possible, offer an alternative.

- *"Deny beams spike. Can give missiles."*
- *"Scanners can go to 150, not 200."*

This keeps the bridge moving rather than simply closing a door.



Post-Combat Recovery

After combat, Engineering should shift from crisis support to recovery, therefore restoring power reserves should become a priority.

Priorities usually include:

- Lowering unnecessary power allocations.
- Cooling overheated systems.
- Supporting repairs.
- Recharging shields where appropriate.
- Conserving and recharging energy reserves.
- Preparing for the next engagement.

Engineering should coordinate with Damage Control to identify what must be restored first.

A battle is not over because the enemy stopped firing. It is over when the ship is ready for the next unpleasantness. For more information, see **Appendix D: Post-Combat Recovery** on page 124.

7.1.4 - Systems and Tools

The Chief Engineer's primary tools are the power, heat, coolant, and system status controls.

Engineering should use them to maintain shipwide capability, not to pursue perfect numbers. The console is a live argument between physics, ambition, and the Captain's latest idea.

The key systems Engineering supports are:

- Reactor.
- Beam Weapons.
- Missiles.
- Manoeuvring.
- Impulse.
- Warp.
- Front and Rear Shield Generators.
- Scanners.
- Cargo Docks.

Engineering has access to Power Management, Damage Control and Systen Overview screens.

For the detailed behaviour of each system, see the relevant **Shared Systems** sections (starting on page 13.)

7.1.5 - Coordination

Captain

The Captain sets intent. Engineering defines what the ship can sustain.

Engineering should inform the Captain when:

- Energy is becoming critical.
- A planned action cannot be supported.
- The ship can only sustain a posture briefly.
- A system is too hot, damaged, or ineffective.
- Disengagement is advisable for technical reasons.

Useful calls:

- *"Captain, we cannot sustain this fight."*
- *"Captain, ship can support one more pass."*
- *"Captain, recommend withdrawal to cool and repair."*

Helm

Helm depends on Impulse, Manoeuvring, and Warp.

Engineering should support Helm during evasive movement, shield-facing manoeuvres, disengagement, and warp travel.

Useful calls:

- *"Spike thrusters confirmed."*
- *"Impulse boosted."*
- *"Warp heating."*
- *"Thrusters heat-limited."*

Helm should not discover propulsion limits by failing to turn.

Weapons: Beams

Beams depends on Shield Generators, Beam Weapons, and sometimes tractor support through Cargo Docks.

Engineering should support shield spikes, beam firing windows, and tractor use where needed, while warning when heat or energy becomes limiting.

Useful calls:

- *"Spike front shields."*
- *"Beams heating."*
- *"Rear shields boosted."*
- *"Cargo Docks boosted for tractor."*



Weapons: Missiles

Missiles depends on Missile power for reloading and may also create indirect demand through non-lethal beam use.

Engineering should support missile spikes when a decisive reload window matters.

Useful calls:

- *"Spike missiles confirmed."*
- *"Missiles reset."*
- *"Energy low; limit reload spike."*

Radar, Navigation, and Comms

Science roles depend heavily on Scanners.

Engineering should support scanner boosts for rapid scanning, probe-supported reconnaissance, and hacking where appropriate.

However, scanner heat must be watched carefully when hacking is active.

Useful calls:

- *"Scanners boosted."*
- *"Scanner heat rising."*
- *"Deny scanner spike; hacking heat too high."*

Damage Control (Manual Engineer)

Damage Control repairs the ship. Engineering keeps it functional while that happens.

The Chief Engineer should coordinate with Damage Control on repair priorities, especially when damaged systems affect current tactical needs.

Useful calls:

- *"Prioritise manoeuvring."*
- *"Warp can wait."*
- *"Repair scanners next."*
- *"Front shields critical."*

Engineering and Damage Control should work as one recovery loop: stabilise, repair, restore, and prepare.

Dock and Drone

Dock and Drone may require Cargo Dock power, drone support, recovery assistance, and tractor capability.

Engineering should support these when they align with Command intent, particularly during drone recovery, shuttle docking, or tractor-heavy operations.

Engineering should also be made aware when hot drones are incoming, as too much heat could damage the Cargo Docks.

Engineering can also request extra power from docked Drones in an emergency.

Standard Callouts

The Chief Engineer may issue:

- *"Confirm. Spike {system}."*
- *"Deny. {reason}."*
- *"Resetting {system}."*
- *"Boosting {system} to {level}."*
- *"{System} overheating."*
- *"{System} heat-limited."*
- *"{System} damaged."*
- *"{System} restored."*
- *"Energy reserves falling."*
- *"We cannot sustain this."*
- *"Coolant committed."*
- *"Recommend disengage."*
- *"Recommend power reduction."*
- *"Ship stable."*

7.1.6 - Authority and Limits

The Chief Engineer has authority to:

- Assign and redistribute system power.
- Allocate coolant.
- Approve or deny spike requests.
- Reset systems after spikes or boosts.
- Warn Command when the ship is operating beyond safe limits.
- Recommend disengagement on technical grounds.
- Support repair priorities with Damage Control.

The Chief Engineer must:

- Support Captain's intent wherever possible.
- Communicate limits clearly.
- Avoid unnecessary power waste.
- Prevent avoidable overheating.
- Reset temporary allocations promptly.
- Prioritise systems according to tactical need.
- Coordinate with Damage Control on damaged systems.
- Avoid treating every request as equally important.



Operational constraints include:

- Limited energy.
- Limited coolant.
- Heat buildup.
- Power redistribution time.
- System damage.
- Hacking effects.
- Conflicting station demands.
- Tactical urgency.
- The persistent belief among bridge officers that "just a little more power" is a plan.

7.1.7 - Summary

There is a difference between a 30 second pause between fights, and a diplomatic conversation that could take 10 minutes. The Chief Engineer needs to know how much time they have to manage the ship's capacity to act.

A good Engineer does not simply provide power. They provide the right power, to the right system, for the right amount of time. They know when to spike, when to boost, when to deny, when to reset, and when to warn the Captain that the ship is being asked to do something expensive.

The essentials are simple:

- Know the Captain's intent.
- Watch heat before it becomes damage.
- Keep coolant deliberate.
- Spike early, then reset.
- Deny unsafe requests clearly.
- Tell officers when systems cannot perform.
- Keep Command aware of endurance.
- Coordinate repairs with Damage Control.
- Preserve capability for the next crisis.

When Engineering performs well, the ship feels powerful without being reckless, responsive without being wasteful, and stubbornly alive despite everyone else's best efforts.

7.2 - Damage Control / Manual Engineer

"Engineering keeps the ship alive. Damage Control keeps it from staying dead."



7.2.1 - Role and Priority

Damage Control is responsible for restoring damaged ship systems during and after an engagement.

Standard UCN guidelines state a Chief Engineer should be incharge of the Damage Control screen. However, many crews feel than Manual Engineers are better placed to handle these repairs.

Owing to this, it is recommended that an agreement be made at the start of each mission as to which officer will take primary responsibility of handling the automated repairs.

In addition, having dedicated officers as part of an "Engineering Team" can be crucial when multiple hands are required to fix problems. It is recommended that other officers are given secondary roles of Engineering Team, so when the crisis occurs, there is no delay in working out who will leave their station to assist.

Where the Chief Engineer's main role is to manage power, heat, coolant, and system effectiveness from the console, Damage Control deals with the physical condition of the vessel itself. They direct repairs, deploy SPANRs, track damaged systems, and ensure the ship regains critical capability before the next disaster arrives with paperwork incomplete.

Priority is therefore:

- Restore systems required for immediate survival.
- Deploy SPANRs where they will have greatest effect.
- Keep the Chief Engineer informed of repair progress.
- Identify systems that are damaged, hacked, or failing.
- Support post-combat recovery and readiness.

Damage Control is not glamorous work. It is often quiet, urgent, and performed while other officers are doing something dramatic with missiles. Nevertheless, a ship with working weapons and no engines is not a warship. It is an ornament with ambitions.

For full mechanics on system damage, overheating, hacking recovery, and repair behaviour, see the relevant **Shared Systems** sections (starting on page 13).

7.2.2 - Core Responsibilities

Repair Prioritisation

Damage Control must decide which damaged systems need attention first.

Not all damage is equally important. A damaged system that is not currently required may wait. A damaged system that prevents movement, defence, awareness, or firepower may decide the fight.

Typical combat priorities are:

- Manoeuvring / Impulse - if the ship cannot move, turn, evade, or present shields.
- Shield Generators - if the ship is under sustained fire.
- Beams / Missiles - if offensive pressure is required.
- Scanners - if awareness, target analysis, probes, or hacking are needed.
- Warp / Docks - if disengagement, recovery, docking, drone operations, or tractor use are required.

These priorities are not fixed. They depend on the situation. A damaged Warp system may be irrelevant in a controlled fight, or absolutely critical if the Captain has just ordered withdrawal and several unpleasant ships are treating that as a suggestion.

SPANR Deployment

Damage Control manages SPANRs and should deploy them deliberately.

SPANRs should be sent where repair matters now, not merely where damage exists. Concentrating repair effort on one critical system is often better

than spreading support thinly across several inconveniences.

Useful questions:

- What system is limiting the ship right now?
- What system will be needed in the next thirty seconds?
- Is this damage preventing Command's intent?
- Is another system more urgent?
- Is a SPANR already repairing something that no longer matters?

In practice, restoring a critical system from 25% or 50% can be far more tactically important than fully repairing a system already out of immediate use. Damage Control should repair for capability first, perfection second.

SPANRs are useful. They are not magic.

> SPANRs can be used to compensate for insufficient coolant. With three SPANRs repairing a system, their combined repair rate can offset minor damage caused by overheating.
>
> In principle, this allows a system at 250% power, with 100% coolant and three SPANRs assigned, to avoid taking damage from overheating.
>
> This is not advised except in an emergency. SPANRs are not a replacement for coolant.

Status Tracking

Damage Control should maintain awareness of which systems are degraded, being repaired, restored, or still waiting.

The Chief Engineer and Captain need to know when capability returns.

Useful calls:

- *"Manoeuvring under repair."*
- *"Front shields repaired in ten."*
- *"Scanners restored."*
- *"Warp still damaged."*

Damage Control should not make the bridge ask repeatedly whether a critical system is back. If the answer matters, report it.

Hacking and System Suppression



Damage Control should understand that a system may be impaired by damage, hacking, heat, or power state.

Not every weak system is physically broken.

If a system appears ineffective despite being repaired, coordinate with the Chief Engineer and relevant Science officer. The problem may be hacking, heat, low power, or some other delightful insult from the universe.

SPANRs may also assist with hacked systems where applicable, so Damage Control should be alert when enemy cyber activity is affecting critical capability.

In addition, Warp Jammers will disable the Warp Drive regardless of its status.

Post-Combat Recovery

After combat, Damage Control shifts from emergency repair to readiness restoration.

Priorities include:

- Restoring damaged critical systems.
- Clearing remaining hacked effects where possible.
- Supporting shield recovery.
- Preparing propulsion for movement or escape.
- Checking whether weapons, scanners, and docks are operational.
- Confirming the ship is ready for further contact.

The end of a fight is not the end of Damage Control's work. It is simply the moment everyone else starts relaxing before they should.

7.2.3 - Execution

Initial Setup

At the start of a mission, Damage Control should identify:

- Number of available SPANRs
- Location of:
  - Spare OCPs
  - OCP Units
  - Quantum Polarisation Hubs

    - Masahirite Crystals
    - Crystal Restabiliser
    - Spare fuel cells
    - Reactor
    - Self destruct mechanism
- Likely mission priorities
- Expected repair priorities during combat
- Whether any systems are especially mission-critical

Damage Control should also know where repair information is displayed and how quickly they can redirect SPANRs when priorities change.

It is recommended that Manual Engineers familiarise themselves with the layouts of their ship. Two standard layouts for typical UCN ships (*Hoplite* Class Destroyer and *Skirmish* Class Frigate) are included in **Appendix H - Standard Deck Plans** on page 137.

During Combat

During combat, Damage Control should keep an eye on the Damage Control screen, and listen for system-related reports from the Chief Engineer, Helm, Weapons, Science, Command, and JUDITH.

Important cues include:

- *"Manoeuvring impaired."*
- *"Shields damaged."*
- *"Beams offline."*
- *"Scanners degraded."*
- *"Warp unavailable."*

Damage Control should act before being explicitly ordered where the priority is obvious. If Helm cannot turn and the ship is under fire, Manoeuvring does not require a committee meeting.

However, when priorities are uncertain, ask the Chief Engineer or Captain.

Useful call:

"Damage Control: manoeuvring and beams both damaged. Confirm priority?"

Repair Focus

Damage Control should avoid constantly moving SPANRs without reason.



Every reassignment has an opportunity cost. If a repair is incomplete, Damage Control will need to decide if it is better to finish it before switching, or if another system is more critical.

Getting Thrusters and Impulse both to 50% might be more preferable than getting one to 100%.

Good repair discipline means:

- Focus on what matters.
- Finish critical repairs where possible.
- Redirect quickly when priorities change.
- Do not chase every damaged system equally.
- Report restoration promptly.

Repairs should support the fight, not merely tidy the ship.

Working with the Chief Engineer

The Chief Engineer sees system power allocation and system damage. Damage Control sees repair state and SPANR allocation.

The two roles should operate as a loop:

- Chief Engineer identifies degraded capability.
- Damage Control assigns repair.
- Damage Control reports progress.
- Chief Engineer restores power or adjusts priorities.
- Both confirm when capability returns.

A repaired system may still be underpowered, overheated, or hacked. A powered system may still be damaged. Between Engineering and Damage Control, someone should know which particular flavour of problem is currently ruining the Captain's day.

Recovery Phase

Once immediate danger has passed, Damage Control should help return the ship to operational readiness.

Useful recovery calls:

- *"Recharging empty fuel cell."*
- *"Critical systems restored."*
- *"Remaining damage is non-critical."*
- *"Warp still unavailable."*
- *"Ship repair status green."*

The Captain should not have to discover during the next emergency that a vital system was still damaged from the last one. That sort of surprise is traditionally known as "poor administration", shortly before becoming "enemy advantage".

7.2.4 - Systems and Tools

Damage Control primarily uses:

- Damage status displays
- SPANR controls
- Repair progress indicators
- System condition reports
- Internal coordination with Engineering
- Physical repairs to the ship's systems

Damage Control does not need to control power or coolant directly, but must understand how damage interacts with system effectiveness. A system that is physically damaged may not perform as expected even if the Chief Engineer assigns power to it.

The role is therefore not just "send repair units". It is maintaining a practical understanding of what the ship has lost, what it is recovering, and what remains unreliable.

For full details on SPANRs and Manual Repair, see **3.11 - System Repair** on page 52.

There is a duplicate eningeering screen in the engineering corridor of most UCN vessels. This can be switched between Damage Control and Power Management. Manual Engineers and Chief Engineers should co-ordinate the screen usage between them

7.2.5 - Coordination

Chief Engineer

Damage Control works most closely with the Chief Engineer.

Coordinate on:

- Current system effectiveness
- Repair priorities
- Overheated systems taking damage
- Hacked systems requiring support
- Whether repaired systems can safely be powered
- Post-combat recovery

Captain



The Captain needs to know when damage affects mission decisions.

Report when:

- A critical system is lost
- A critical system is restored
- The ship cannot disengage
- The ship cannot continue fighting effectively
- Repairs will take significant time
- Recovery requires a pause or withdrawal

Helm

Helm depends on Impulse, Manoeuvring, and Warp.

Damage Control should prioritise propulsion when movement, evasion, shield-facing, or withdrawal depends on it.

Weapons

Weapons depends on Beams, Missiles, Shield Generators, and sometimes Cargo Docks for tractor support.

Damage Control should report when these systems are degraded or restored.

Science

Science depends heavily on Scanners, and may also be affected if hacking pressure is being used or received.

Damage Control should report scanner repair status clearly, particularly if Radar, Navigation, or Comms require scanning, probe support, or hacking.

Dock and Drone

Dock and Drone uses Cargo Docks for drone recovery, tractor support, and internal dock handling.

Damage Control should support Cargo Dock repair when drone recovery, shuttle operations, or tractor use is mission-critical.

Standard Callouts

Damage Control may issue:

- *"{System} damaged."*
- *"{System} under repair."*

- *"{System} restored."*
- *"Repair complete in {time}."*
- *"Critical systems restored."*
- *"Remaining damage non-critical."*
- *"Confirm repair priority?"*
- *"Warp unavailable."*
- *"Manoeuvring impaired."*
- *"Scanners restored."*
- *"Ship repair status green."*

7.2.6 - Authority and Limits

Damage Control has authority to:

- Deploy SPANRs to damaged systems.
- Reassign SPANRs as priorities change.
- Report damaged or restored systems to the bridge.
- Recommend repair priorities.
- Support hacked systems where applicable.
- Advise Command when damage limits the ship's options.

Damage Control must:

- Follow Captain and Chief Engineer priorities.
- Keep critical repair information flowing.
- Avoid spreading repair effort without purpose.
- Avoid redirecting SPANRs unnecessarily.
- Report restoration of critical systems promptly.
- Understand when damage affects current tactical capability.
- Support post-combat recovery before declaring the ship ready.

Operational constraints include:

- Limited SPANRs.
- Repair travel time.
- Repair duration.
- Multiple damaged systems.
- Ongoing incoming damage.
- Heat-induced damage.
- Hacking effects.
- Conflicting station demands.

7.2.7 - Summary

Damage Control restores the ship's ability to function.

A good Damage Control officer understands that not all damage matters equally. They prioritise repairs according to survival, mission need, and Command intent. They deploy SPANRs deliberately, report progress clearly, and ensure the bridge knows when critical capability has been lost or restored.



The essentials are simple:

- Know what is damaged.
- Know what matters now.
- Repair critical systems first.
- Concentrate SPANRs where they matter.
- Report progress without being asked.
- Coordinate constantly with the Chief Engineer.
- Do not waste repair effort on cosmetic inconvenience.
- Confirm when the ship is ready again.

When Damage Control performs well, the ship recovers faster than the enemy expects.

[[Behind the Curtain]]

Bridge Command policy is clear: **do not touch another officer's console without explicit permission.**

The Manual Engineer has no fixed console and moves around the ship fixing physical problems. Crews should therefore agree in advance how repairs will be handled.

Some Manual Engineers welcome help. Others prefer to handle repairs themselves unless asked. Neither approach is wrong, but expectations should be clear before launch.

If another officer is beside a damaged component, it may be quicker for them to swap a crystal themselves - but they should check first. That repair is part of the Manual Engineer's role and enjoyment.

In short: agree the crew policy first, then offer help rather than assuming control. "Do you want me to handle this?" is usually enough.

Manual Engineers may also handle cargo, NPCs, or fill missing crew spaces. As always, have the conversation before the mission so everyone is on the same page.

7.3 - Dock and Drone

"A small craft in the right place can ruin a very large plan."

7.3.1 - Role and Priority

Dock and Drone is responsible for the launch, control, recovery, and management of the ship's drone complement, as well as the handling of dock-related operations.

The role extends the ship's reach beyond its own hull. Drones can scout, distract, harass, intercept, deploy ordnance, support wider sensor coverage, and force enemies to divide their attention. Used well, they allow the main ship to influence more of the battlespace than it could alone.

However, drones are not designed to operate alone. They are ship assets. Their use must support Command's intent, avoid friendly fire, and remain coordinated with Operations, Engineering, and Science.

Priority is therefore:

- Support the main ship through effective drone use.
- Deploy drones where they create tactical or strategic advantage.
- Avoid obstructing friendly manoeuvres or weapons fire.
- Recover drones safely where practical.
- Manage docks, recovery, and heat-related risks.

Dock and Drone is a focused role. The officer may spend long periods controlling small craft with very little margin for error. Even so, they must not become isolated from the bridge. Drone action is only useful if it supports the wider fight.

7.3.2 - Core Responsibilities

Drone Deployment

Dock and Drone controls when and where drones are launched. Drones may be used to:

- Scout ahead of the ship.
- Harass hostile vessels.
- Draw fire away from the main ship.
- Deploy mines or other ordnance.
- Intercept incoming threats.
- Chase fleeing targets.
- Extend influence into areas the ship cannot safely enter.
- Support operations inside or around nebulae.

A drone does not need an immediate target to be useful. Scouting ahead, screening an approach, covering ground, or standing ready for rapid response are all valid uses. However, every launched drone should still have a purpose. Drones left idle may be slow to recover, difficult to protect, or abandoned if the ship must warp or reposition suddenly.

"Hold this area and report contacts" is a task. "Launch it because we might need it" is merely optimism with a flight plan.

Tactical Drone Control

Once launched, drones must be actively controlled.

The Drone officer should remain aware of:

- The Captain's intent.
- Helm's current target and manoeuvre.
- Missile firing lines.
- Area-of-effect weapons.
- Enemy beam arcs.
- Drone heat and survivability.
- Recovery routes back to the ship.

Drones can be used aggressively, but not carelessly. A drone in the wrong place can block fire, trigger unsafe detonations, or draw enemies into an unfavourable position for the main ship.

Dock and Recovery Management

Recovering drones is part of the role, not an afterthought.

Dock and Drone should monitor:

- Which docks are available.
- Which drones need recovery.
- Whether a drone is overheated.
- Whether the Thermic Dock is required.
- Whether recovery will place strain on Engineering.
- Whether the ship is moving in a way that makes docking difficult.

A drone that survives the battle but cannot be recovered cleanly is only slightly less annoying than a drone that nobly explodes while achieving nothing.

Heat Awareness

Drones can return hot, and dock handling may place thermal strain on the ship. The Thermic Dock is designed to take a hot drone, and dissipate the heat slowly.

Dock and Drone must coordinate with Engineering when recovering overheated drones, particularly if the Thermic Dock is occupied or multiple drones are returning in quick succession.

Useful call:

"Engineering, hot drone returning."

This gives Engineering time to prepare for any dock-related heat or power impact.

Dock Selection and Recovery Behaviour

Drone recovery is not always as orderly as officers might prefer.

When a drone returns to the ship, the docking system will assign it to an available bay according to internal docking logic. This assignment may not always appear obvious from the bridge. A returning drone may not select the bay the operator expected, and may not always choose the most tactically convenient or thermally appropriate dock.

[[Behind the Curtain]]

At time of writing, dock selection is effectively random. A returning drone will not reliably choose the most sensible bay, including the Thermic Dock.

To improve the odds, fill the dock slots above the Thermic Dock first, starting with the one immediately above it. This does not guarantee a hot drone will use the Thermic Dock, but leaves the docking logic fewer unhelpful options.

In practice: prepare the docks before recovery.

This matters most when recovering overheated drones. A hot drone entering the wrong bay may transfer heat into the dock system, creating avoidable strain for Engineering. For this reason, Dock and Drone should monitor recoveries carefully, warn Engineering before recovering overheated craft, and avoid assuming that the ship will automatically place a drone exactly where the officer wants it.

Where possible, Drone officers should manage recovery deliberately: clear the approach, confirm dock availability, and coordinate with Engineering if heat is a concern.

In short: docking is reliable, but not always intelligent. Treat recovery as an active operation, not a background process.

Ordnance Discipline

Some drones may carry or deploy ordnance. This creates tactical opportunity, but also significant risk.

Dock and Drone must understand:

- Blast radius.
- Friendly proximity.
- Missile safety calls.
- The ship's intended movement.
- Whether friendly drones or craft are inside the danger area.

Area weapons are not improved by enthusiasm. They are improved by distance, timing, and the absence of friendly assets in the blast zone.

7.3.3 - Execution

Initial Setup

At the start of a mission, Dock and Drone should confirm:

- Number and type of drones available.
- Drone loadouts, if applicable.
- Dock layout and recovery points.
- Location and use of the Thermal Dock.
- Any ordnance carried by drones.
- Expected mission use: scouting, combat, recovery, or support.
- Whether the Captain wants drones used conservatively or aggressively.

The Drone officer should also confirm with Missiles how drone ordnance and ship-launched ordnance will be coordinated.

If drones are carrying area-effect weapons, this should be explicitly understood before combat begins. "I assumed everyone knew about the nuke" is rarely a sentence that improves a post-mission report.

During Combat

In combat, Dock and Drone should use drones to support the ship's current tactical needs.

This may mean:

- Pressuring the same target as the main ship.
- Distracting a secondary hostile.
- Drawing fire away from damaged shields.
- Forcing an enemy to turn.
- Delivering ordnance into a better position.
- Keeping eyes on a fleeing or hidden target.
- Interfering with enemy movement.

Dock and Drone should listen carefully for calls from Helm and Weapons. If Missiles calls "Clear the Line", drones should move out of the firing path immediately. If Missiles calls "Danger Close - bearing {number}", drones should be moved clear of the blast area unless there is a very good reason not to.

A drone is a useful asset. It is not an excuse to make Missiles nervous.

Scouting and Forward Operations

Outside direct combat, drones can act as forward scouts or mobile observers.

This is especially useful when:

- Entering uncertain areas.
- Approaching nebulae.
- Searching for hidden contacts.
- Investigating hazards.
- Supporting remote probe operations.
- Monitoring flanks or escape routes.

Drones extend the ship's presence, but their operational range is influenced by scanner effectiveness. Dock and Drone should coordinate with Science and Engineering when extended drone range or improved remote awareness is needed. The upper screen on Dock and Drone stations can be toggled to the long-range sensors if required. This is especially useful to help navigate drones back to the mothership.

Useful call:

"Request scanner boost for extended drone operations."

Recovery

After combat, Dock and Drone should recover surviving drones where practical.

Recovery should be orderly:

- Identify returning drones.
- Prioritise damaged or overheated drones.
- Confirm dock availability.
- Warn Engineering of heat concerns.
- Recover drones without obstructing ship movement.
- Confirm drone status once docked.

If the ship is still manoeuvring aggressively, recovery may need to wait.

7.3.4 - Systems and Tools

Dock and Drone primarily uses:

- Drone control interface.
- Tactical display.
- Dock controls.
- Ordnance controls, where available.
- Tractor beam controls, where assigned.

The role is powered through the Cargo Docks system. Increased Cargo Docks effectiveness improves dock handling and related support functions. It will also affect tractor beam performance.

The Drone officer should understand the practical limits of their display. Drone tactical awareness is local. The wider situation must come from Radar, Navigation, Comms, and the Captain.

For full mechanics on drones, scanner scaling, tractor behaviour, and ordnance characteristics, see the relevant Shared Systems sections.

Energy Dock

Drones can syphon energy from the mothership to recharge, or redirect energy to the mothership to assist in an emergency.

7.3.5 - Coordination

Captain

Dock and Drone should follow Command intent.

Confirm whether drones are to be used for:

- Scouting.
- Combat pressure.
- Distraction.
- Ordnance delivery.
- Defensive interception.
- Conservation for later use.

Useful calls:

- *"Drone deployed for scouting."*
- *"Drone engaging secondary target."*
- *"Drone available for mine deployment."*
- *"Drone recovery in progress."*

Helm

Dock and Drone must remain aware of the ship's movement.

Coordinate with Helm when:

- Drones are near the ship.
- Recovering drones.
- Deploying mines or area weapons.
- Using drones to influence enemy movement.
- Operating near hazards.

Useful calls:

- *"Drone close to bow."*
- *"Recovering drone, hold course if able."*
- *"Drone deploying mine ahead of target."*

Weapons: Missiles

Missiles and Drones must coordinate carefully around firing lines and area-effect weapons.

Dock and Drone must respond quickly to:

- *"Clear the Line."*
- *"Danger Close."*
- *"Hold drone position."*
- *"Move drone clear."*

Dock and Drone should also report when drone ordnance is being prepared or deployed.

Useful calls:



- *"Drone mine ready."*
- *"Drone clear of blast radius."*
- *"Drone obstructing line, moving."*

Weapons: Beams

Beams may coordinate with Dock and Drone for tractor use and target control.

If tractor beam control is shared, crews should agree who has priority before combat begins. During combat, Beams will often have priority for defensive or tactical use. Outside combat, Dock and Drone may use tractor control for recovery, repositioning, or object handling.

Useful call:

- *"Request tractor control for drone recovery."*

Engineering

Dock and Drone must coordinate with Engineering on dock power, drone recovery, drone energy and heat.

Useful calls:

- *"Request Dock power."*
- *"Hot drone returning."*
- *"Multiple drones returning."*

Engineering should not discover a thermal problem only after Dock and Drone has posted it through the wall like a gift.

In addition, tranferring energy to the Drone must only be done with Engineering's authorisation.

Radar and Navigation

Radar and Navigation provide wider awareness for drone operations.

Coordinate when:

- Sending drones beyond immediate tactical view.
- Scouting ahead.
- Operating in nebulae.
- Searching for hidden contacts.
- Supporting probe or scan coverage.
- Maintaining awareness of drone position relative to objectives.

Useful calls:

- *"Navigation, mark location for drone pickup."*
- *"Radar, drone has contact."*
- *"Drone scouting objective area."*

Communications

Comms may coordinate drone use with allied assets, shuttle operations, or Command instructions.

This is especially important if drones are operating near friendly ships, stations, shuttles, or civilian craft.

Useful call:

- *"Comms, warn shuttle that nuke drone is active."*

Standard Callouts

Dock and Drone may issue:

- *"Drone deployed."*
- *"Drone engaging {target}."*
- *"Drone scouting {area/sector}."*
- *"Drone drawing fire."*
- *"Drone damaged."*
- *"Drone overheated."*
- *"Drone returning."*
- *"Drone recovered."*
- *"Hot drone returning."*
- *"Request Dock power."*
- *"Request scanner boost for drone range."*
- *"Drone clear of line."*
- *"Drone in blast radius."*
- *"Drone mine deployed."*
- *"Drone lost."*
- *"Tractor control requested."*

7.3.6 - Authority and Limits

Dock and Drone has authority to:

- Launch and control drones.
- Select drone positioning in support of Command intent.
- Recover drones when safe and practical.
- Use drone ordnance where authorised.
- Request scanner or dock support.
- Request or operate tractor control where appropriate.
- Warn other stations when drone position affects safety or firing lines.

Dock and Drone must:



- Respect Captain's intent.
- Avoid obstructing Helm or Weapons.
- Respond immediately to missile safety calls.
- Avoid careless use of area-effect ordnance.
- Coordinate hot recoveries with Engineering.
- Avoid sacrificing drones without tactical purpose.
- Keep the bridge informed when drones affect the wider engagement.

Operational constraints include:

- Limited drone numbers.
- Drone heat.
- Dock availability.
- Recovery time.
- Scanner-dependent range.
- Local tactical display limits.
- Friendly fire risk.
- Ordnance blast radius.

7.3.7 - Summary

Dock and Drone extends the ship's reach.

A good Drone officer uses small craft to create options: scouting ahead, distracting hostile vessels, drawing fire, delivering ordnance, recovering assets, and shaping the fight beyond the main ship's immediate position.

The essentials are simple:

- Deploy drones with purpose.
- Keep them clear of friendly fire.
- Coordinate with Helm and Weapons.
- Warn Engineering before hot recoveries.
- Use ordnance carefully.
- Recover drones where practical.
- Support Command intent, not personal adventure.
- Remember that every drone is useful only while it serves the ship.

When Dock and Drone performs well, the enemy is forced to fight more than one problem at once. Their attention is divided, their movement is pressured, and their plans become untidy. This is, in many respects, the highest calling of a drone: to be small, irritating, and tactically impossible to ignore.



Appendices

Appendix A: Shuttle Operations

A.1 - Overview

Most UCN vessels carry a shuttle, which is smaller, more lightly crewed, and far less durable in a sustained engagement, making it easy to underestimate. This is a mistake.

A shuttle is fast, agile, flexible, and capable of affecting a battle far beyond what its size suggests. Properly used, it can scout ahead of the main vessel, conduct scans, deploy mines, harass larger ships, exploit blind spots, recover personnel, and provide a second point of pressure in combat.

The shuttle should therefore be treated as a tactical asset, not a novelty.

A.2 - Primary Uses

The shuttle is most valuable when it is given a clear purpose. It should not simply be launched because someone wants to fly it, although history suggests this remains a powerful and recurring temptation.

Common shuttle tasks include:

- Reconnaissance and forward scouting.
- Mine-laying and hit-and-run attacks.
- Flanking larger targets.
- Exploiting enemy blind spots.
- Conducting scans away from the main ship.
- Acting as an independent tactical screen.
- Rescue or evacuation duties.
- Drawing attention away from the mothership.
- Supporting the main ship during complex engagements.

The shuttle's speed and manoeuvrability make it particularly effective at mine deployment and close-range harassment. Officers have reported success using the shuttle to conduct mine hit-and-run attacks and to fight craft significantly larger than itself when properly coordinated.

A.3 - Tactical Value

A deployed shuttle gives the crew something extremely valuable: a second position in the battlespace.

This allows the mothership to act in ways it could not manage alone. A shuttle can approach from a different direction, threaten an exposed shield facing, force an enemy to divide attention, or place ordnance where the main ship cannot safely go.

It may also be able to move into enemy blind spots, particularly against ships with limited beam coverage. With careful coordination, the mothership may even assist positioning through tractor beam use, though this requires considerable care and should not be attempted casually.

The shuttle is not merely "extra firepower." Its true value lies in making the enemy account for another moving piece.

And enemies, like junior officers, are rarely at their best when asked to think about two things at once.

Shuttle Tactical Capabilities

Most UCN shuttles have two forward firing lethal beams with two ordnance tubes, one forward firing and one rear facing.

A standard UCN shuttle loadout is:

- 8 Torpedoes.
- 8 HVLIs.
- 8 Mines.

The shuttle's ordnance can be restocked by docking with supply probes or other allied vessels, but not from the mothership.

However, if the shuttle is docked while the mothership itself is being resupplied, the shuttle will be resupplied at the same time.

A.4 - Scanning and Information Sharing

One of the shuttle's most useful capabilities is information support.

Scan results are shared between the shuttle and mothership. If the shuttle scans an enemy ship, that information becomes available to the main ship, and vice versa. This allows the shuttle to contribute meaningfully even when it is not directly attacking.

This is especially useful when:

- The mothership is busy or under pressure.
- Weapons needs target analysis completed quickly.
- Radar is saturated with multiple contacts.
- A target is outside the main ship's ideal position.
- The shuttle can approach from a safer or more useful angle.

If the main ship has an idle or spare tactical officer, assigning them to shuttle scan work can be a useful way to increase the crew's information-gathering capacity.

A scanned enemy is a known enemy. A known enemy is far easier to kill, disable, avoid, bargain with, or blame for whatever just exploded.

Note, owing to the peer-to-peer encryption used by probes, any probes launched by the shuttle will *not* share its information with the mothership.

A.5 - Mine-Laying

The shuttle is particularly well suited to mine-laying.

Its agility allows it to pass through or around the combat area, place mines in useful positions, and disengage before the enemy can respond. A well-flown shuttle can use mines to punish pursuit, disrupt enemy formation, or deliver decisive damage against larger vessels.

Mine deployment from the shuttle requires strong coordination between the shuttle's Helm and Weapons/Generalist officer. The shuttle must understand:

- Where the target is moving
- Where the mine will become dangerous
- Whether friendly ships or drones are nearby
- How quickly the shuttle can clear the blast area
- Whether the mothership intends to manoeuvre through the same space

Mines are powerful, but they are not polite. Once placed, they do not care whose idea it was.

A.6 - Crew Requirements

The ideal shuttle crew depends on mission complexity, crew availability, and officer experience.

Four Officers

Standard UCN guidelines have a four-officer crew, often with a ships XO acting as shuttle commander. Whilst this is a legitimate crewing option, many crews feel the forth officer is superfluous.

Three Officers

A three-officer shuttle crew is the most comfortable arrangement. Below are the shuttle roles, with their mothership equivalent.

- Shuttle Helm (Helm)
- Shuttle Generalist (Radar, Weapons, Comms, Navigation)
- Shuttle Engineering (Chief Engineer)

This allows each officer to focus on their own workload. Helm flies, Generalist manages targeting and ordnance, and Engineering handles power, coolant, repairs, and system stability.

This is strongly recommended for dangerous combat operations.

Two Officers

A two-officer shuttle crew is workable, but demanding.

A common arrangement is:

- Forward officer: Helm and tactical awareness.
- Rear officer: Engineering, weapons, scans, and general support.

This requires officers who are comfortable multitasking and communicating clearly. Several

experienced officers have reported that two-person shuttle operation is possible, though it is generally considered the sensible lower limit for sustained combat use.

One Officer

One officer can operate the shuttle in extremis. This is not recommended as standard practice.

Solo shuttle operation involves constant movement between stations, reduced awareness, delayed reactions, and significant heat-management challenges. It can be done, but the officer should expect to disengage periodically to cool systems, stabilise the craft, and contemplate the questionable chain of decisions that led them there.

Solo operation is therefore best reserved for emergencies or low-risk tasks.

A.7 - Internal Coordination

The shuttle crew must communicate constantly.

Because the shuttle is smaller and less forgiving than the mothership, errors compound quickly. Helm needs to know when weapons are ready. Weapons needs to know where Helm intends to pass. Engineering needs to know when the shuttle is about to manoeuvre hard, fire repeatedly, or enter danger.

Good shuttle communication includes:

- Helm warning before attack runs.
- Weapons calling mine or torpedo readiness.
- Engineering warning of heat or power limits.
- Clear calls before disengagement.
- Immediate reporting of damage or system loss.
- Confirmation before dangerous ordnance deployment.

The shuttle's strength is speed. Its weakness is that everything happens quickly, including mistakes.

A.8 - Coordination with the Mothership

The shuttle has limited strategic awareness compared with the main bridge and may not know the Captain's wider intent unless someone tells it.

For this reason, shuttle tasking should usually be managed through Comms, in line with existing asset-control duties. Comms on the mothership is responsible for managing friendly assets and should relay orders, ordnance deployment, status updates, and requests to the shuttle crew.

Clear shuttle tasking should include:

- Objective.
- Target or area of operation.
- Expected level of risk.
- Whether the shuttle may engage independently.
- Restrictions on mines, EMPs, or nuclear proximity.
- Recall conditions.
- How often status updates are expected.

Example:

"Shuttle, scout ahead to sector K12. Avoid engagement. Scan contacts and report."

Or:

"Shuttle, support attack on hostile destroyer. Harass from aft arc. Mines authorised, danger close calls required."

The shuttle crew should report:

- Launch and docking status.
- Contact sightings.
- Scan completions.
- Ordnance use.
- Damage or heat concerns.
- Disengagement.
- Need for support or recovery.

A shuttle that is not kept informed may accidentally fly into the mothership's plan. A mothership that is not kept informed may accidentally catch the shuttle in an area effect explosion.

A.9 - Tactical Employment

The shuttle is most effective when used to create opportunity rather than win the battle alone.

Good shuttle use includes:

- Striking when the enemy is focused on the mothership.

- Placing mines along likely enemy paths.
- Scanning targets before the main ship closes.
- Forcing enemies to turn away from the mothership.
- Drawing fire briefly, then disengaging.
- Finishing damaged or isolated targets.
- Supporting pursuit of fleeing vessels.
- Providing early warning from a forward position.

Poor shuttle use includes:

- Charging unsupported into the main enemy formation.
- Remaining in close combat after heat or shields become critical.
- Dropping mines without warning the main ship.
- Flying through friendly firing lines.
- Forgetting blast radii exist.
- Treating agility as invulnerability.
- Assuming the mothership knows what the shuttle is doing.

A shuttle survives by choosing when to be present. It should not loiter in danger unecessarily.

A.10 - Damage, Repair, and Recovery

The shuttle is more fragile than a full starship, but it also has advantages.

Its smaller hull can often be repaired quickly by the parent vessel. This makes it possible to use the shuttle aggressively in some circumstances, provided the crew understands when to disengage and recover.

This does not mean the shuttle should soak damage.

It means the shuttle can take calculated risks, withdraw, recover, and re-enter the fight if the situation allows.

A damaged shuttle should report immediately:

- Hull condition.
- Shield status, if applicable.
- Propulsion damage.
- Weapon or scanner loss.
- Whether it can return under its own power.
- Whether recovery assistance is required.

The mothership should avoid forgetting the shuttle exists once it leaves the bay.

A.11 - Heat and Power Management

The shuttle's systems require active management, especially during combat.

Small crews may be tempted to "set and forget" coolant, but sustained manoeuvring and weapons use can still create serious heat issues. Reports from shuttle crews indicate that solo or short-crewed operation often requires periodic disengagement to cool systems and recover control.

Engineering aboard the shuttle should prioritise:

- Manoeuvring / propulsion during attack runs.
- Weapons during repeated firing.
- Scanners when conducting active scan work.
- Cooling periods after hard manoeuvres or combat passes.
- Keeping enough reserve to escape.

A shuttle that cannot manoeuvre is in immediate danger. A shuttle that cannot fire is still useful. A shuttle that cannot leave is a rescue operation with ambitions.

A.12 - Suggested Shuttle Doctrine

The shuttle should generally follow these principles:

- **Have a definitive plan before launch.** - Launching without purpose creates distraction.
- **Stay in communication.** - The shuttle is useful only while its actions are understood.
- **Exploit mobility, not durability.** - Do not trade damage with larger ships.
- **Use mines deliberately.** - Mine-laying is powerful, but must be coordinated.
- **Scan when not shooting**. - Information gathered by the shuttle benefits the whole crew.
- **Disengage early.** - Waiting until systems fail is not bravery. It is paperwork.
- **Warn before danger-close actions** - Especially mines or anything involving large explosions.
- **Do not assume the mothership sees what you see.** - Report it.

- **Do not assume the shuttle sees what the mothership sees.** - Tell it.
- **Return before you become a problem.** - A recovered shuttle can launch again. Debris has fewer options.

[[Behind the Curtain]]

As much as Shuttle Operations have a tangible benefit, it does split the crew into two ships. If this is done poorly, it might mean:

- The shuttle crew have no idea what the plot is.
- The Captain has no idea what the shuttle is up to.

There is a balance to having a shuttle be both independent enough that the people playing the shuttle crew get to have fun, but are still part of the same crew and story as the main ship.

Too strict control from the mothership to the shuttle means the shuttle crew may be sat bored until "released". Too lax control might feel that the shuttle is a completely independent ship, and the crew might miss out on critical story points.

Sadly, this balance cannot be taught in a manual! But with sufficiently busy Comms, your crew can work it out along the way.

A.13 - Recommended Callouts

From Comms / Command to Shuttle

- *"Shuttle, launch and scout ahead to {sector}."*
- *"Shuttle, hold position and observe."*
- *"Shuttle, scan {target}."*
- *"Shuttle, support attack on {target}."*
- *"Shuttle, mines authorised."*
- *"Shuttle, disengage and return."*
- *"Shuttle, danger close. Clear {bearing / area}."*

From Shuttle to Mothership

- *"Shuttle launching."*
- *"Shuttle clear."*
- *"Shuttle engaging {target}."*
- *"Shuttle scanning {target}."*
- *"Mine deployed."*
- *"Danger close, mine deployed near {target / bearing}."*
- *"Shuttle taking damage."*
- *"Shuttle disengaging."*
- *"Shuttle returning to dock."*
- *"Shuttle requires assistance."*

A.14 - Summary

The shuttle is a force multiplier.

It extends the ship's reach, adds a second tactical position, improves scanning options, enables aggressive mine deployment, and gives the crew flexibility that a single hull cannot provide.

However, it is not independent by default, and it is not expendable. Its effectiveness depends on clear tasking, disciplined communication, careful heat management, and a shared understanding of what the shuttle is trying to achieve.

Used well, the shuttle can change the shape of an engagement.

"We were out on shuttle detail. Standard mission. Close enough to support, far enough to stay out of harm's way… or so we thought.

No warning. No call from Comms. Just a flash on scopes, an orange icon moving far too damned fast, and the sudden realisation of what was heading our way.

A shuttle's a fine piece of kit, but it is not outrunning a nuke detonation. We tried anyway. It didn't matter.

The blast caught us. Systems screamed, radiation alarms lit up the cockpit, and for a moment I was quite certain that was it.

It wasn't. We made it back. Barely. Never seen a shuttle on 1% hull before, nor a mechanic angrier when our sorry little craft docked.

The Captain picked up the callsign "Fallout" after that mission. Me? Why d'ya think they call me "Toast"?

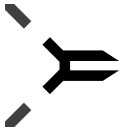
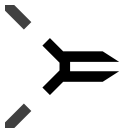
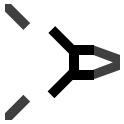
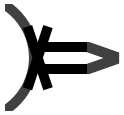
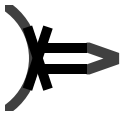
So cadets, remember this:

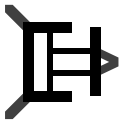
If there are friendlies in the area and you're firing a nuke… gorram warn us!"

- Lt. Cmdr. Ray "Toast" Hodson

Appendix B: Radar icons & typical ship information

The following radar icons are in order of ship size. Be aware that there is a great deal of overlap, and so this scale is not always 100%

Class	Icon	Hull (Average)	Shield Projectors (Average)	Shield Strength (Average)	Speed	Notes
Drone		40	1	20	Very High	**Behaviour:** Task-specific **Threat:** Mines, distraction, scouting **Action:** Threat commensurate with amount
Heavy Drone		125	0	0	Very High	**Behaviour** Task-specific **Threat:** Mines, distraction, scouting **Action:** Threat commensurate with amount
Starfighter		40	1	40	Very High	**Behaviour**: Swarm, harass, intercept, distract. **Threat:** Target saturation; type-dependent weapons. **Action:** Ascertain subclass: *Interceptor*: fastest, lighter firepower *Gunship:* heavier beams/HVLIs. *Bomber:* slower, heavier ordnance.
Shuttle		100	2	100	Moderate	**Behaviour:** Task-specific **Threat:** Mines, distraction, scouting **Action:** Ignore unless relevant
Freighter		100	2	50	Low to Moderate	**Behaviour:** Task-specific **Threat:** Usually low direct combat threat **Action:** Ignore unless mission-relevant
Frigate		75	2	50	Moderate to High	**Behaviour:** Fast, flanking, opportunistic **Threat:** Positioning advantage, can overwhelm through coordination **Action:** Track movement, support Helm
Assault Frigate		250	2	200	Moderate to High	**Behaviour:** Fast, flanking, opportunistic **Threat:** Positioning advantage, significant offensive/defensive capabilities **Action:** High threat target

Class	Icon	Hull (Average)	Shield Projectors (Average)	Shield Strength (Average)	Speed	Notes
Destroyer		210	4	225	Moderate	**Behaviour:** Closes distance, trades damage confidently **Threat:** Heavy shields (often 4–5 segments), High ordnance (nukes/EMP likely) **Action:** High threat target
Cruiser		500	4	400	Moderate	**Behaviour:** Highly variable **Threat:** Can be missile-heavy, beam-heavy, or specialist - Often misjudged **Action:** Assess role quickly (missile vs beam)
Battlecruiser		500	4	450	Low to Moderate	**Behaviour:** Minimal manoeuvre, overwhelming presence **Threat:** Massive shield pools. Sustained beam pressure or extreme durability **Action:** Avoid or commit fully with multiple ordnance strikes. No half-measures.
Battleship		800	4-6	550	Low	**Behaviour:** Minimal manoeuvre, overwhelming presence **Threat:** Massive shield pool. Sustained beam pressure or extreme durability **Action:** Only attack if no other option presents itself

Ordnance icons/behaviour

Icon	Ordnance	Behaviour
	Torpedo (Tracker)	Guided. Flies straight initially, then homes once the target is within its lock-on range approximatyley 1.7U[7].
	HVLI (Slug)	Unguided. Fires in a straight line and does not correct course. Requires good alignment and spacing; reaches full effectiveness after roughly 2 seconds in flight.
	Nuke	Guided area-effect weapon. Homes only at close range ~1U[7]. Large blast radius, dangerous to friendly assets and potentially) other ordnance.
	EMP	Guided area-effect weapon. Homes only at close range ~1U[7]. Large blast radius, dangerous to friendly assets and potentially) other ordnance. Damages/disrupts shields rather than carrying damage through to hull.
	Mine	Armed 10 second after deployment. Stationary when armed unless influenced by tractor beam. 1s fuse after an object comes within 0.6U. 1U blast radius (note - mine blast radii overlap, so mines will need to be laid at least 1.2U apart.)

[7] effective lock-on range varies by target size - examples given are for a target 0.5U wide

Appendix C: Target Priority Assessment

Target priority is not a fixed list. A heavy hostile may be dangerous, but a swarm of drones, a missile boat, or a fast craft threatening a convoy may be the more urgent problem. Officers should therefore assess priority by effect, not by class.

In combat, the tactical lead should ask: which enemy can we affect now, which enemy is hurting us or the mission most, which enemy will become dangerous if ignored, and which target best matches our current position, weapons, and objectives?

Many times, this information will need to be obtained from other stations. Indeed, sometimes Radar can take over as Tactical Lead from Helm or Captain, as they'll have this information quicker.

A destroyer slowly closing at long range may be less urgent than three drones already inside the defensive perimeter. Conversely, a small fighter may be safely ignored if allied assets are handling it and the real threat is preparing a missile strike.

Target priority should therefore be reviewed throughout the engagement. The correct target at the start of the fight may not remain correct once shields fail, friendlies move, ordnance is launched, or the mission objective changes. A good crew does not simply attack the biggest contact. It attacks the contact whose removal most improves the situation.

Question	Why it matters
What is the current mission objective?	Target priority must serve the mission, not merely the fight.
Which enemy threatens the mission objective most?	A hostile attacking the convoy, station, shuttle, or rescue target may matter more than the one shooting at you.
Which enemy threatens our survival most?	If the ship is about to lose shields, propulsion, or weapons, that threat may need immediate attention.
Which enemy is causing the most immediate harm?	Current damage output often matters more than theoretical danger.
Which enemy will become dangerous next if ignored?	A distant ship, flanking force, missile boat, or fast craft may not matter yet, but may dominate the next phase.
How quickly can we engage effectively?	A target outside arcs, range, firing line, or tactical position may not be the right immediate choice.
Which target is best suited to our current weapons and position?	Beams, missiles, mines, drones, hacking, and tractor use may each favour different targets.
Which target is vulnerable right now?	Weak shields, exposed facings, poor positioning, or damaged systems can create short-lived opportunities.
Where will unengaged enemies move if left alone?	Ignored enemies may flank, pursue friendlies, reinforce, flee, or enter stronger firing positions.
Are friendly assets already handling part of the fight?	Allied vessels, drones, shuttles, stations, or mines may already be containing a threat.
Are friendly assets at risk if we use area weapons?	Nukes, EMPs, mines, and drone-heavy engagements may make some targets unsafe to attack.
Can hacking, tractor, drones, or manoeuvre reduce the threat without destroying it?	Some threats can be delayed, displaced, disrupted, or contained rather than killed immediately.
Will changing target waste pressure already applied?	Constant switching can leave several enemies damaged but none removed. Very democratic. Not very lethal.
What happens after this target is destroyed or disabled?	A good target choice improves the next stage of the fight, not just the next firing window.

Table C-1: Threat Priority Questions

Appendix D: Post Combat Recovery

Combat does not end when the last hostile disappears from the Tactical Display. It ends when the ship is once again ready to survive the next engagement.

Post-combat recovery is the process of returning the vessel from combat posture to operational readiness. This includes restoring shields, reloading weapons, recovering drones, cooling overheated systems, repairing damage, refilling power cells, and ensuring the ship is positioned for repair, resupply, or further tasking.

A ship that wins a fight but fails to recover properly has not finished the engagement. It has merely scheduled the next crisis.

D.1 - Calling the All Clear

Radar should call the end of immediate combat by declaring: *"All clear."*

This indicates that no hostile contacts are visible on long-range scanners and that the current engagement appears to have ended.

"All clear" does not mean the ship is safe. It means no active hostile is currently detected. Mines, unexploded ordnance, damaged systems, low shields, depleted cells, and hostile reinforcements remain capable of causing trouble without the courtesy of appearing on the enemy roster.

Once "all clear" has been called, the ship should move into post-combat reset unless the Captain orders otherwise.

D.2 - Recovery Priorities

After combat, all stations should work together to restore the ship's fighting condition.

The main priorities are:

- Confirm the battlespace is safe enough to recover.
- Recharge shields.
- Reload missile tubes.
- Recover, cool, and rearm drones.
- Cool overheated systems.
- Repair damaged systems.
- Refill power cells.
- Dock with a friendly asset where possible.
- Confirm readiness for further combat.

These actions do not always occur in strict order. A ship with empty shields and a hot Warp Drive may need Engineering's attention before Missiles gets the perfect tube arrangement. A ship surrounded by mines may need Helm and Radar focused before anyone relaxes into administrative tidiness.

Recovery is a coordinated process, not a queue.

D.3 - Immediate Safety Check

Before stations begin their own resets, the bridge should confirm that the ship is not about to drift into something regrettable.

Radar should continue monitoring for new contacts, warp signatures, hazards, or enemy reinforcements.

Helm should maintain safe position, avoid mines and debris, and ensure the ship does not casually wander back through the battlefield. Victory does not disarm mines. It merely gives them a quieter room in which to embarrass you.

Navigation should update the Strategic Map, removing stale information where appropriate and marking any remaining hazards, damaged assets, or useful resupply locations.

Comms should confirm whether Command, friendly assets, stations, shuttles, or mission contacts have new instructions or support available.

The Captain or XO should ensure that the crew understands the immediate recovery priority.

Useful call:

"Post-combat reset. Radar maintain watch, Engineering restore power, Weapons reload and recharge, Drones recover."

D.4 - Shield Recovery

Beams and Engineering should coordinate to recharge shields.

Shield regeneration is slow under normal shipboard conditions and should be treated as recovery, not combat defence. Standard UCN vessels recharge shields at 0.3 points per second, increasing linearly with shield generator power; at 200% this becomes 0.6 points per second. External docking support may improve this significantly.

Beams should report the state of both shield facings clearly i.e.:

"*Front shields at 18%, rear shields at 72%.*"

Engineering can then assign power and coolant intelligently. Shield generator heat comes from power allocation rather than shield activation, so sustained high-power recharge must still be managed properly.

Beams may need to leave their console briefly and speak directly with Engineering, especially if the bridge is noisy or multiple systems are recovering at once. This is not a failure of discipline; it is communication with legs.

Engineering should normally avoid committing all coolant to one shield generator unless there is a specific reason. Coolant is often more efficient when split across multiple hot or overpowered systems, particularly when both shields need recovery and other combat systems are still warm. Remember, coolant needs to be used.

Don't leave it in a place it's not being useful.

Once shield levels are restored, Beams should confirm:

"*Shields restored.*"

Engineering should then decide whether shield generator power should remain elevated, return to standard, or be reduced for energy recovery.

D.5 - Missile Reloading

Missiles should reload tubes according to tactical preference and expected mission need.

Missiles has discretion over which weapons should be loaded, unless the Captain has given specific instructions. A good post-combat loadout should reflect the situation ahead, not merely replace whatever was fired last.

Missiles should consider:

- Whether the ship expects another engagement soon.
- Whether the next threat is likely to require EMP, trackers, slugs, mines, or heavier ordnance.
- Whether nuclear authorisation is available or likely.
- Whether ordnance reserves are becoming low.
- Whether drones have used shared munitions.

Missiles should inform Helm when the tubes are ready, since Helm may need to know whether the ship is prepared for close engagement, area denial, or long-range fire.

Useful calls:

- "*Trackers and slugs ready.*"
- "*EMP reserve low.*"
- "*Nukes expended.*"

Missiles should inform the Captain if any weapon type is running low enough to affect future tactical options. The Captain and Comms can then request resupply if a suitable asset is available.

D.6 - Drone Recovery

Dock and Drone should recover surviving drones as soon as operationally sensible.

Drone recovery is part of the role, not a decorative epilogue. The manual already stresses that Dock and Drone should monitor available docks, drone recovery needs, drone heat, thermal dock requirements, Engineering strain, and whether the ship's movement makes docking difficult.

If multiple drones are returning, Dock and Drone should stagger recovery where possible. This is especially important if any drone is hot, since a returning overheated drone may create thermal strain on the docking system.

Useful call:

- "*Engineering, hot drone returning.*"

This gives Engineering time to prepare for dock-related heat and power impact. The existing Dock and Drone guidance makes clear that hot drones and thermic dock use require coordination with Engineering, particularly when several drones are returning close together.

Dock and Drone should also keep the thermic dock clear wherever possible. If a hot drone docks, it may automatically enter the thermic dock, but dock selection is not always as orderly as one might wish. A returning drone may not choose the bay the operator expected, including the thermic dock. In plain language: the docking system is reliable, but not especially bright.

Once drones are recovered, Dock and Drone should confirm their condition:

- *"Drones recovered, one hot, two ready."*
- *"Drone One destroyed, Drone Two rearming."*

Drones should generally not recharge their drones immediately after combat unless Engineering directs it. Drone recharge can rapidly drain the ship's power cells, and power recovery usually matters more than topping up a drone battery for neatness.

D.7 - Heat Recovery

Engineering should review system heat across the ship once immediate defensive recovery has begun.

All ship systems generate passive heat above 100% power, while some systems generate additional heat through use, including Warp, Combat Manoeuvre, Hacking, and Beams. Heat above safe limits damages systems, reduces effectiveness, and risks failure. Common post-combat heat sources include:

- Beam weapons after sustained fire.
- Impulse or Manoeuvring after evasive action.
- Warp after rapid repositioning.
- Scanners after repeated hacking.
- Cargo Docks after drone recovery or tractor-heavy operations.
- Shield generators after prolonged overpowered recharge.

Engineering should reduce unnecessary power, assign coolant to priority systems, and avoid leaving temporary combat boosts in place.

If a system is damaged and not immediately needed, Engineering may reduce its power while Damage Control repairs it. Depowering a damaged system does not slow repair, and can save energy or reduce heat strain while restoration is underway.

D.8 - Repairs

Damage Control should begin or continue repair work as soon as combat pressure allows.

The priority is not always "most damaged first." It is "most important next." A ship preparing to dock may need Impulse and Manoeuvring restored first. A ship expecting another attack may need Shields, Beams, Missiles, or Scanners. A ship needing to leave the area may need Warp above all else.

Engineering and Damage Control should work as one loop:

Stabilise. Repair. Restore. Prepare.

Damage Control should also check for minor system degradation. A system at 95% may appear almost healthy, but it is still not fully restored. Small damage left unattended has a nasty habit of becoming relevant at the least dignified possible moment.

Once critical repairs are complete, Damage Control should report what has been restored and what the next priority is.

D.9 - Power Cell Recovery

Engineering should restore the reactor to a recovery posture once shields, heat, and urgent repairs are under control.

The goal is to refill power cells quickly without leaving the ship helpless if new enemies appear.

To improve power recovery, Engineering may reduce power to systems that are not currently needed at full capacity, including:

- Shields.
- Beams.
- Missiles.
- Warp, if stationary.
- Cargo Docks, if drones are recovered.

Engineering should not normally remove all power from these systems unless the ship is in a secure position or docked. A sudden return to combat may require shields, weapons, or movement at short notice.

A useful default is to reduce non-essential combat systems to approximately 50%. This produces meaningful savings while keeping the ship ready to respond. Scanners should usually remain at 100%, and may sometimes be kept slightly boosted if the ship is still near hostile space. The ship cannot recover from a threat it fails to notice.

Once immediate power draw is under control, Engineering should overclock the reactor as far as is safe and useful.

Damage Control should monitor spare power cells and swap discharged cells into charging positions as installed cells become full.

Useful calls:

- *"Cells refilling."*
- *"Swapping full cell."*
- *"All cells full."*

D.10 - Docking with Friendly Assets

If mission priorities allow, Helm should seek out a friendly asset for repair, rearmament, recharge, or mission support.

Not every friendly asset can provide every service. Some may repair hull. Some may refill probes or ordnance. Some may recharge power. Some may do very little beyond occupying space and sounding reassuring on Comms.

Comms and the Captain should confirm what support is available. Helm and Navigation should then coordinate approach.

Useful calls:

- *"Comms, confirm this station can rearm us."*
- *"Navigation, mark nearest repair asset."*
- *"Helm proceeding to dock."*
- *"Docking approach."*

Docking can sometimes occur in unexpected ways, depending on ship and asset configuration. Some larger friendly vessels may be able to receive ships that appear, at first glance, far too large to dock sensibly. Helm should therefore communicate in plain English rather than relying on assumptions.

If docking is successful, each station should confirm what service is actually being provided:

- *"Missiles restocking."*
- *"Shields recharging."*
- *"Repairs underway."*
- *"Power cells refilling."*

D.11 - Using Drones as Reserve Storage

Once shipboard power cells are full, Engineering may consider using drones as ancillary energy storage.

This should be directed by Engineering, not initiated by Drones as a routine preference. The purpose is not simply to refill drone batteries because they are low. The purpose is to divert excess reactor output into another reserve once the ship's own cells are already full.

This should be done slowly and deliberately. Rapid drone recharge can deplete ship power cells quickly, which rather defeats the purpose unless one is conducting an experiment in avoidable disappointment.

Useful call:

- *"Drones, begin slow recharge under Engineering direction."*

Dock and Drone should acknowledge and monitor drone state, but Engineering should control the timing and priority.

D.12 - Final Readiness Check

Once recovery appears complete, the Captain or XO should call for a readiness check.

Each station should respond with a concise status:

- *Radar: "All clear, scanners normal."*
- *Navigation: "Route plotted."*
- *Comms: "No pending traffic."*
- *Helm: "Manoeuvring ready."*
- *Beams: "Shields restored, beams ready."*
- *Missiles: "Tubes loaded, reserves noted."*
- *Drones: "Drones recovered and ready."*
- *Engineering: "Ship stable, cells full."*
- *Damage Control: "Repairs complete."*

If any station is not ready, say so clearly. False readiness is worse than delay.

Once confirmed, the Captain may return the ship to standard cruising posture, continue the mission, dock for further support, or prepare for the next engagement.

D.13 - Summary

Post-combat recovery is not downtime. It is the bridge turning survival back into readiness.

The essentials are simple:

- Radar calls "all clear," but keeps watching.
- Helm keeps the ship safe from hazards.
- Beams and Engineering restore shields.
- Missiles reloads and reports reserves.
- Drones recovers craft carefully and manages heat.
- Engineering cools systems and restores power.
- Damage Control repairs what the ship needs next.
- Comms and Navigation find support, resupply, and the next objective.
- The Captain or XO confirms when the ship is ready again.

A disciplined crew exits combat with a plan. An undisciplined crew exits combat by relaxing, drifting into a mine, and giving the accident investigators a short but colourful report.

Recovery is part of the fight. Treat it accordingly.

"We came out of a routine engagement-nothing dramatic. Shields held, systems intact, and Engineering had managed the load cleanly throughout. Textbook work.

Once the last contact dropped off scopes, the Engineer reset power, cleared heat… and, satisfied with a job well done, stood down.

What he didn't do was recharge the reserve batteries.

We discovered that oversight about ten minutes later, when a routine scan turned into a brief contact, and suddenly there was nothing in reserve for spikes. No surge for shields, no burst for thrusters-just a very polite, very underpowered response to a situation that didn't quite warrant it.

We weren't in real danger. Just… sluggish. Inelegant. The Captain handled it. The contact disengaged. No harm done.

The Engineer, however, did not escape unscathed.

To this day, no one calls Lieutenant Harris by his given name.

Around the fleet, he is known-universally and without mercy - as "Half-Charge."

Engineers never sleep."

- Lt. Cmdr. Ray "Toast" Hodson

Appendix E: Power Spikes

A Spike is a short-term increase of power to a system, normally to maximum output. In standard bridge use, a Spike should be understood to mean raising the requested system from 100% power to as high as possible (**300%** in the below examples), holding it only for the required tactical window, and then resetting it immediately.

Power does not change instantly. A system takes approximately **7 seconds** to climb from 100% to 300%, and another **7 seconds** to return from 300% to 100%. This delay is critical. A spike requested at the moment of need will often arrive too late to matter.

Spikes are most useful when a short period of maximum performance is worth the heat and power cost. Common uses include:

- Strengthening shields before incoming ordnance.
- Increasing beam rate of fire during a firing window.
- Reloading missiles quickly before a planned strike.
- Improving manoeuvring or impulse response during evasive action.
- Extending scanner coverage or drone control range for a specific operation.

A spike should be requested early, confirmed clearly, and reset as soon as the tactical purpose has passed.

E.1 - Heat Limits

At 300% power, systems heat rapidly. From cold, with no coolant support, a single full-power spike may be held for approximately **5 seconds** before it must be reset if the crew wishes to finish below **100° heat**.

During the ramp up/ramp down period, the system will gain **~26° heat.** Therefore, to ensure the system doesn't overheat, an Engineer should reset the power to 100% when the temperature is at 70°

Engineering should not wait until the system is any higher than 70° before resetting unless the situation is genuinely desperate.

Remember, some systems will generate heat when in use, so this will need to be taken into consideration when calculating heat issues.

Coolant

Coolant takes slightly longer than power to redistribute (approximately one second more for a full distribution). Therefore, adding coolant to the system immediatley *before*, and leaving it in the system for the duration of the spike allows a 300% spike to be held for up to **20 seconds** before overheating. If full coolant has been applied, the system can reach **95°** heat before being reduced to 100% power.

Spike Power	Coolant	Safe hold at spike power	Reset at heat reading
150%	0%	~67 sec	~95°
	25%	~7m 40s	~95°
	50%	Indefinite	N/A
	75%	Indefinite	N/A
	100%	Indefinite	N/A
200%	0%	~26 sec	~90°
	25%	~44 sec	~95°
	50%	~1m 47s	~95°
	75%	Indefinite	N/A
	100%	Indefinite	N/A
250%	0%	~12 sec	~85°
	25%	~18 sec	~90°
	50%	~27 sec	~95°
	75%	~44 sec	~95°
	100%	~1m 41s	~95°
300%	0%	~5 sec	~70°
	25%	~8 sec	~80°
	50%	~11 sec	~90°
	75%	~15 sec	~95°
	100%	~22 sec	~95°

Table E-1: Power Spike Hold Time Table

The figures in **Table E-1** assume the system starts cold, power and coolant are raised *at the same time*, and coolant remains assigned during reset.

E.2 - Prepared Spikes

Where incoming damage is predictable, officers should use a prepared spike call rather than a sudden request.

A prepared spike warns Engineering that a spike is about to be needed, giving them time to select the correct system and stand ready on the power control before the exact timing window arrives.

Example:

- *"Engineering, prepare to spike front shields."*
- *"Engineering ready."*
- *"Spike on my mark. Three, two, one, mark."*
- *"Spiking front shields."*
- *"Reset front shields."*

This is especially useful against incoming HVLIs, torpedoes, nukes, EMPs, or any other attack where Radar, Helm, or Weapons can see the impact coming. It reduces delay, avoids Engineering having to find the correct control under pressure, and makes the spike more likely to reach full effect at the correct moment.

For urgent situations, a direct call may still be required:

"Spike front shields now."

However, where time allows, **prepare first, mark second** is usually more reliable than simply calling for power at the last moment.

E.3 - Defensive Shield Spikes

Shield spikes are most effective against predictable burst damage. The purpose is not merely to raise shields, but to have the shield generator at maximum effectiveness before the impact lands.

Because a shield generator takes about **7 seconds** to reach 300%, Engineering should aim to begin the spike around **8 seconds** before impact, allowing roughly one second of margin for human delay.

Recommended defensive spike call timings:

These distances are approximate. Relative motion matters. If the ship is closing with the incoming ordnance, call earlier. If the ship is moving away, the window may be slightly longer. In combat, slightly early is usually preferable to perfectly late.

E.4 - Standard Callouts

Useful spike calls should include the system and, where relevant, the reason.

Incoming Threat	Call shield spike when...
HVLI / Slug	Approximately **4U out**
Torpedo / Tracker	Approximately **1.5–1.6U out**
Nuke	Approximately **1.5–1.6U out**
EMP	Approximately **1.5–1.6U out**, unless shield dipping is planned

Table E-2: Incoming Ordnance Range Spike Timing

Examples:

- *"Engineering, prepare to spike front shields. HVLIs inbound."*
- *"Engineering, spike front shields on my mark."*
- *"Engineering, spike rear shields. Nuke inbound."*
- *"Engineering, spike beams. Firing window opening."*
- *"Engineering, spike missiles. Prepare follow-up strike."*
- *"Engineering, reset shields."*
- *"Engineering, reset beams."*

Engineering should acknowledge clearly:

- *"Confirm. Preparing front shields."*
- *"Confirm. Spiking front shields."*
- *"Deny. Front shields too hot."*
- *"Resetting beams."*

E.6 - Spikes vs Boosts

A boost is a longer term increase in power allocation, needed for when the circumstances arise. **Table E-1: Power Spike Hold Time Table** (page 129) can also be used to work out the Coolant vs Power requiments for a boost.

Not all spikes will need to be to 300%, or held for as long as possible. As always communication is key to ensure power and coolant are not wasted on spikes being called unnecessarily, or held for too long.

Appendix F: Warp Power Configurations

The following presets provide standard warp configurations for long-distance travel. They are designed to give Helm and Engineering a shared starting point when crossing known distances.

These assume a low-demand transit posture:

- Scanners at 100%.
- Manoeuvring at 100%.
- Impulse at 0%.
- All other non-essential systems at 50%. [8]
- No active beams/shields/hacking/tractor

The preset calculations have three safety limits:

- The Reactor must remain at 0° heat.
- The Warp Drive must not exceed 75° heat.
- The ship should arrive with at least 500MJ remaining from a 1500MJ reserve.

Included at the bottom is an "infinite" warp setting. This does not lose energy, or add heat to any system. As such, it is the fastest "base" speed if you do not wish to arrive at your destination at a deficit of any sort.

Settings have deliberately been rounded down to the nearest 5%. This makes them easier to set accurately under pressure and gives Engineering a small margin for error. A careful Engineer may choose to push Warp slightly above the listed value if they are watching heat and energy closely, but the table values should be treated as the recommended safe settings.

The below settings have been calibrated to travel the distances as fast as possible within the stated safety limits, assuming the ship has a base Warp 1 speed of 60U/min.

Distance	Warp Level	Warp Power	Warp Coolant	Reactor Power	Reactor Coolant	Top Speed	Travel Time	End Warp Heat	Energy Used
5 sectors	Warp 4	**210%**	100%	100%	0%	8.40U/s	~21s	~74°	~125MJ
10 sectors	Warp 3	**135%**	100%	100%	0%	4.05U/s	~57s	~74°	~225MJ
15 sectors	Warp 2	**160%**	100%	100%	0%	3.20U/s	~1m 39s	~73°	~315MJ
20 sectors	Warp 2	**155%**	100%	100%	0%	3.10U/s	~2m 14s	~74°	~415MJ
25 sectors	Warp 2	**150%**	100%	100%	0%	3.00U/s	~2m 52s	~74°	~515MJ
30 sectors	Warp 2	**145%**	100%	100%	0%	2.90U/s	~3m 32s	~72°	~615MJ
35 sectors	Warp 2	**145%**	100%	100%	0%	2.90U/s	~4m 07s	~72°	~715MJ
40 sectors	Warp 2	**145%**	100%	100%	0%	2.90U/s	~4m 40s	~68°	~805MJ
45 sectors	Warp 2	**140%**	100%	100%	0%	2.80U/s	~5m 26s	~66°	~905MJ
50 sectors	Warp 2	**140%**	100%	100%	0%	2.80U/s	~6m 02s	~65°	~1000MJ
∞	Warp 1	**125%**	55%	170%	45%	1.25U/s	-	0°	0MJ

Table F-1: Warp/Reactor Configurations for fastest Warp Travel

[8] Sure, all systems off is better, but that means people on those consoles can't do anything at all during a long warp!

Appendix G: Advanced Combat Manoeuvres

A UCN vessel does not survive combat by firepower alone. It survives by being in the right place, facing the right way, at the right moment.

The manoeuvres in this appendix are not basic ship-handling instructions. They assume Helm understands movement, shield facings, weapon arcs, combat manoeuvre, and the consequences of discovering a mine with one's hull.

Most are simple in concept, but require clear coordination between Helm, Weapons, Engineering, Radar, and Command. They should be treated as practical tools, not rigid scripts. Ship class, enemy behaviour, damage, heat, power, and available space will all affect execution.

Advanced manoeuvres are used to preserve shields, control range, break firing arcs, avoid ordnance, exploit openings, or create time for the crew to act.

Move with purpose. Call clearly. Coordinate early.

G.1 - About Face

G1.1 - Purpose

An About Face is a rapid turn used to present the aft of the ship to the enemy. It is primarily a defensive manoeuvre, used to protect a weakened forward shield, reset the engagement, or begin a controlled withdrawal.

G.1.2 - Conditions

Use when the forward shield is failing, rear shields are stronger, or the ship needs to open distance while keeping protection toward the threat.

Do not use if the rear shield is weaker, there are hazards behind the ship, or the turn would break an essential firing line at the wrong moment.

G.1.3 - Coordination

- Helm calls and executes the turn.
- Beams confirms shield status
- Engineering may spike manoeuvring or rear shields.
- Radar watches enemy movement and incoming ordnance.
- Missiles confirms that no firing line or area weapon will be spoiled by the turn.

G.1.4 - Execution & Callouts

1. "*Prepare About Face.*"
2. Beams confirms*: "Rear shield stronger" or "Forward shield failing.*"
3. Helm requests support if needed: "*Spike manoeuvring.*"
4. Helm turns through 180 degrees, avoiding over-rotation and maintaining awareness of hazards.
5. Beams confirms shield presentation: "*Aft presented.*"
6. Engineering resets as required: "*Reset manoeuvring.*"

G.1.5 - Risks

The manoeuvre can expose a weaker shield if called too late or without confirmation. It may also disrupt beam arcs, spoil missile shots, turn the ship into mines or hazards, or leave Helm fighting the turn while incoming fire continues.

An About Face is not an escape by itself. It creates the conditions for survival; the crew must still decide whether to withdraw, re-engage, or exploit the new position.

G.1.6 - Variants

- **Defensive About Face** - present the stronger aft shield.
- **Withdrawal About Face** - turn and open distance.
- **Baited About Face** - draw pursuit into mines, drones, or friendly arcs.

G.1.7 - Summary

The About Face is a deliberate defensive rotation, not a panic spin. Done early, it preserves the ship. Done late, it merely lets the enemy admire the other end.

G.2 - Breakthrough

G.2.1 - Purpose

A Breakthrough is a rapid forward pass through or past hostile ships, used to present the rear shield, escape a collapsing forward shield, or force enemies to turn and reacquire firing arcs.

Where an About Face turns the ship around, a Breakthrough uses the warp drive to move the ship through the engagement and makes the enemy do the turning.

G.2.2 - Conditions

Use when the forward shield is weakening but still strong enough to survive the pass, when the rear shield is fresher, or when a slow-turning ship needs to change shield facing faster than it can rotate.

It is especially useful against large or slow enemies, when breaking through a formation, or when surrounded and an About Face would not meaningfully present a safer shield.

Do not use if the forward shield is already failing, the route is cluttered, collision risk is high, mines or hazards are ahead, or Engineering cannot support a brief warp burst.

G.2.3 - Coordination

- Helm selects the line and executes the pass.
- Beams confirms front and rear shield state.
- Engineering prepares front shield, manoeuvring, and warp support.
- Radar calls enemy position, formation, and hazards.
- Missiles may prepare a mine drop or hold fire if the firing line becomes unsafe.

G.2.4 - Execution & Callouts

1. "*Prepare Breakthrough.*"
2. Beams confirms shield state: *"Front shield holding" / "Rear shield stronger."*
3. Helm calls: *"Spike front shields. Spike manoeuvring. Reset warp."*
4. Helm reduces impulse, turns to a clear line past or between hostiles, and avoids committing until the route is safe.
5. Once aligned: *"Reset manoeuvring. Warp burst on my mark."*
6. Helm engages a short warp burst, passing beyond the enemy and forcing them to turn toward the ship's rear shield.
7. If appropriate, Helm may drop warp and call: *"Fire mine."*
8. Missiles deploys the mine once able, remembering that the mine travels approximately 1U behind the ship before arming.
9. Once clear, Helm either continues opening distance to allow the front shield to recover, or calls an About Face to bring forward weapons back onto the enemy.

G.2.5 - Risks

A Breakthrough deliberately carries the ship through enemy firing arcs. The front shield may take several hits before the ship clears the formation, so this manoeuvre generally requires a stronger forward shield than an About Face.

Collision risk is significant. Helm may not be able to judge enemy size perfectly from the Tactical Display, and a warp-speed collision can damage shields, damage the hull, or stop the ship dead in precisely the sort of location one would rather not be stationary.

The manoeuvre can also spoil missile lines, outrun friendly support, or leave the ship exposed if the rear shield is not actually ready to receive fire.

G.2.6 - Variants

- Blind-Spot Breakthrough - pass a large or slow enemy and fight from behind its main firing arcs.
- Formation Breakthrough - cut between enemy groups to make them turn inward and bunch together.
- Mine Breakthrough - deploy a mine after the pass to punish pursuit or inward turns.
- Escape Breakthrough - use the pass to break encirclement and continue withdrawing. Can potentially be used to escape incoming ordnance.

G.2.7 - Summary

The Breakthrough is a controlled shove through the enemy line, used when turning is too slow or too predictable. Done well, it resets the fight and leaves the enemy turning after you.

G.3 - Special Delivery

G.3.1 - Purpose

Special Delivery is an advanced mine-delivery manoeuvre used to carry an armed mine toward a predictable enemy position at warp speed. Rather than leaving a mine behind and hoping the enemy obliges, the ship uses the tractor beam to hold the mine in place while Helm delivers it directly to the target area. This requires an overpowered Docks, and an underpowered Warp to ensure the mine remains in the tractor beam.

It is best used against stationary targets, slow targets, or enemies whose future position can be judged with confidence. In short: one does not post a mine to an enemy who insists on moving house.

G.3.2 - Conditions

Use when the enemy is stationary, committed to a predictable course, or unlikely to evade before the mine arrives. The technique can be used from considerable distance, potentially up to 10U or more, provided the target position is reliable and the ship can maintain tractor control.

Do not use if the enemy is manoeuvring unpredictably, the route is cluttered, the ship cannot safely enter warp, or Cargo Docks cannot be powered and sustained. Do not attempt it near friendly craft unless all relevant stations understand exactly where the mine is going.

G.3.3 - Coordination

- Helm lines up the delivery run and controls warp.
- Beams/Dock and Drone operates the tractor beam and holds the mine.
- Missiles deploys the mine on command.
- Engineering boosts Cargo Docks and supports warp power as required.
- Radar confirms target position and warns if the target begins moving.

G.3.4 - Execution & Callouts

1. *"Prepare Special Delivery."*
2. Radar confirms: *"Target stationary"* or *"Target course predictable."*
3. Helm aligns the ship on a straight delivery line toward the target position.
4. Engineering boosts Cargo Docks to approximately 230%, with 100% coolant.
5. Engineering drops Warp power to 48%.
6. Tractor operator sets tractor to max range (2U) to Hold the mine at 1U.
7. Helm calls: *"Drop mine."*
8. Missiles deploys the mine.
9. Tractor confirms: *"Mine held."*
10. Helm engages Warp 1 and maintains the delivery line. With Cargo Docks around 230%, the tractor should be able to hold the mine at Warp 1 for at least 10–15U, though Engineering should be ready to increase power if control begins to slip.
11. As the ship approaches the target position, Helm and Tractor coordinate release or positioning so the mine arms and detonates near the intended target.
12. Once delivered: *"Mine away"* / *"Clear delivery line."*
13. Helm then breaks away, continues past, or prepares for follow-up fire.

G.3.5 - Risks

This manoeuvre depends on prediction. If the enemy moves unexpectedly, the ship may deliver a mine to a tactically irrelevant patch of space, which is dramatic but not especially useful.

Helm cannot alter course at *all*, or the mine will be lost out of the Tractor.

Tractor control is the critical failure point. If Cargo Docks are underpowered, damaged, overheated, or hacked, the mine may fall behind, drift out of control, or arm in a place the crew did not intend. At warp speed, small errors become large errors very quickly.

The ship is also deliberately carrying an explosive hazard while moving at speed. Friendly assets, drones, shuttles, and allied ships must be kept clear of the delivery line.

G.3.6 - Summary

Special Delivery turns a mine from a passive trap into a delivered threat. Done well, it places the explosion exactly where the enemy is about to regret being. Done poorly, it is merely high-speed explosives management, which is not a phrase any insurer enjoys.

G.4 - Longbow Shot

G.4.1 - Purpose

A Longbow Shot is a long-range ordnance attack using Radar guidance, precise alignment, and tractor-assisted launch velocity to strike targets beyond the normal Tactical Display range.

The principle is simple: Radar sees farther than Weapons can comfortably fight. Helm and Missiles use Radar's bearing to line up the shot, then the tractor beam pushes the launched ordnance forward, extending its reach and reducing travel time.

G.4.2 - Conditions

Use when the target is outside Tactical Display range but visible to Radar, stationary, or already locked before leaving local tactical view.

This is especially useful for:

- HVLIs, which can continue travelling beyond Tactical Display range.
- Homing ordnance, where lock can be maintained after the target leaves Tactical Display.
- Opening an engagement before the enemy expects contact.
- Pressuring retreating or approaching enemies at standoff range.

Do not use if Radar cannot provide a reliable bearing, the target is manoeuvring unpredictably, friendly assets are near the firing line, or Helm cannot hold a stable launch heading.

G.4.3 - Coordination

- Radar provides bearing and range.
- Helm aligns the ship precisely on the firing bearing.
- Missiles locks or fires the ordnance.
- Beams/Dock and Drone operates the tractor beam, usually set to 2U Push.
- Engineering may boost Cargo Docks to increase tractor push strength (although this will provide negligable benefit owing to the mass of ordnance)

G.4.4 - Execution & Callouts

1. *"Prepare Longbow Shot."*
2. Radar calls: *"Target bearing 042."*
3. Helm turns to the bearing and stabilises: *"On bearing."*
4. For homing missiles, Missiles locks target while available and confirms: *"Lock held."*
5. For HVLIs, Helm and Radar refine the line: *"Bearing steady" / "Adjust left two degrees."*
6. Tractor operator sets tractor beam forward, 2U range, Push.
7. Engineering boosts Cargo Docks if required: *"Boost Cargo Docks."*
8. Missiles fires on command: *"Longbow away."*
9. Tractor pushes the ordnance forward as it leaves the ship, increasing effective range and helping it reach the target sooner.
10. Radar continues to call corrections or impact assessment: *"Shot tracking" / "Target manoeuvring" / "Impact expected."*

G.4.5 - Risks

This manoeuvre depends on alignment and prediction. HVLIs do not guide, so even a small bearing error at long range can turn a lethal shot into an expensive gesture at empty space.

Homing ordnance are more forgiving, but still depend on lock, travel time, and target behaviour. If the enemy manoeuvres, breaks lock conditions, or changes course sharply, the shot may fail.

G.4.6 - Summary

The Longbow Shot lets the ship fight beyond what Weapons can directly see, using Radar as the eyes and tractor-assisted ordnance as the reach. Done well, the enemy is hit before they believe the engagement has begun. Done badly, the crew has merely fired very confidently at maths.

G.5 - Monkey at the Zoo

G.5.1 - Purpose

Monkey at the Zoo is a pursuit-breaking manoeuvre in which a ship retreating astern deploys a mine, passes beyond it, then uses its forward tractor beam to push the armed mine towards the pursuing enemy.

The manoeuvre allows the ship to continue withdrawing while presenting its bow towards the threat and turning a standard mine deployment into a mobile barrier or close-range attack.

G.5.2 - Conditions

Use when withdrawing from an enemy while travelling in reverse, with the ship's bow facing the pursuing vessel. This is especially useful for:

- Discouraging or disrupting pursuit.
- Forcing an enemy to turn away or alter course.
- Delivering a mine towards an enemy without closing directly.
- Creating additional distance while continuing to threaten the pursuer.

Do not use if friendly assets are between the ship and the enemy, the pursuing vessel is manoeuvring unpredictably, or Helm cannot maintain a straight withdrawal.

G.5.3 - Coordination

- Helm retreats at full reverse impulse while keeping the bow towards the pursuing enemy.
- Missiles prepares and deploys the mine
- Engineering increases Impulse power to at least 110% immediately before or after deployment (unless already going faster than 6U/min).
- Beams or Dock and Drone configures the tractor beam to heading 0°, maximum range of 2U, and Push.

Command confirms that the firing line is clear of friendly craft.le benefit owing to the mass of ordnance)

G.5.4 - Execution & Callouts

1. *"Prepare Monkey at the Zoo."*
2. Helm confirms the enemy is directly ahead and begins withdrawing at full reverse impulse: *"Retreating astern."*
3. Tractor operator sets the beam forward at heading 0°, 2U range, Push: *"Tractor configured."*
4. Engineering raises Impulse power to 110% (if required): *"Impulse boosted."*
5. Missiles deploys the mine: *"Mine away."*
6. The ship continues travelling in reverse and passes beyond the mine before it arms.
7. Once the mine lies ahead of the ship, the tractor beam begins pushing it towards the pursuing enemy.
8. Helm maintains bearing and reverse speed: *"Holding line."*
9. Radar or Weapons monitors the pursuing vessel: *"Target continuing pursuit" / "Target altering course."*
10. Once the mine has detonated, been avoided, or passed outside tractor control: *"Mine clear."*
11. Engineering resets Impulse power when directed (if required).

If circumstances permit, Helm may reduce or reverse the withdrawal and advance behind the mine, continuing to push it towards the enemy. This should only be attempted once the mine's position and the enemy's course are clearly understood.

G.5.5 - Risks

Helm must hold a steady bearing. Turning may cause the mine to leave the tractor beam or drift back within danger range.

Friendly assets must remain clear of the mine's path and blast radius.

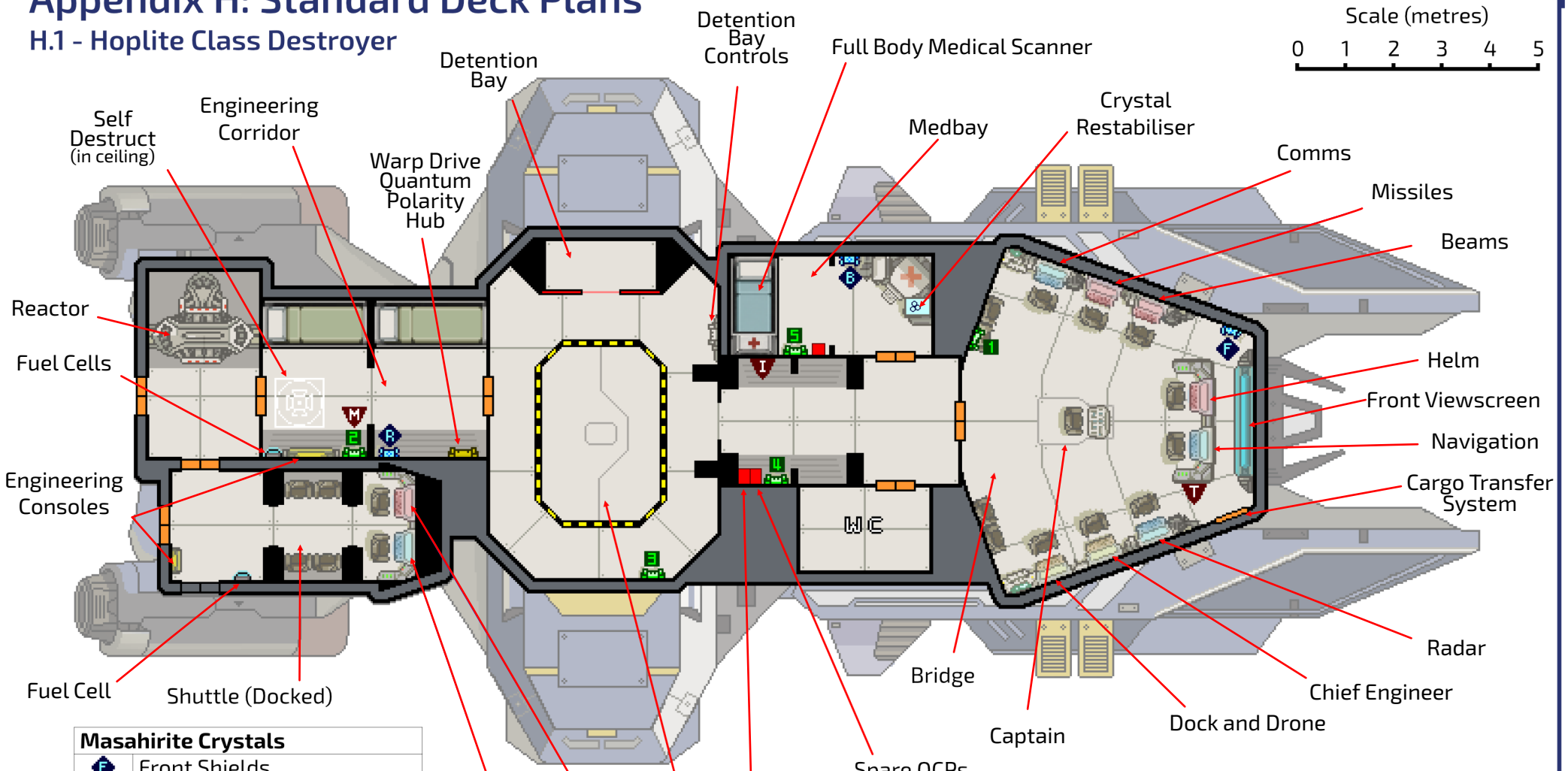
G.5.6 - Summary

Monkey at the Zoo turns a retreat into a moving mine attack, forcing the pursuer to evade or abandon the chase.

Done badly, the crew has simply placed an armed mine directly in front of itself.

Appendix H: Standard Deck Plans

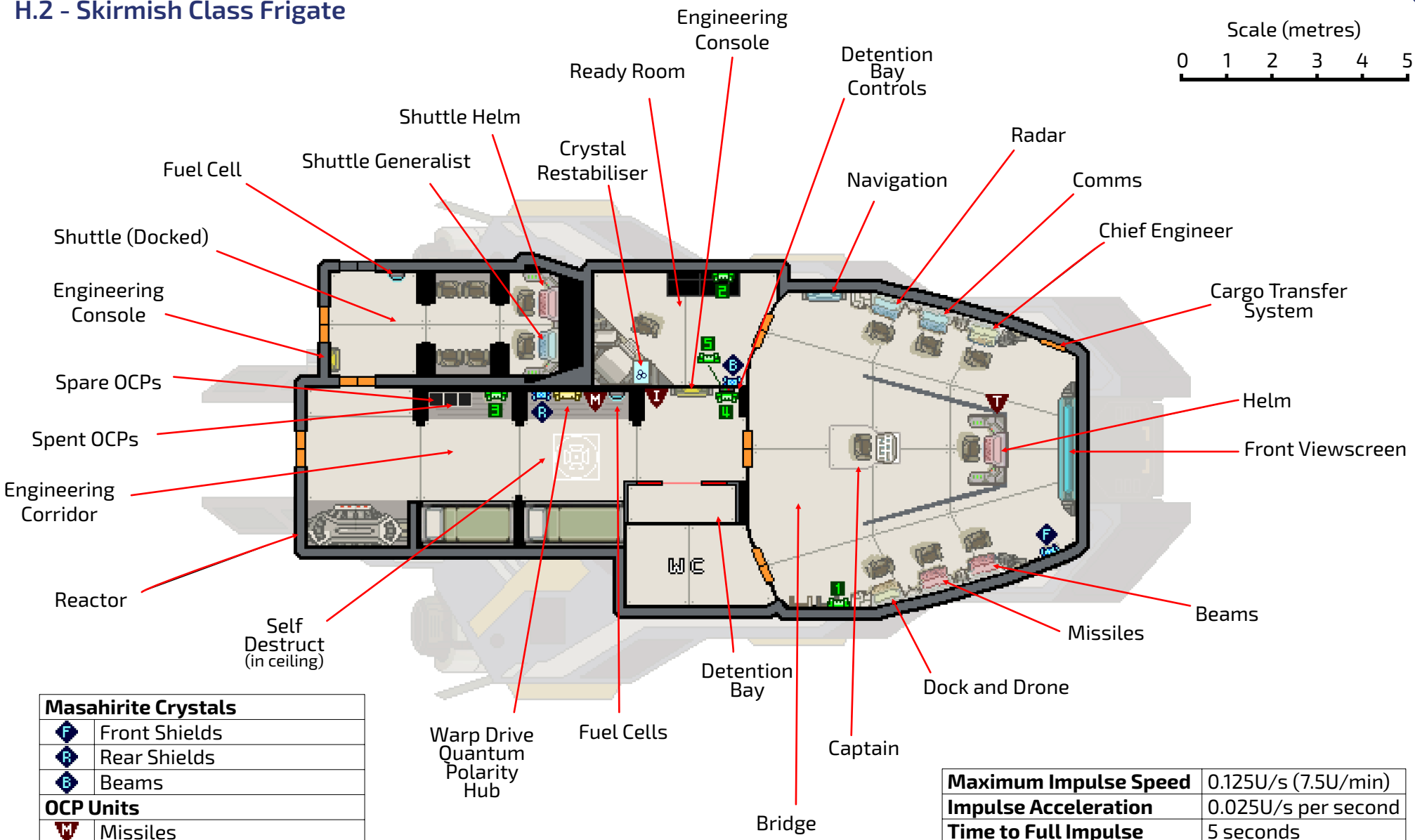
H.1 - Hoplite Class Destroyer



Masahirite Crystals	
F	Front Shields
R	Rear Shields
B	Beams
OCP Units	
M	Missiles
T	Manoeuvring Thrusters
I	Impulse
Quantum Polarisation Switches	
1 2 3 4 5	

Maximum Impulse Speed	0.10U/s (6U/min)
Impulse Acceleration	0.02U/s per second
Time to Full Impulse	5 seconds
Turning Rate	10°/s
Warp Speed	1U/s per Warp level
Shields	200:200
Hull Strength	250

H.2 - Skirmish Class Frigate



Masahirite Crystals	
F	Front Shields
R	Rear Shields
B	Beams
OCP Units	
M	Missiles
T	Manoeuvring Thrusters
I	Impulse
Quantum Polarisation Switches	
1 2 3 4 5	

Maximum Impulse Speed	0.125U/s (7.5U/min)
Impulse Acceleration	0.025U/s per second
Time to Full Impulse	5 seconds
Turning Rate	14°/s
Warp Speed	1U/s per Warp level
Shields	200:200
Hull Strength	200

Appendix I: Battlespace Examples

I.1 - Scanners at 100% Effectiveness

All of these images are from the same moment in time with Scanner effectiveness at 100%.

Note how even though T1 and T2 are inside the nebula, as they are within 5U of the ship (or probe), they are visible on all displays. However, Targets T4 and T5 are not visible to the long-range sensors, owing to the nebula blocking signals.

T3 is not visible to any sensors.

The Probe View (**Figure I-3**) shows T4 and T5, as they are within 5U of the probe.

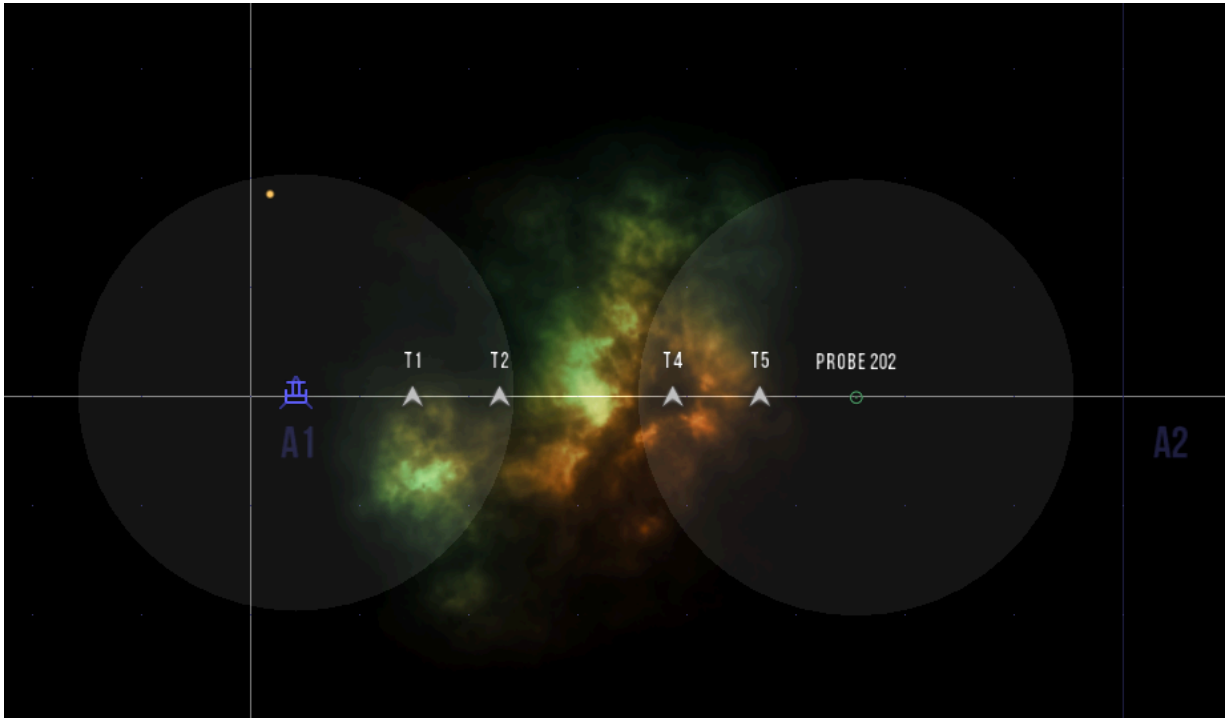


Figure I-1: Strategic Map with launched probe

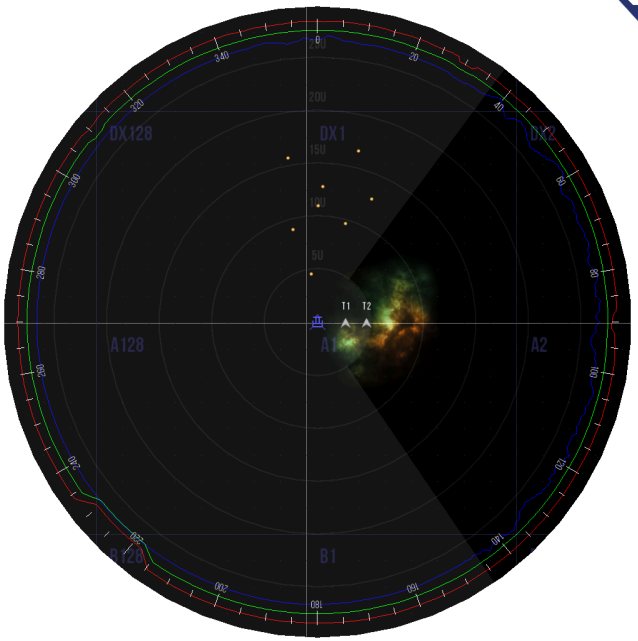


Figure I-2: Long Range Sensors

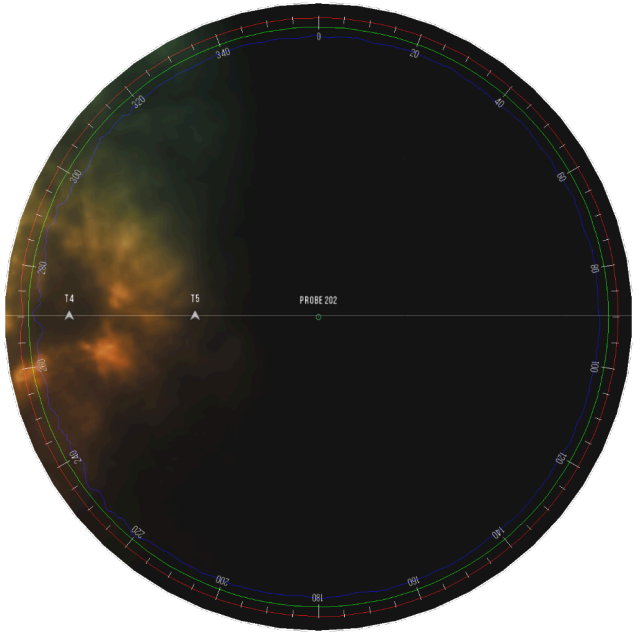


Figure I-3: Probe View

I.2 - Scanners at 230% Effectiveness

With scanners increased to 230% effectiveness, all five targets are now visible on the Strategic Map.

Note: even though the white circle on **Figure I-5** has increased, the Long-Range sensors still cannot see T3, T4 or T5. [9]

However, the Probe View has now been automatically zoomed out with the additional power, and so the probe view can see all five targets.

This means the Probe view ignores the blocking effects of a nebula at all times.

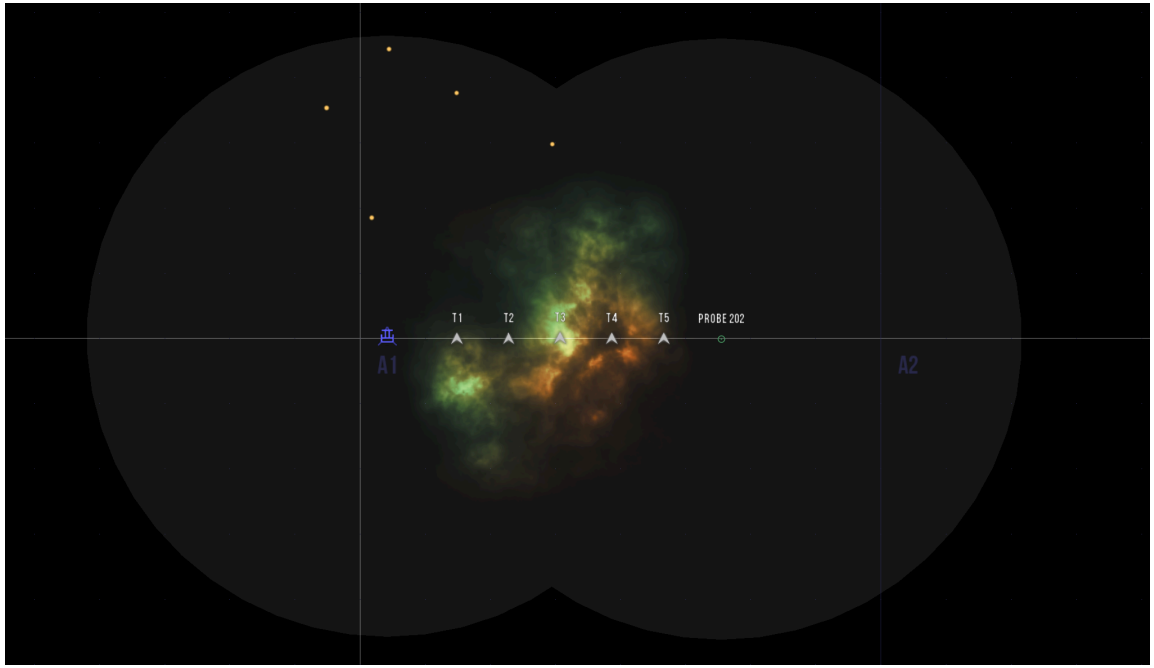


Figure I-4: Strategic Map with launched probe - 230% Scanners

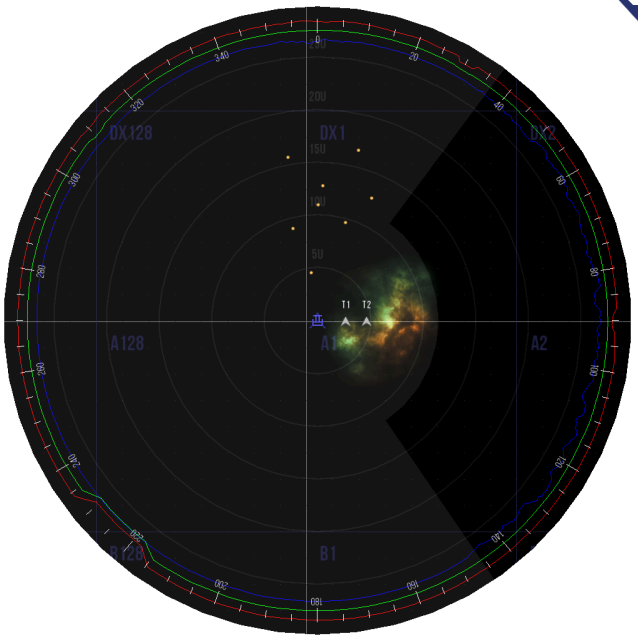


Figure I-5: Long Range Sensors - 230% Scanners

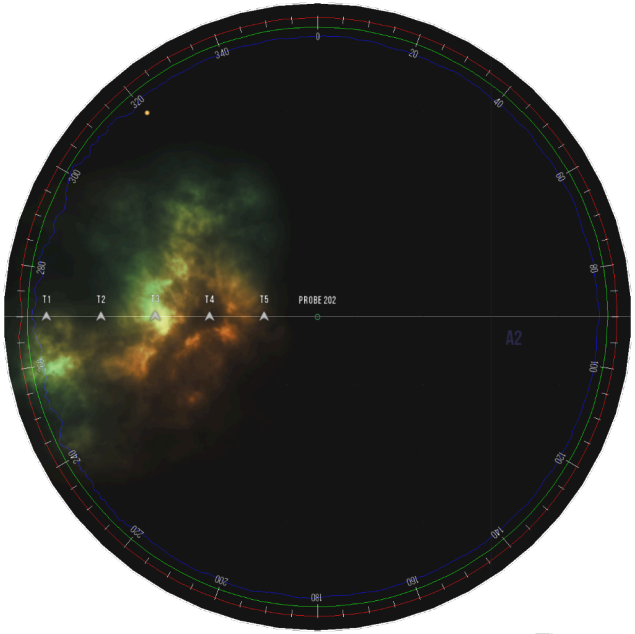


Figure I-6: Probe View - 230% Scanners

[9] This may be a bug in the software, and has been reported.

Afterword

Thank you for reading.

If you have made it all the way to the end, I am genuinely grateful. I hope this manual has helped you understand Bridge Command a little better.

If, against all probability, you have opened the book here first, then welcome! An unconventional approach, certainly, but not *entirely* out of keeping with Bridge Command.

This guide began with a simple desire: to understand Bridge Command a little better. After my first mission, I wanted to return not merely as an audience member with a console, but as an officer who understood what the ship was doing, why it was doing it, and how I could help the people around me do their jobs well.

That curiosity became research into all the deep dives and tactical discussion that came before me. The research became notes. The notes became tables, diagrams, procedures, conversations about terminology, and rather more discussion on warp power levels than any reasonable person should admit to enjoying.

Eventually, it became this.

For all the numbers, tactics, and technical detail contained in these pages, none of them are what makes a crew good.

A good crew communicates. They trust one another. They listen when someone else has better information or a better idea. They remain calm when the plan collapses, adapts when the ship catches fire, and remembers that every other person on the ship is there to enjoy the experience too.

You do not need to memorise this manual. You do not need to use every procedure, adopt every term, or agree with every conclusion. Bridge Command will change. Empty Epsilon will change. Some of the information here will eventually become inaccurate, obsolete, or rendered magnificently irrelevant by an update no one saw coming.

That is perfectly all right.

Whilst I'll do my best to keep this manual (at least the digital version) of this up-to-date, you can trust the community as a whole more than one person's obsession with writing it all down in a huge manual!

This remains an unofficial, fan-created guide, and *nothing* within it should be mistaken for official Bridge Command doctrine or fact. It exists only because of the generosity, patience, knowledge, stories, testing, and encouragement of the many people who contributed to it, and because Bridge Command itself created something worth caring about this much. My sincere thanks go to them all.

Take what helps. Question what does not. Test things for yourself. Develop procedures that suit *your* crew, and do not mistake familiarity with this manual for authority over *anyone* else's game.

The finest moments in Bridge Command don't always come from executing a procedure exactly as planned. They come when a crew understands one another well enough to improvise together: the last-second tractor deflection, the desperate warp escape, the probe that finds what everyone else missed, or the quiet officer who speaks at precisely the right moment and changes the course of the mission.

That is what this manual was written to support.

Preparation, not perfection. Confidence in your crew, not mastery of a console.

So close the book. Take your station. Listen to the officers around you, help them where you can (if they ask!), and remember that the objective is not to complete the mission...

It is to have a story worth telling afterwards.

Go Beyond.



- Callum “Face” Fraser
30th June 2026